

Science Refresh for Adults

Michael Gannotti

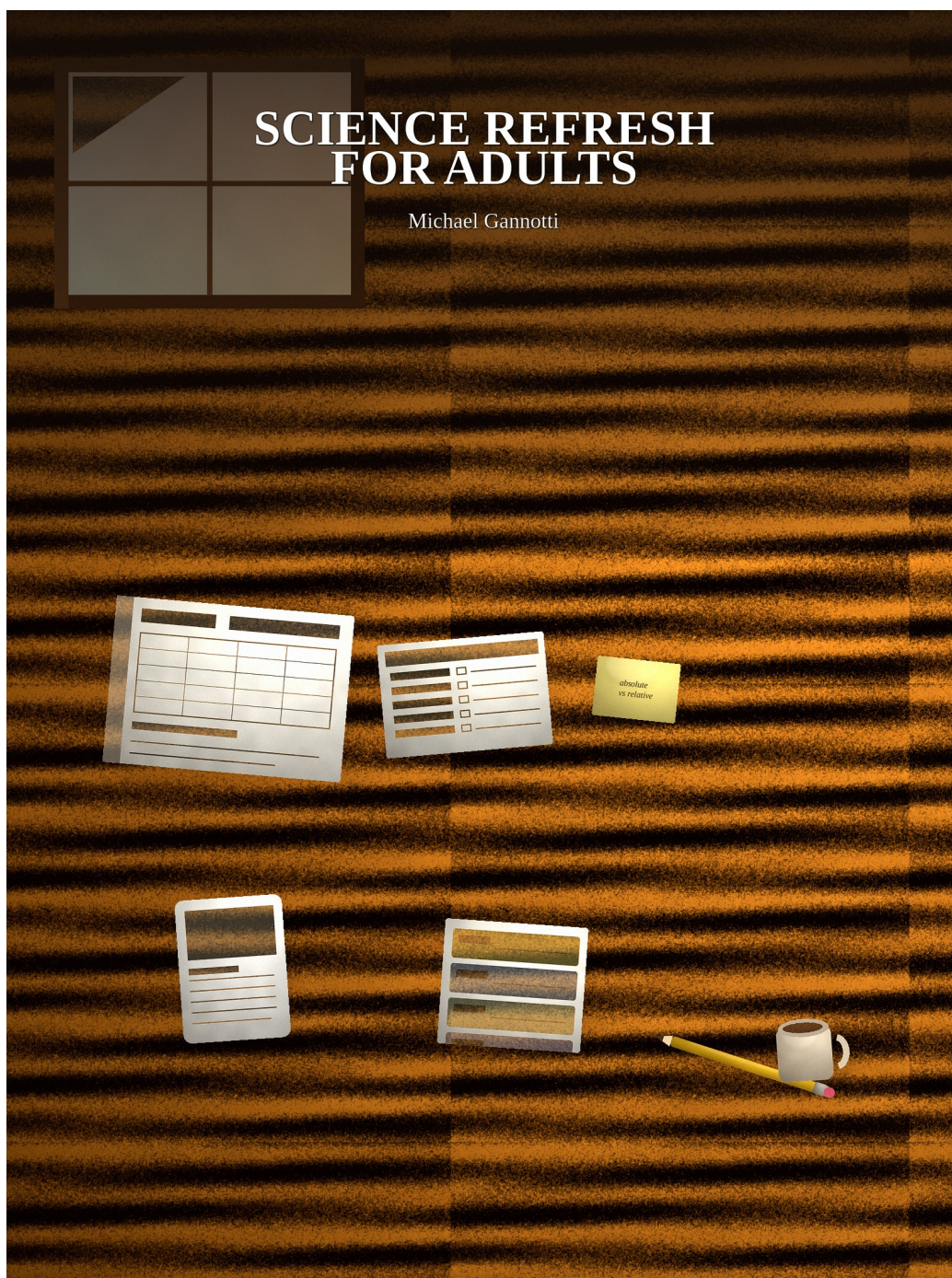


Figure 1: Cover

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This illustrated edition is the teaching-manual of 22 September 2026. Endnotes run in one series. Chapter-opening images are original still-lives, not portraits of persons or children. Documents still constrain the facts.

Contents

- Welcome
- How to Use This Book This Week
- If You Only Remember Five Things
- The Science Refresh Hour
- Chapter 1 — What science literacy looks like for adults
- Chapter 2 — Observe and record
- Chapter 3 — Feed vs study
- Chapter 4 — Systems looking
- Chapter 5 — Fair tests
- Chapter 6 — Life, matter, motion, Earth soft as Hour vehicles
- Chapter 7 — Talking and writing a denser claim
- Chapter 8 — Keeping track
- Chapter 9 — Resources
- A Note on Sources

Welcome

This book exists because usable science literacy is ordinary adult work — a feed card that says “study shows,” a workplace slide with a graph and a tidy caption, a pharmacy label with a rate (units only), a weather page, a grocery produce look, a museum exhibit systems diagram, a park transect you walk on a Saturday, a kitchen picture labelled not-for-credit. There is you, the object in front of you, and today’s looking move. You already have a life. The hour is the work.

I wrote this for a capable, busy, willing adult. You may be on a train, at a desk, on a shift, in a trade bay, a clinic, a warehouse, a retail counter, or at a kitchen table after work. Many adults feel rusty when they sit down to science they last saw in school. That feeling is common. It is not a verdict, and it is not proof of a science gene. This book will make you fluent enough in today’s looking move to hear a caption offered as the study, relative risk without a base rate, invent-data, a generated gel filed as a result, a model précis offered as an observation, a feed-first reflex, kitchen fizz treated as Chemistry credit, and a chatbot claim with no trial of yours — and to ask yourself one good question before you grab the feed or the chatbot.

You do not need to be a scientist. You do need to hear a caption offered as the study, and to open the evidence first.

Ordinary stakes live on the go and in deeper sits: a workplace “study says” slide at ordinary stakes, a science-news card, a weather glance, a grocery structure–function look, a park transect, a museum caption, an absolute-versus-relative sticky from a named public figure, a denser claim with a warrant and a named limit. Life and work motivate. They do not replace the looking move.

The cover says *Science Refresh for Adults*. The work is this commute, this desk, this weekend sit. A job and a degree are not a bicycle you now ride on every

evidence surface and every feed.¹ You already meet science in pockets of life. That is not the same as having rebuilt usable looking — evidence before feed summary; observe and record light; a systems boundary; a fair look soft; a denser claim that names a limit. Adulthood is not a science birthday.

A large adult skills survey places U.S. literacy near the international average — mean 258 versus about 260 — with about twenty-eight in a hundred U.S. adults at Level 1 or below on literacy, and about thirty-four in a hundred at Level 1 or below on numeracy (mean 249).² That survey has **no science domain**. It is adjacent reading and numbers stock. It is why dense methods prose needs a short crumb and why absolute risk needs a denominator. It is not a grade for you, and it is not this book's pretest. Process items from adult surveys — about sixty in a hundred finding a control group useful, about half identifying a hypothesis, about sixty-six in a hundred saying science is continually tested and updated — are also population context, not a proficiency ladder and not the finish line of an hour.³ The abstract, the table, or the labelled set on the table tomorrow is the work.

What a good Science Refresh Hour looks like

You sit down already knowing today's looking move. An object is on the table — a feed card you will treat as a later telling until design, absolute risk, sample, and a named limit are said; a weather page; a pharmacy label (units only); a workplace study claim; a blank observation card; a systems-boundary sketch. You retrieve a little. Then you open evidence before the feed summary — or you observe before you explain. Then you look: one measurement with units, one honesty card, one lateral hop off a caption, one systems boundary, one fair-look redesign soft, or one denser claim crumb with a warrant and a limit. Then you try a parallel item unaided. Then you ask one good self-question and wait before the key, the feed,

¹Willingham, "Critical Thinking" (*American Educator*, 2007): thinking is not a bicycle skill transferable across every domain. A job and a diploma are domains — not all evidence and feed surfaces. Access date for URLs in these notes: 22 September 2026.

²NCES/IES, *Highlights from PIAAC 2023* (ages 16–65): U.S. literacy mean **258** vs international about **260**; numeracy mean **249** vs international about **263**; Level 1 or below literacy **28%**, numeracy **34%**. Survey, not a trial of this book. Adjacent reading and numbers stock. **No science domain**. Not teen NAEP Science. Not mashed into a "science score."

³NSF Science and Engineering Indicators / Pew American Trends Panel process items (2020 fieldwork, reported in Indicators 2024 contexts): control-group usefulness about **60%**; hypothesis identification about **50%**; science continually tested/updated about **66%** (about **34%** unchanging or unsure). Labelled adult process neighbour, not a proficiency ladder and not this book's pretest.

or the chatbot. Then you leave a short sticky or dated line you could almost show a colleague. *The Science Refresh Hour* will teach this in full — micro-session, desk sit, weekend deep sit, talk-to-yourself shape, on-the-go and deeper try-its, the RAMP safety spine, and the AI rules once for adults. Later chapters will not reinvent it.

Adults can rebuild usable science literacy. Place by looking-move strip, not by birthday and not by diploma year.⁴ A person who once passed a survey class is recalling a domain. They have not finished the dual job of holding a feed until evidence is named and writing a denser claim that points at *this* table with a named limit. We practice on surfaces you already meet.

If you used *Science for Little Thinkers*, *Science for Young Minds*, *Science for Emerging Adults*, or *Science for Homeschooling* at a kitchen table, one pointer each, then this desk and this commute. Those books keep a parent in the room, a grade band, or a transcript edge. This book does not. You are the learner. This is not a reprint of those lesson banks.

History Refresh for Adults owns primary-before-feed and civic looking. One pointer for that architecture and the rust rule — then stay on evidence before feed. *Math Refresher for Adults* owns numeracy rust and the absolute-versus-relative neighbour; one pointer, then science. *Critical Thinking for Adults* owns myside and testimony craft; one pointer, then science. Looking at evidence is not the same as Wineburg sourcing, and it is not a PIAAC “science score.”

What you will actually get

Each teaching chapter does seven jobs, always in the same order. You will learn why this week’s looking move is worth the struggle, and understand it yourself. You will get a session you can run this week: named try-its across **commute/lunch**, **desk-or-shift**, and **weekend deep sit**, and a talk-to-yourself box. Tools stay optional. “Done enough” places by skill, not age or diploma year.

The seven teaching chapters follow the looking-move strip, not twelve thin survey labels: what science literacy looks like for adults (evidence before feed); observe and record; feed versus study; systems looking; fair tests / can we tell for sure?;

⁴National Mathematics Advisory Panel, *Foundations for Success* (2008), Finding 15 — adult flip, continuity from *Math Refresher for Adults* and *History Refresh for Adults*: place by skill strip, not “too old.” Diploma year is not current looking skill.

life, matter, motion, and Earth soft as Hour vehicles; talking and writing a denser claim. Keeping track and resources come last.

Today you can put a feed card, a weather page, a labelled table, or a workplace study claim on the table, name whether you are looking at evidence or at a later telling about evidence, try a parallel item, and leave one exit sticky.

What this book will not do

This book will not hand you 180 days of worksheets. Coverage is a map. Depth is the hour.⁵

It will not treat rust as destiny, a feed summary as the study, or a three-quiz process win as graduation. It will not use teen science-test percentages as if they measured working adults. It will not invent Science Refresh for Adults I, Emerging Adults Science I, or a certificate of science literacy. The promise is a path through ordinary science looking — not a fake diploma.

It is not a reprint of the school-age science manuals named above, and it is not a sequel to *History Refresh for Adults*, *Math Refresher for Adults*, or *Critical Thinking for Adults*. It is not a curriculum sales pitch or a guaranteed score. It is not a culture-war pamphlet, a party platform, or a campaign speech. It is not a weekly AP free-response dump. It is not kitchen theater sold as Chemistry credit.

It is not medical advice, dosing advice, or lending advice dressed as science literacy. Pharmacy-label try-its stay on units and rates. Live controversy stays off this table. Climate, energy, and health *topics* may appear only as looking-move surfaces — who measured, what was compared, what the abstract can settle — never as a pamphlet curriculum.

The promise

If you only remember one sentence, remember this: a path through ordinary adult science looking is the promise. A fake certificate is not.

⁵Parallel to Wineburg's coverage critique in history (light-rail coverage leaves few traces): for this book, depth is the Science Refresh Hour, not a 180-day survey dump. NAS/AAAS literacy vision and the NRC *Framework* practices are a map, not an adult mandate and not a worksheet calendar.

Age is not placement. Diploma year is not a window that closed.

Three things have to hold. You understand today's looking move well enough to hear a wrong turn — including from yourself. You retrieve, look at evidence, and try a parallel item before any tool finishes the attempt. Any helper — including an AI tool — stays a supplement you host and constrain, not a partner for the first attempt, not invent-data, not a generated gel filed as a result, not a model summary offered as the observation or the study, not a secret friend, and not a paste of proprietary workplace data into a public chat.

You can do this. You do not have to know next week's move today. You have to know this week's move well enough to sit still while you struggle, then ask one good self-question. Start here. Read *How to Use This Book This Week*, then the five things, then *The Science Refresh Hour*. Open the chapter that matches the looking move in front of you. You will have something to try on the commute, and something to try in a deeper sit. We can do this.

How to Use This Book This Week

Start at the looking move in front of you, not on page one because a diploma year said so.

This book is a handbook you open to this week’s looking-move strip, not a novel. Teaching chapters run the strip from evidence-before-feed through a denser claim. Keeping track and resources come last. Context bands — **commute/lunch**, **desk-or-shift**, and **weekend deep sit** — live *inside* each chapter. They tell you the grain of the same idea, not where *you* sit by age.

This book does not replace a transcript if you need credit, or a method that already holds. If something is working, keep it. The title is the skill you are rebuilding — not Science Refresh for Adults I.⁶

Pick the chapter by looking move, not by birthday

Open the chapter you think is right. Skip to **What “done enough” looks like** at the end of the *previous* chapter, or to **If it isn’t clicking** in the one you opened. If you can already do those checks unaided, you are too early in that chapter. If the checks from two chapters back are still failing, drop back.

If every feed caption still feels like the study, you are still in the first looking strip, whatever your diploma.⁷ If you cannot yet cover a card until design, absolute risk,

⁶Dated looking note; stranger-readable table with units; feed honesty card; denser CER with warrant and named limit; looking-move strip log; optional CLEP Biology or Chemistry score report. Never Science Refresh for Adults I, Emerging Adults Science I, or a chatbot science-literacy certificate. Access date: 22 September 2026.

⁷Place by looking-move strip and life demand (workplace study claim, weather page, grocery look, museum visit, feed card), not by diploma year. NMAP Finding 15 adult flip, continuity from *Math Refresher for Adults* and *History Refresher for Adults*.

sample, and a named limit are said, stay. “I once took AP Bio” is not placement. A binge-watched documentary series is not placement either.

If you can already do those checks unaided, skip ahead. A chapter label is a scope, not a birthday.

The strip, in order: evidence before feed → observe and record → feed versus study (evidence hygiene) → systems boundary → fair look / can we tell for sure? → Bio/Chem/Phys/Earth soft as vehicles → denser claim with warrant and named limit. Optional CLEP Biology or Chemistry sits at the edge when credit is the goal. It is not the spine.

How you use a teaching chapter

Each teaching chapter has the same seven headings, always in this order, so you always know where the explanation, the session, the named try-its, the talk-to-yourself box, and the tools box live.

Why this matters tells you what this looking move unlocks. **Understand it yourself** gives one everyday picture, one precise picture, and three to five wrong answers you should be able to hear — including from yourself.

How to try it this week assumes the session shape from *The Science Refresh Hour*. It will not rebuild the hour. It will give you this week’s named try-its — at least one on the go and one in a deeper sit — and a talk-to-yourself box with an exact opening question.

Practice that actually builds it names try-its with time, materials, safety, and the move. Weather pages, grocery aisles, and feeds *are* the practice. They do not replace the looking structure. **Tools, including AI** is optional. The rules live once in *The Science Refresh Hour*. **What “done enough” looks like** is unaided work on the strip — not a chatbot diploma.

You do not have to read the whole chapter tonight. You do have to read *this week’s* move before you put the object on the table.

The five-minute warm-up

Five minutes. Phone face down — or at least not on a feed, a key, or a chatbot. Object in reach.

1. Read today's looking move until you can say it in one sentence.
2. Look at one object yourself. Ask, out loud, whether you are holding evidence or a later telling about evidence — and what you notice before you explain. Stay off any tool until you have looked.
3. Glance at the “wrong answers you should be able to hear.” Name the one you would have given last year.
4. Write one sentence you will actually ask yourself. Not a speech. Example: “Am I treating a feed summary as science — or holding for design, absolute risk, sample, and a named limit?” Or: “What do I notice that I didn't expect — before I explain?”
5. Close the book to the try-it. You are ready.

If you are learning the move *while* you are stuck mid-look, you will grab the feed too soon. Prepare first.

This week's shape

Pick **one** focus looking move. Run three to five micro-sessions plus one desk sit when life allows. Optionally one weekend deep sit. Banks in the later chapters are menus, not calendars. Borrow one support bank if you need it. End the week with an exit sticky on Friday — one sentence a colleague could almost follow.

Space across days when you can. A massed Sunday dump feels productive. Spacing usually wins more for retention.⁸ If you miss Tuesday, resume from the last sticky.

Point to **Keeping track** for the file. Point to **Resources** for the next door: library, museum, and nature first; OpenStax and Khan as free digital backbones; optional CLEP if credit is the goal.⁹

⁸Dunlosky and colleagues (2013): practice testing and distributed practice rated high utility — technique ratings, not an adult-science-refresh programme evaluation. Useful study, not a promise that every adult will see the same result.

⁹Library, museum, and nature first. OpenStax Biology / Chemistry / College Physics / Astronomy; Khan Academy science; CK-12. CLEP Biology and Chemistry **\$97** each plus fees; remote

On the go and in deeper sits

Every teaching chapter names try-its across commute/lunch, desk-or-shift, and weekend deep sit. The talk-to-yourself box travels. The live fight does not. If a workplace claim is too hot today, pick a public abstract, a labelled composite, a grocery aisle, a weather page, or a pamphlet you already hold.

Sister science books stay at their tables. This book does not reprint those lesson banks. Pictures show objects. Notes live at the back.

Safety is ordinary plus RAMP when risk enters. No public quiz that humiliates you or a colleague. No undercover audit of a workplace. No photographing strangers without consent. Noticing is not a sting. No proprietary paste into public tools. No unauthorized chemical mixing. Kitchen pictures stay food-safe and labelled not-for-credit. Stop if you cannot minimize. A science try-it uses a public abstract, a labelled composite, a grocery aisle, a park, a museum caption, or a pamphlet you already hold. Pharmacy-label examples stay on units and rates — not medical advice.

When to skip ahead

Skip ahead when this chapter's "done enough" checklist is already true *unaided*. Slow down when the same caption-as-study grab repeats after a clear look at methods, or when a finished-looking chatbot claim still stands in for your attempt. **If it isn't clicking** will give you three likely causes and a next move.

A stranger-readable record is a dated looking note, a table with units, a feed honesty card, or a denser CER with a named limit. Never Science Refresh for Adults I. Never a chatbot certificate. Optional CLEP is a bridge when credit is the goal — not the spine.

You hold the account. Try unaided first. The full allow-and-ban list lives in *The Science Refresh Hour*.

For this week: pick the chapter by looking move. Do the five-minute warm-up. Run the hour. Space across days. Stop peeking sooner than you want to.

proctoring **\$30** on Quick Reference 2026–27. Institution sets credit. Sitting is not finishing this book. AP is not the adult default.

If You Only Remember Five Things

Keep this page. The chapters will add wording, objects, and this week's questions. They will not replace these.

1. Evidence before feed summary. A feed card, a podcast host claim, a press release, an influencer graph, a workplace “study says” slide, or a model précis is a later telling until design, absolute risk, sample, and a named limit are named — or until your own table is on the page.¹⁰ Open the abstract, the methods crumb, or the table first. Then judge the card. Absolute risk before a relative headline: “double what? Out of how many?”

2. Observe before explain — write what you measured so a stranger could almost read it. Notice what you did not expect before you tidy it into a story. Light measure with units. Fill an honesty card: my measurement, labelled secondary, screenshot, AI-generated, or kitchen picture. A stranger-readable note is the floor.¹¹ Invented neat rows are not a look. A generated gel filed as a result is not a look either.

3. Name a system boundary — and what the model leaves out. “Everything is connected” without a drawable edge is a slogan. Point to what is inside, what goes in and out, and one thing you left outside on purpose.¹² Scale shift when needed. A museum exhibit systems diagram or a plant-pot sketch is practice — not a tour

¹⁰Focus lock of this book: evidence before feed summary. Absolute-versus-relative neighbour from Gigerenzer: a relative “100%” scare can be absolute **1→2 per 7000**; a “25% risk reduction” headline is often misread against roughly **1 in 1000**. Access date: 22 September 2026.

¹¹America’s Lab Report (NRC, 2006): isolated labs are weak; fake constructed data is not a lab experience. Honesty about what the object is — measurement, secondary, screenshot, AI-generated, kitchen picture — is the adult floor.

¹²NRC *A Framework for K–12 Science Education* (2012): systems and system models as a crosscutting concept — map for adult looking, not an adult mandate and not a full DCI dump.

that replaces the looking move.

4. A denser claim needs evidence, a warrant, and a named limit. “According to *this* table (or labelled set), ...” is the floor. Then one warrant — why that evidence counts under a scientific idea. Then one limit — what this claim does not settle.¹³ Fluency without locatable evidence is theater. A thesis without a table on the page is a speech. Insufficient evidence is a valid exit when you cannot tell for sure.

5. Tools after you try. A helper, including an AI tool, is a supplement you host. It does not sit in the chair during the first attempt. A model summary is not the observation or the study. A generated gel is not a result. A kitchen picture is not Chemistry credit. You hold the account. COPPA’s under-13 line does not apply here — that is not a licence for a secret friend.¹⁴ A path through ordinary science looking is the promise. A fake diploma is not. There is no Science Refresh for Adults I.

You look, wait, and try before the tool. You hold the account. Five things. Then sit down and try.

¹³McNeill / IQWST claim–evidence–reasoning aged with a named limit for adults. Middle-school origin labelled; denser adult exit adds the limit. Fade scaffolds. Fair-look redesign soft (“can we tell for sure?”) is the neighbour when confound is the issue.

¹⁴COPPA under-13 does not apply — not a companion licence. Full AI rules live in *The Science Refresh Hour*. Never Science Refresh for Adults I, Emerging Adults Science I, or a chatbot science-literacy certificate. Kitchen $\neq$ Chemistry credit; RAMP when heat or chemicals enter.

The Science Refresh Hour



Figure 2: A desk still-life: RAMP sticky, honesty card, micro/desk/deep stickies, lab notebook, commute mug — no people, no logos, no readable UI.

The hour has a shape. Learn it once. Later chapters will give you today's looking move. They will not rebuild this hour. When a chapter says "run the session," it means this.

You do not need a school bell. You need the object on the table, a try that could come out otherwise, you doing the looking and the recording, a record you could almost show a colleague, talk-to-yourself about whether this card is study or later telling and what you notice before you explain, and stop. Three shapes carry the week: a **micro-session** on the commute or at lunch (~15–20 minutes), a **desk sit** (~25–40 minutes), and a **weekend deep sit** when needed (~45–75 minutes). Space them across days when you can. Cramming one Sunday feels productive. Spacing feels harder and usually wins more for retention.¹⁵ Three to five micros plus one desk sit in a week, with an occasional deep sit, is the honest ambition — not under-running of a proven adult commute science trial, because there isn't one at this grain.

Sit down having already done the five-minute warm-up from *How to Use This Book This Week*. The object changes. The shape does not.

1. Micro-session (~15–20 min, commute/lunch)

One looking move. Brief feedback. No high-stakes countdown timer as default rigor. Interruptible without shame. If the session becomes scrolling, it is not a session — set a fifteen-minute clock, one object, stop.

Retrieve yesterday's sticky (1–2 min). Close the book. Write or say yesterday's move in one line — an absolute-risk sticky, an Observe line, an honesty label, a boundary word, or a limit sentence. Prefer retrieve over reread.¹⁶

One evidence flash or feed honesty card — caption covered (4–7 min). Name what you are holding: abstract crumb, table, press release, feed card, model précis, screenshot, or AI-generated. A feed card is a later telling until design, absolute risk, sample, and a named limit are said — or until your own table is on the page.

One try or evidence look — talk-to-yourself (5–8 min). One measurement with units, one lateral hop off a caption, one absolute-versus-relative sticky, or one “this

¹⁵Dunlosky and colleagues (2013): practice testing and distributed practice rated high utility — technique ratings, not an adult-science-refresh programme evaluation. Useful study, not a promise that every adult will see the same result. Access date for URLs in these notes: 22 September 2026. Session minutes in this chapter are pedagogical design, not a national adult-science-minutes table. There is no adult commute science RCT at this grain — named sessions are the ambition.

¹⁶Retrieve over reread is the same house preference as *Math Refresher for Adults* and *History Refresh for Adults*. Looking-move stickies are the science object.

card is a later telling because...” line. Hands off the key. Hands off the chatbot. If you stall: ask yourself one question, wait, hint, then look — not the reverse.

Denser CER crumb + exit sticky (3–4 min). One claim crumb that points at *this* evidence, or one limit sentence. One line dated: the move, one check, one wrong twin you now hear. Put the phone away. RAMP only if a later try this week needs heat or chemicals.

2. Desk sit (~25–40 min)

Start from a real artifact when you can: a workplace study claim at ordinary stakes, a pharmacy label rate soft (units only — no medical advice), a weather page, a grocery look, a museum caption, a kitchen picture with an honesty label. Order: open question or system first; run RAMP if heat or chemicals will enter; short look; one try/record or one evidence look; talk-to-yourself denser CER with a named limit; exit sticky.

Open question / system + RAMP if needed (5–8 min). Honesty card. Observe column only first if the object is dense. If kitchen or materials with risk: Recognize–Assess–Minimize–Prepare before you begin. Stop if you cannot minimize.

Short look (10–15 min). One evidence look, one stranger-readable table with units, one systems-boundary sketch, or one fair-look redesign soft.

One move from another strip (5–10 min). A denser CER crumb with warrant and limit, an absolute-versus-relative sticky, or a feed-versus-abstract hop.

Exit ticket (3–5 min). A dated note a colleague could follow. Name tomorrow’s retrieval flash.

Proprietary names and unpublished employer numbers stay off public consumer tools. Strip to a public twin or use a tool your employer allows. Open *your* acceptable-use page.

3. Weekend deep sit (~45–75 min, when needed)

One central looking-move strip. Short background — five to eight minutes at most. Longer try: two-source feed versus study plus a claim with a limit; park

transect plus systems plus CER; museum exhibit systems plus a short write; fair-test redesign across two ordinary materials. Fading scaffolds: first pass guided; second pass you withhold the gloss. A claim with warrant and named limit. Stop while a look remains.

A deep sit that becomes “finish the survey textbook” is a coverage dump wearing a weekend hat. One move deeply beats seven skimmed.

Open with a diagnostic wrong when you can — the caption treated as the study, relative risk without absolute, invent-data, a generated gel filed as a result, a model précis filed as observation, feed-first, kitchen fizz as credit, confounded “proof,” chatbot CER with no trial. Run the chapter’s “done enough” checklist. Schedule next week’s three micro cues on a calendar — not vibes.

The talk-to-yourself shape, once

Later chapters fill the exact opening. The shape lives here.

Opening question. Authentic. About today’s strip. No recitation answer already written on your page. Use one locked opening when the chapter matches, or this stem: Before I look up the answer or trust the feed — what do I notice, what claim can I defend from *this* evidence, and what limit must I name?

Locked openings you will meet: Am I treating a feed summary as science — or holding for design, absolute risk, sample, and a named limit? What do I notice that I didn’t expect — before I explain? Is this card the study or a later telling — and what would I open next? Where is the boundary of this system — and what did I leave outside on purpose? Can I tell for sure it was the variable I named — what else differed? What looking move am I practicing on this Bio / Chem / Phys / Earth soft vehicle — and what does a picture try not earn as credit? What do I claim — which evidence from *this* table or labelled set shows it — and what does this claim not settle?

Three to five follow-ups. Pick; you do not need all five every Tuesday. Meaning: What do you mean, *later telling*? Representation: What does the table, figure, or abstract show before the caption? Alternative: What else could explain these numbers? Check: What can this evidence not tell me? Revise or diagnose: You may change your mind — where could someone go wrong?

How to wait alone. After you ask yourself, cover the answer or the feed. A slow

three is a convention, not a sacrament — Stahl’s think-time is classroom-origin; Rowe’s wait-time is science class; neither is an adult science RCT.¹⁷ After you stop mid-reason, wait again before peeking. If the silence is hard, look at the table, the boundary sketch, or the clock — not at the phone’s answer key or the feed. Rescue at one second is still rescue. Abandonment for forty minutes of scrolling is not a micro-session.

What a stuck silence usually means. The question was vague; wait-time after the question was zero; you are searching for a word; you are guessing what a textbook would want; the question was a recitation item in disguise; you offered a keyword where a looking move was asked for; you think “science person” means leave; you think a finished-looking chatbot CER is already a check; anxiety is eating working memory — shorten the session, do one Observe win or one absolute-risk sticky, come back tomorrow. Your job is to pose, wait, and point back at the object — not to fill the silence with the right answer from a model summary.

You do not need five questions stacked. You do not need to perform Socrates. You do not need a live controversy. You do not photo-to-key mid-attempt.

The on-the-go try-it shape, once

Time. Fifteen to twenty minutes, then stop.

Materials. A saved feed card about a study, a weather page, a pharmacy label (units only), a workplace “study says” slide at ordinary stakes, a grocery produce look, an absolute-versus-relative sticky from a named public figure.

Safety. Curious, not courtroom. No public humiliation of yourself or a colleague. No proprietary paste into public tools. No culture-war sting. Noticing is not a sting. Phone is allowed as a *surface* — not as an answer key during the attempt. No medical advice on label rates.

The move. Name whether you are holding evidence or a later telling. Name design, absolute risk, sample, or limit — or mark unknown. One honesty card. One exit sticky.

¹⁷Stahl (1994), ERIC ED370885, think-time — classroom-origin. Rowe (1972) wait-time is science class. Adult translation: pause-before-lookup / pause-before-feed. Not an adult science RCT.

Types you will meet, not a lesson bank: feed-card honesty; absolute-versus-relative sticky; one Observe line; pharmacy label rate soft; weather glance without story first.

The deeper try-it shape, once

Desk or weekend. Same safety. More room for stranger-readable tables, systems boundaries, fair-test redesign, denser CER with limit, museum exhibit systems, park transect.

Types you will meet: stranger notebook table with units; systems-boundary sketch; confound-versus-fair redesign; denser CER four-box with warrant and named limit; museum systems postcard; park transect card; dated note a colleague could follow.

These are not the hour: a gotcha audit of a workplace; a live-controversy field trip; a 180-day worksheet dump; invent-data filed as measurement; a generated gel filed as a result; a model summary offered as the observation or the study; kitchen theater sold as Chemistry credit.

Safety spine, once (RAMP)

When a try involves heat, chemicals, glass, or sharp tools, run **RAMP** before you begin: **Recognize** the hazards; **Assess** the risks; **Minimize** the risks; **Prepare** for emergencies.¹⁸ Adult present to self. Kitchen pictures stay food-safe and labelled not-for-credit. Kitchen is not Chemistry credit. No unauthorized mixing. If home cannot meet the ceiling, point to a lab-path neighbor — community-college or supervised lab — and return to looking moves you can run safely. Stop if unsafe. Ordinary noticing — weather, grocery, feed, museum — needs ordinary courtesy, not a RAMP sheet, and still needs no sting and no proprietary paste.

¹⁸ACS RAMP (Recognize–Assess–Minimize–Prepare). NSTA and Flinn safety pointers as neighbors. Kitchen ≠ Chemistry credit; no unauthorized mixing; lab-path neighbor when home cannot minimize.

The AI rules, once

Later chapters will point here. You talk to the page, the trial, and the check. You may use a tool. A chatbot is never the only partner during the first attempt. An agent is not the science partner for the first attempt. The unaided try is still yours. You hold the account. COPPA's under-13 line does not apply to an adult reader. That is not a science birthday and not a licence for unsupervised chat or a secret friend.¹⁹

The science is on the table, the phone note, or the page: a named textbook, OpenStax, agency, or museum page or study abstract; a real look, measure, and record; a blank data table you fill from *your* measuring; a claim or lab note pointing at the table; a labelled schematic put away before the try; a predict–measure–claim or CER cycle; a short science op-ed or science memo *you* drafted first; absolute-versus-relative risk framing practiced after a try. You attempt first. You hold the answer key on **YOUR** side.

After you have tried, you **may**:

- Explain this idea *to yourself* from a **named** lesson, textbook page, or study abstract (title, page, today's micro-goal).
- Ask for **extra isomorphic practice**, with answers on **YOUR** page (practice sheet plus key) — blank observation card, blank data table, CER frame.
- Use a labelled **SCRIPT** only **after you tried** — a short spoken or written sequence for *you* to say, not a paragraph to copy as your lab note or science op-ed.
- Ask for **revision questions** after *your* claim, lab note, science op-ed, or science memo draft — not a rewrite as the only draft, and not “fix to publishable.”
- Ask for a **hint** after an attempt. A hint is a question or a named idea, not the finished paragraph.
- Ask a tool to **diagnose work you already did** (crop to paper or note — no face).
- Use **discussion questions** you vet and ask yourself.
- Use a **labelled schematic** put away before the try — never filed as “what I saw.”
- Practice **absolute versus relative risk prompting** after you drafted a risk

¹⁹COPPA covers operators directed to children under 13. It does not gate this book. Holding the account is not a companion licence.

claim from named numbers.

You **may not**:

- Paste tonight’s worksheet, blank data table, lab packet, or employer sheet and ask the tool to complete it (**photo-to-key**).
- Ask “write my lab report / science op-ed” or “fix to publishable” as the only draft.
- Invent an observation or a data table, or generate a gel, micrograph, sky photo, spectrum, or “lab photo” and treat it as a result.
- Park an unsupervised chatbot as your only partner during the attempt.
- Treat “I am an adult / I hold the account / COPPA is off” as a permission slip or companion licence.
- Put a model summary on the table as the observation or as the study.
- Use spoken rewrite or a voice toy to replace looking or measuring at today’s named page or abstract.
- Invent research or trust a fluent citation without opening the paper.
- Detector-grade your own unaided write-up. Detectors have mislabelled a large share of human writing as machine-written in published tests; a detector score is not a grade for your practice note.²⁰
- Paste proprietary employer data into a public consumer tool. Strip proprietary rows to invented isomorphic prompts on YOUR practice sheet, or use an enterprise tool your employer allows. Open *your* acceptable-use page. This book will not invent a federal workplace chatbot statute for you.
- Hand yourself an **agent** (Hermes Agent, Grok Bot, or any cloud-computer worker) as a science partner for the first attempt.
- Print a certificate of science literacy from a chatbot.
- Run live-exam AI during a timed credentialing sitting.

One mechanism sentence, labelled OFF-AGE for science literacy, then teach.

In a high-school math field experiment (grades 9–11, nearly 1,000 students, Turkey), an unguarded chatbot made practice look better and left students worse on the test with the window closed.²¹ **Not an adult science RCT. Not this science**

²⁰Liang, Yuksekgonul, Mao, Wu, and Zou, *Patterns* (2023): average false-positive rate **61.22%** on human TOEFL essays (arXiv 2304.02819). A detector score is not honesty.

²¹Bastani and colleagues, *PNAS* (2025): Turkish high-school math, grades 9–11, nearly 1,000 students. GPT Base +48% assisted practice, −17% closed exam; GPT Tutor +127% practice, unaided $\approx$ control. **OFF-AGE for science literacy. Not an adult science RCT.** *Math Refresher for Adults* may label the same paper ON-AGE-adjacent for math. This book does not mash that label. Numbers here are the paper’s, not “what AI does to adult science literacy.”

band. *Math Refresher for Adults* may call the same paper ON-AGE-adjacent for math. This book does not. The lunch-break rule it supports: the model may prepare you and the next looking page; it may not do your looking, measuring, recording, claim, lab report, or science op-ed when those are the lesson.

Names, in ordinary language. Hermes 4 is a chat model. Hermes Agent is a different runtime — a computer that can keep going. Grok is a chat assistant. Grok Bot is a cloud-computer agent. A public-school partnership is a third Grok. Game artist makes pictures. None of these is a science refresher class. There is no Grok Bot education SKU. A chatbot answers in a window. An agent has tools and can keep going. Neither is a partner for the first attempt.

Not a secret friend. A tool that talks in the first person is still a tool. It does not get a private channel or a seat in the chair during the first attempt.

Language models invent citations. Open the source.²² Works cited only for what you opened.

You can practice this entire book with no AI.

A life of the habit

Growing fluency is more miles on the same road. Revisit the same strip next Tuesday. Mix last week's sticky into this week's warm-up. Space across days. Stop when the exit sticky is honest. Placement is by looking-move strip, not diploma year. Optional CLEP Biology or Chemistry is a bridge when credit is the goal — sitting an exam is not finishing this book.²³ The honest pitch is *often enough that the question is familiar*.

Anxiety around science you last saw in school is common. It can eat working memory. Short sessions, early Observe wins or absolute-risk stickies, and no countdown timer as default rigor are posture, not a therapy manual. If a session is eating you, shorten it, win one looking item, return tomorrow.

²²Walters and Wilder, *Scientific Reports* (2023): GPT-3.5 **55%** / GPT-4 **18%** fabricated citations; real citations still err on dates. Open the source.

²³CLEP Biology and Chemistry: **\$97** each plus fees; remote proctoring **\$30** on Quick Reference 2026–27. Institution sets credit. AP is not the adult default. Sitting is not finishing this book. Never Science Refresh for Adults I.

Raising looking skill is not foremost a campaign to raise general support for science.²⁴ Teach looking because it is how evidence and denser claims work. Money, workplaces, health leaflets, and civic life motivate. They do not replace the looking move.

A Tuesday, said plainly

Here is an illustration, not a reported workplace. Someone opens a saved “study shows” card on the train, covers the caption, names whether they are holding a later telling, waits before a tidy recap, drafts one absolute-risk sticky from a named denominator, and leaves a dated line: *card compressed design; absolute still unknown*. Later chapters will say: run the session as in *The Science Refresh Hour*.

²⁴NAS (2016) *Science Literacy*: science literacy is more than facts; Conclusion 11 — raising literacy is not foremost the means to raise general support for science. Teach looking. Do not sell trust-rescue as the hour’s success metric.

Chapter 1

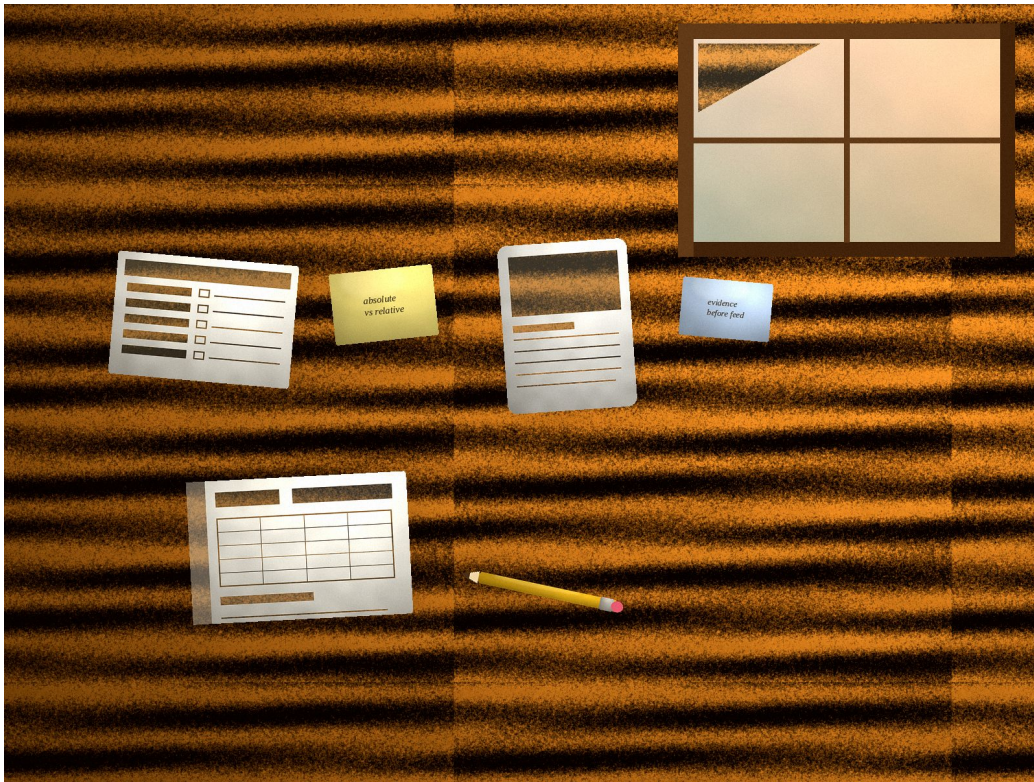


Figure 3: Desk still-life: honesty card, absolute-vs-relative sticky, described feed card, lab notebook — no people, no logos, no readable UI.

What science literacy looks like for adults

You open a feed. A card says a study shows something sharp. Your thumb is already halfway to share. Or a workplace slide ends with “the research is clear,” and the room nods. Or a leaflet quotes a risk that sounds huge until you ask, “Out of how many?”

This chapter names what usable science literacy looks like when you are the adult on the go — and when the default archive is a phone. The hour is not a fact dump. It is evidence before feed summary. Place yourself by looking-move strip, not by diploma year. Rust is ordinary. A caption is not yet a study.

You are the learner. There is no parent section and no student section. The teacher beside you on the train is this page: calm, specific, and willing to wait while you struggle before you rescue yourself with a summary.

Why this matters

Here is the move in one breath: when a card offers science, you hold for design, absolute risk, sample, and a named limit — or you mark them unknown — before you treat the card as settled. That hold is science literacy for adults on the go. Everything else in this book either deepens it or rides on it.

Adults meet science as feed, workplace slide, leaflet, weather page, grocery look, museum caption, and park path — not as a weekly AP free-response set. That is not a demotion. Those are the surfaces where looking either happens or fails.

Money, workplaces, health leaflets, and civic life motivate. They do not replace the looking move. A pharmacy label try-it stays on units and rates. A workplace “study says” slide stays at ordinary stakes. Live controversy stays off this table. Climate, energy, and health *topics* may appear later only as looking-move surfaces — who measured, what was compared, what the abstract can settle — never as a pamphlet curriculum.

What unlocks next is simple and hard: you can hold a loud card until you name design, absolute risk, sample, and a limit — or until you mark those unknown. You can tell evidence from a later telling about evidence. You can refuse the story that a diploma finished the job, or that a three-item process quiz graduated you, or that teen school-test percentages somehow measured working adults.

Science literacy here means usable looking and denser claim craft on ordinary stakes. The National Academies' adult-facing report on science literacy describes content knowledge, understanding of scientific practices, and understanding of science as a social process — and it warns that raising literacy is not the foremost way to raise general support for science.²⁵ AAAS *Science for All Americans* sketches a citizen who uses scientific ways of thinking for individual and social purposes, with strengths and limits named.²⁶ The NRC *Framework* practices are a map you may still practice as an adult — not a mandate, not a franchise, not a 180-day dump.²⁷

Why struggle for this? Because without the hold, every later move collapses into slogan. Observe and record need a mind that has not already filled the blank with the caption. Systems looking needs a boundary, not a feed mood. Fair tests need “can we tell for sure?” instead of confident confounds. Denser claims need locatable evidence and a named limit. Evidence before feed is the hinge this book hangs on.

Many adults feel rusty when they sit down to science they last saw in school. That feeling is common. It is not a verdict, and it is not proof of a science gene. A job and a degree are domains you once practiced. They are not a bicycle you now ride on every evidence surface and every feed.²⁸ You already meet science in pockets of life. That is not the same as having rebuilt usable looking.

You do not need to be a scientist. You do need to hear a caption offered as the study, and to open the evidence first.

This is useful guidance for adult looking, not a promise that every adult will see

²⁵National Academies of Sciences, Engineering, and Medicine, *Science Literacy: Concepts, Contexts, and Consequences* (2016): science literacy includes content, practices, and science as social process; foundational literacy/numeracy necessary but not sufficient; Conclusion 11 — available research does not support the claim that increasing science literacy will lead to appreciably greater support for science in general. Access date for URLs in these notes: 22 September 2026.

²⁶American Association for the Advancement of Science, *Science for All Americans* (Project 2061, 1989): citizen literacy vision — interdependent human enterprises with strengths and limitations; scientific ways of thinking for individual and social purposes. Map for adults, not a K–12 pacing guide.

²⁷National Research Council, *A Framework for K–12 Science Education* (2012): practices and crosscutting concepts as **map**, not adult mandate, not statute, not franchise.

²⁸Willingham, “Critical Thinking” (*American Educator*, 2007): thinking is not a bicycle skill transferable across every domain. A job and a diploma are domains — not all evidence and feed surfaces.

the same result as any named survey or study.

Understand it yourself

Everyday picture. A science-news card lands in your feed. Bold caption. Soft graph. Soft “study shows.” You feel briefly informed. That feeling is not yet looking. Looking starts when you cover the caption, ask what design and numbers would have to be true, and open a methods crumb or abstract — or mark unknown — before you treat the card as settled.

Picture the same moment at a desk. A colleague forwards a chart with a tidy heading. The room is already using the heading as a decision. Your job, for yourself, is not to win the room. Your job is to notice that the heading is a later telling until someone can point at design, denominator, sample, and a limit.

Precise picture. Evidence before feed summary means the feed, podcast host claim, press release, influencer graph, workplace “study says,” or model précis is a *later telling* until design, absolute risk, sample, and a named limit are on the table — or until your own labelled table is. The Science Refresh Hour is the named session where you practice that hold: retrieve a sticky, flash one evidence or honesty move, look briefly, try a parallel item, leave an exit sticky. Place by looking-move strip (evidence before feed → observe/record → systems → fair look → denser claim), not by birthday and not by diploma year.

Design, here, means how the comparison was built — who was compared to whom, what was held still, what kind of study it was. Absolute risk means the change with a denominator (out of how many?), not only a relative percent that can sound huge when the baseline is tiny. Sample means who or what was measured, how many, and over what window. A named limit means what this evidence cannot settle.

Adult skill surveys help explain why dense methods prose and absolute-risk denominators feel heavy — without inventing a science grade for you. PIAAC 2023 for U.S. adults ages 16–65 reports literacy mean **258** (international about **260**), numeracy mean **249** (international about **263**), with about **28%** of U.S. adults at Level 1 or below on literacy and about **34%** on numeracy. That survey has **no science domain**. It is adjacent reading and numbers stock. It is not this book’s pretest.²⁹ Process items from NSF Indicators / Pew American Trends Panel con-

²⁹NCES/IES, *Highlights from PIAAC 2023* (ages 16–65): U.S. literacy mean **258** vs interna-

texts report about **60%** of U.S. adults correctly recognizing the usefulness of a control group, about **50%** correctly identifying a scientific hypothesis, and about **66%** saying scientific findings are meant to be continually tested and updated (about **34%** choosing unchanging truths or unsure).³⁰ Those are population context for process familiarity — neighbours, not a full literacy scale and not a three-quiz graduation.

Wrong answers you should be able to hear — including from yourself:

1. **“I already took science / I have a degree, so I’m done.”** Diploma year is a domain you once practiced. Unused looking rusts. Adulthood is not a science birthday. Place by what you can do on today’s card, not by the year on a transcript. A forty-five-year-old who opens the abstract before the caption can deepen systems and denser claims; a thirty-year-old who treats every “study says” card as settled needs evidence-before-feed work first — regardless of diploma year.
2. **“The caption is enough — the feed is the study.”** A caption is a later telling. Until you name design, absolute risk, sample, and a limit — or mark them unknown — you have mood, not evidence. Virality is not corroboration. A confident tone is not methods.
3. **“I’m not a science person, so I can’t.”** The phrase names a feeling many adults carry. It is ordinary. It is not a fixed trait this book affirms as destiny. Looking moves can be practiced on weather pages, grocery aisles, and feed cards you already meet. Translate once: “I am practicing evidence before feed today.” Then return to the blanks.
4. **“Teen NAEP Science / PISA scores prove what adults are.”** Those assessments target school grades or ~15-year-old students. They are not the adult measure. PIAAC is adjacent adult stock with no science domain. Framework language from teen assessments may inspire questions if labelled — never paste country ranks as your adult score.³¹

tional about **260**; numeracy mean **249** vs international about **263**; Level 1 or below literacy **28%**, numeracy **34%**. Survey, not a trial of this book. **No science domain.** Adjacent reading and numbers stock — not mashed into a “science score.”

³⁰NSF Science and Engineering Indicators / Pew American Trends Panel process items (2020 fieldwork, reported in Indicators 2024 contexts): control-group usefulness about **60%**; hypothesis identification about **50%**; science continually tested/updated about **66%** (about **34%** unchanging or unsure). Labelled adult process neighbour, not a proficiency ladder and not this book’s pretest.

³¹NAEP Science assesses U.S. school grades (4/8/12). PISA science literacy targets students

5. **“If I ace a short process quiz, I’ve graduated science literacy.”** Control, hypothesis, and updating items are useful practice flashes. They are not the hour’s finish line. Support-for-science is not the success metric of this hour either.³² The abstract, the table, or the labelled set on the table tomorrow is the work.

Five-minute warm-up before the try. Cover one science card’s caption for sixty seconds. Write four blanks: design? absolute risk (out of how many)? sample/population? named limit? Leave blanks empty if unknown. Uncover the caption. Notice what the card still has not given you. That gap is the work of this chapter — not a failure. If the card has a soft graph, ask one more: axes visible? baseline clear? Then stop. Warm-up ends while a look remains.

How to try it this week

Session shape follows the Science Refresh Hour you met in the front matter. This week’s move is identity: usable science literacy as evidence before feed, placed by strip not diploma.

Warm-up retrieval (2–3 min). Say aloud or write: “A caption is a later telling until design, absolute risk, sample, and a limit are named — or marked unknown.” Retrieve one wrong twin you heard from yourself recently (diploma-as-done, caption-as-study, “not a science person” as destiny, quiz-as-graduation).

Short model — an evidence look, not a lecture (3–5 min). Take one public science card or workplace study claim (strip identifiers if workplace). Cover the caption. Name what you would need. Open one abstract sentence or methods crumb if findable; otherwise write “unknown — cannot evaluate yet.” Speak one sentence to yourself: “This card claims ; ***I have checked*** / marked unknown ____.”

Your attempt (6–10 min). Parallel card. Same four blanks. No sharing before the hop. No pasting the card into a chatbot as the first move. If you feel the urge to scroll for a second card “for balance,” write the urge on the sticky and stay with one object.

One good self-question. Use the locked opening in the talk-to-yourself box below.

roughly age 15. Neither is the adult measure for this book.

³²NAS 2016 Conclusion 11 (see note 25): do not sell the hour as polarization rescue or trust-rescue by quiz.

Wait before the feed or the key.

Mixed practice / exit sticky (2–3 min). Write one sentence: “Today’s object was evidence / feed-about-evidence / unknown because ____.” Tomorrow’s flash: cover a caption for thirty seconds and name one blank you always forget.

Context bands

Commute / lunch (~15–20 min). Micro evidence flash. One card. Cover caption. Four blanks. Exit sticky. FEED-01 warm and AS-01 honesty tick fit here. If the train is crowded, the notebook line can be four words: design / absolute / sample / limit — with checkmarks or “?”.

Desk or shift (~25–40 min). Real artifact: a workplace “study says” slide (ordinary stakes; strip identifiers), a science-news claim, or a leaflet risk line. Ask for citation or abstract if workplace. Name one limit. Do not weaponize against colleagues. FEED-02 soft and AS-02 methods crumb fit here. If shift noise is high, whisper the self-question once, then write.

Weekend deep sit (~45–75 min when needed). One-question evidence look across two surfaces: the card and one abstract or reputable explainer. Write what the card can settle and what it cannot. Optional SAFE-02 stop-authority flash if you also plan any kitchen picture later in the book — practice stopping before risk, not escalating. Stop while a look remains; do not turn the deep sit into a Sunday dump of twenty cards.

First try-its this week

FEED-01 Caption Is Not the Study (warm). Time 15–20 min. Materials: one science newsfeed or social card; screenshot OK if honesty-labelled. Safety: none beyond ordinary. The move: cover caption; note image or graph; uncover; ask what evidence would be needed; open abstract/methods crumb or reputable explainer before sharing. Exit: “Caption claims ; **evidence I checked** .” Refuse sharing before the hop; refuse virality as corroboration. Band: commute/lunch micro.

AS-01 Honesty Card Drill. Time 10–15 min. Materials: today’s object. Safety: none. The move: tick my data / labelled secondary / feed screenshot / generated / kitchen picture / unknown. Exit: card completed. Refuse skipping the card on “ob-

vious” feeds. Band: micro. This crumb returns in Chapter 2 with thicker honesty layers; this week it trains the reflex to label what the object *is*.

AS-02 Methods Crumb or Mark Unknown. Time 15–20 min. Materials: one science claim. Safety: none. The move: write a methods crumb or explicitly “unknown — cannot evaluate yet.” Exit: crumb or unknown. Refuse assuming rigor from confident tone. Band: micro or desk.

SAFE-02 Stop Authority Drill (warm). Time 10–15 min. Materials: written stop rule; one hypothetical unsafe prompt labelled hypothetical. Safety: practice stopping, not escalating. The move: script “I stop if I cannot name a hazard, cannot minimize, or feel pressured to mix unknowns.” Rehearse saying stop aloud once. Exit: stop sentence written; conditions listed. Band: micro or desk. Pair lightly this week so safety stays in view before later kitchen pictures. Kitchen theater is never Chemistry credit in this book.

How to fade help. First day: cover caption with your hand. Second day: cover with a sticky and write the four blanks from memory. Third day: name the four without looking at a template. When a tool wants to summarize the study for you, put the summary away until you have filled the blanks yourself. Fade means fewer scaffolds, not faster grabbing.

When to stop looking up. After one methods crumb or one honest “unknown,” stop. Do not scroll into a second outrage card “for balance.” One parallel try is enough for a micro-session. Looking up becomes rescue when it finishes the attempt for you.

Talk-to-yourself box

Opening (locked): “Am I treating a feed summary as science — or holding for design, absolute risk, sample, and a named limit?”

Follow-ups (pick three to five; wait alone after each):

1. Meaning: “What do I mean by *later telling* — who told this after the measurement?”
2. Representation: “What does the table, figure, or abstract show before the caption?”
3. Alternative: “What else could explain these numbers if the caption is soft?”
4. Check: “What can this evidence not tell me?”

5. Revise: “I may change my mind — where could someone go wrong on this card?”

How to wait alone. After you ask, cover the answer or the feed. A slow three is a convention, not a sacrament — classroom think-time translated for adults as pause-before-lookup / pause-before-feed.³³ Look at the four blanks or the clock, not at the phone’s answer key. Rescue at one second is still rescue. Abandonment for forty minutes of scrolling is not a micro-session. If silence is hard, put the phone face down and write one blank in larger letters.

Stuck silence usually means: the question was vague; wait-time was zero; you are searching for a vocabulary word instead of a looking move; you offered “science says” where a methods crumb was asked; you think “science person” means leave; you think a finished-looking chatbot précis is already a check; anxiety is eating working memory — shorten the session, do one absolute-risk sticky or one honesty tick, come back tomorrow. Your job is to pose, wait, and point back at the object — not to fill the silence with a model summary.

Exact wording you can say to yourself: “Before I look up the answer or trust the feed: what do I notice, what claim can I defend from *this* evidence, and what limit must I name?”

Practice that actually builds it

Practice lives on the feed you already meet and on ordinary work objects. Weather pages and grocery aisles will deepen later chapters. This week’s practice is the hold — evidence before feed summary — until it feels like a habit of the thumb.

Named try-its below come from the FEED, AS, and SAFE banks. Banks are menus, not calendars. You do not owe every item this week. You owe honest attempts on objects you can see.

On the go — commute / lunch

FEED-01 (named above). Run it on a real card from your actual feed. Honesty-label a screenshot if you save it. If the card has no graph, still cover the caption; words alone can pretend to be a study.

³³Rowe (1972) wait-time in science class; Stahl (1994) think-time — classroom origin. Adult translation: pause-before-lookup / pause-before-feed. Not an adult science RCT.

FEED-07 Click Restraint Before Science Outrage (warm). Time 10–15 min. Materials: timer; one tempting science-outrage card. Safety: emotional self-care — skip if flooded. The move: wait 60–90 seconds; write what you already know and what you need; then click only if still worth it. Exit: “I waited; I needed ____.” Refuse instant outrage reward as the session. Band: micro. This is not moral theater. It is working-memory protection so the four blanks can exist.

AS-01 (named above). Tick the honesty card before you argue with the card in your head. A feed screenshot is not your measurement. A generated image is not a result. Mark it.

Desk or shift

FEED-02 Workplace “Study Says” Soft. Time 15–25 min. Materials: one Slack, email, or meeting claim that cites a study. Safety: workplace confidentiality — strip identifiers in your notes; no undercover audit; no pasting proprietary employer data into a public consumer tool. The move: ask for citation or abstract; note sample/population if findable; write one limit; do not weaponize against coworkers. Exit: citation status + one limit. Refuse gotcha culture as science literacy. Band: desk / shift. Noticing is not a sting.

AS-03 Sample Size / Population Soft (crumb). Time 15–25 min. Materials: abstract or graph you can open. Safety: none. The move: who/what was measured? N? Time window? Write a generalization limit. Exit: population limit sticky. Refuse one anecdote as population. Band: desk. If N is missing, write “N unknown — cannot evaluate generalization yet.” That sentence is progress.

NSF-style process flash (labelled practice, not pretest). Time 10 min. Materials: three short prompts you write yourself — control usefulness, hypothesis identification, updating. Safety: none. The move: answer unaided; check against a reputable explainer after. Exit: “Practice flash — not a literacy score.” Refuse scoring yourself as graduated. Population context earlier in this chapter is why these flashes exist; they are not a ladder you climb to a certificate.

Weekend deep sit

One-question evidence look. Time 45–60 min. Materials: one viral or workplace claim + one abstract or agency page on the same claim when findable. Safety: ordinary; strip workplace identifiers. The move: write the four blanks on the card;

open the abstract; list compressions (proves vs suggests; absolute missing; sample unnamed). Exit: two-column sticky — card vs evidence. Optional: SAFE-03 Kitchen Picture Honesty Card only if you also want a food-safe dissolve or melt *picture* labelled not-for-credit — this week it is honesty practice, not Chemistry credit.³⁴ Write: “This is a picture because ____.” Write what a supervised lab path would add. Then stop.

Incorrect example to diagnose (composite illustration, not a reported workplace). A slide says “Study shows productivity up 40% with new schedule.” No citation. No N. No absolute baseline (40% of what?). No limit. Someone in the room says, “Science says we’re done discussing.” Diagnose: caption-as-study; relative-without-base; sample unknown; limit unnamed. Repair move: FEED-02 + AS-02 — request abstract; write “unknown” where needed; one limit sentence: “This claim cannot settle schedule policy until design and baseline are named.” Your repair is for *your* looking file, not for shaming a colleague.

Talk-to-yourself box (practice depth)

Reuse the locked opening. After a stuck silence, point at the object: “Show me the denominators. Show me who was measured. Show me what this cannot settle.” If you hear yourself say “I’m not a science person,” translate once: “I am practicing evidence before feed today.” Then return to the blanks.

Blocked for a new move: do not jump ahead to denser CER four-boxes as this week’s identity. Mixed practice: when the card is soft, hold; when you have your own labelled table later in the book, claim from *that* table — still with a limit. Chapter 3 will thicken feed hygiene (press vs abstract, absolute vs relative with named figures). Chapter 2 will thicken observe-before-explain. This chapter owns the identity lock so those chapters have somewhere to stand.

Exit sticky bank (pick one)

After any micro this week, leave one sticky in ordinary words:

- “Caption claimed ; ***I held for .***”
- “Object type: feed-about-evidence / evidence / unknown.”
- “Absolute: ____ per ____ (or unknown).”

³⁴SAFE-03 Kitchen Picture Honesty Card (memo 02 bank): label picture / not Chemistry credit; food-safe only. Kitchen ≠ Chemistry credit.

- “Tomorrow’s flash: cover caption 30 seconds.”

A sticky is a retrieval seed, not a diary novel. If you cannot show the sticky to a future-you tomorrow and recover the move, add three words of context.

Space across days when you can. A massed Sunday dump of twenty cards is not the hour. Three to five micros plus one desk sit in a week is an honest ambition — there is no adult commute-science RCT at this grain; named sessions are the aim.³⁵

What the looking-move strip is (and is not)

This book’s strip is a short ladder of looking moves: evidence before feed; observe and record; systems boundary; fair look; denser claim with warrant and limit. Bio, Chem, Phys, and Earth soft appear later as *vehicles* that carry one move — not as three transcript years to restart from scratch. You place yourself by which move still wobbles, not by which survey class you once passed.

The strip is not a personality test. It is not a percentile. It is not NSF/Pew process items scored as a full literacy scale. It is not teen NAEP Science or PISA pasted onto working adulthood. It is a practical way to answer: what should I practice this week so tomorrow’s card does not own me?

If you used *Science for Emerging Adults* at a kitchen table with a parent in the room, one pointer: that book owns ages 15–18 systems looking and denser CER with transcript edges. This book ages the grain into adult self-study — no parent/student split, no AP as weekly spine. If you used *History Refresh for Adults*, one pointer for commute/desk/weekend architecture and primary-before-feed — then stay on evidence before feed; looking at evidence is not history sourcing with science nouns. If you used *Math Refresher for Adults*, one pointer for rust-is-ordinary and absolute-versus-relative as a numeracy neighbour — then science; PIAAC numeracy stays a neighbour, not a science score, and Bastani stays labelled OFF-AGE for science literacy here. *Critical Thinking for Adults* owns myside and testimony; one pointer, then science.

³⁵Dunlosky et al. (2013) rate practice testing and distributed practice as high-utility learning techniques — technique ratings, not an adult-science programme evaluation. Dose honesty: no adult commute science RCT at this grain; named sessions are the ambition.

Absolute-risk sticky (micro practice you can keep)

Time: 5–10 minutes. Materials: one headline that shouts a relative change (“doubles,” “cuts by half,” “100% increase”). Safety: none; no medical decisions from this sticky. The move: write “Double what? Out of how many?” Fill absolute change if the card or abstract gives numbers; otherwise mark unknown. Exit: one sticky with relative claim and absolute (or unknown). Refuse treating relative alone as settled. Chapter 3 will thicken this with named public figures (for example, relative “100%” that is absolute **1→2 per 7000**). This week, the sticky trains the blank so the identity lock has a number shape.

A week that is honest (labelled hypothetical composite)

Illustration only — not a reported diary. Monday micro: FEED-01 on a morning card. Tuesday micro: AS-01 honesty tick on a workplace forward (identifiers stripped). Wednesday desk: FEED-02 request for abstract + one limit sentence. Thursday micro: FEED-07 wait before an outrage card. Friday: process flash labelled practice, then stop. Saturday optional deep: card vs abstract two-column sticky. Sunday: rest or one absolute-risk sticky — not a dump. If life deletes two days, keep one micro. Named sessions beat guilt calendars.

If it isn’t clicking

Diagnostic 1 — Diploma nostalgia. You keep rehearsing the class you once passed instead of looking at today’s card. The nostalgia feels like competence. It is memory of a domain, not current looking. Next move: put the diploma story on a sticky and set it aside for the session. Run FEED-01 once. Success is four blanks, not a transcript replay. If nostalgia returns mid-session, nod to it and point back at the caption cover.

Diagnostic 2 — Quiz-prep reflex. You want a score, a percentile, or a three-item process win to feel finished. The reflex is understandable; school rewarded it. Next move: label any process flash as practice, not pretest. Done-enough this week is the hold, not a fake certificate. Teen school-test percentages stay off your self-score. If you catch yourself googling “science literacy test for adults” as a finish line, close the tab and cover one real card instead.

Diagnostic 3 — Shame spiral. “Not a science person” becomes a stop sign.

Shame eats the working memory you need for denominators. Next move: shorten. One absolute-risk sticky (Double what? Out of how many?). One honesty tick. Stop while a look remains. Return tomorrow. Anxiety is ordinary; early wins beat a countdown timer as default “rigor.” This is not a therapy manual — it is session design. Correct a wrong reason without crushing the attempt — applied to yourself. “Never correct yourself” is not this book’s rule. Name the twin (caption-as-study), then try the parallel card.

When to slow down: if you cannot name whether today’s object is evidence or feed-about-evidence. When to go ahead: if you can cover a caption and fill or mark the four blanks without grabbing a summary first. When to ask a human: a colleague for a workplace citation path (without gotcha); a librarian for abstract access; a tutor or class if denser methods prose is the real goal — not because you failed a gene test.

If none of the three diagnostics fit, check wait-time. If you asked the locked opening and peeked at one second, the silence never had a chance. Cover again. Wait a slow three. Then write one blank only.

Tools, including AI

Keep this short. The full AI allows and bans live once in *The Science Refresh Hour*. Point back there. This chapter will not become an AI policy paper.

After you have tried the hold unaided: you may ask a tool to explain an idea *to yourself* from a named abstract title and today’s micro-goal; you may ask for a labelled SCRIPT of questions for *you* to ask yourself; you may ask revision questions after *your* honesty card or four blanks — not a rewrite of the study as your only draft; you may ask for isomorphic practice prompts with answers kept on *your* page.

You may not: paste the blank worksheet or employer sheet for completion (photo-to-key); treat a model summary as the study or as an observation; invent research or trust a fluent citation without opening the paper; park an unsupervised chatbot as the only partner during the first attempt; print a certificate of science literacy from a chat window; paste proprietary employer data into a public consumer tool; hand yourself an agent as a science partner for the first attempt.

Labelled OpenStax sections, Khan Academy explainers, museum pages, and

agency abstracts are free surfaces *after* the try — they check or explore; they replace thinking when they finish the attempt for you. Photo-to-key stays banned during the attempt. COPPA’s under-13 line does not apply to you — that is not a companion licence and not a science birthday.

One mechanism sentence, labelled **OFF-AGE for science literacy**, then teach: in a high-school math field experiment (grades 9–11, nearly 1,000 students, Turkey), an unguarded chatbot made practice look better and left students worse on the test with the window closed.³⁶ Not an adult science RCT. Not this science band. The lunch-break rule it supports: the model may prepare you and the next looking page; it may not do your looking when looking is the lesson.

What “done enough” looks like

Before you move on, check:

- ☐ I can cover a feed until design, absolute risk, sample, and a named limit are named — or marked unknown.
- ☐ I can refuse diploma-as-done and teen NAEP/PISA-as-adult-measure as my self-score.
- ☐ I can name whether today’s object is evidence or feed-about-evidence.
- ☐ I ran at least one on-the-go try (FEED-01 or FEED-07 or AS-01) and one desk try (FEED-02 or AS-02/AS-03).
- ☐ My talk-to-yourself opening matched the locked stem for this chapter.
- ☐ I did not treat a process flash or a chatbot précis as graduation.
- ☐ I can say, in one plain sentence, why support-for-science is not this hour’s success metric.

Placement is by looking-move strip, not diploma year. A “science literacy certificate” is a receipt someone else might print — not the work — and this book never invents one. Optional CLEP Biology or Chemistry appears later as a bridge when credit is the goal — not this chapter’s spine and not a fake diploma from a chat window.

Revisit next Tuesday: cover one new card for sixty seconds. Space across days.

³⁶Bastani et al. (2025), *Proceedings of the National Academy of Sciences*: Turkish high-school math field experiment; GPT Base about **+48%** on practice / **–17%** unaided; GPT Tutor about **+127%** / approximately control. **OFF-AGE for science literacy**. Not an adult science RCT. Keep Math Refresher’s adjacent label separate from this book’s OFF-AGE for science literacy tag.

Mix last week's sticky with a parallel card. Life-of-the-habit tips live in the Hour, in this checklist, and later in Keeping track — not in a ninth teaching chapter.

When “done enough” feels blurry, ask one sorting question: did I practice the hold on a real object, or did I only read about the hold? Reading this chapter counts as orientation. Covering a caption on a card you actually meet counts as practice. Both matter. Only practice moves the strip.

If you are already fluent at opening abstracts before captions, deepen toward sample limits and absolute denominators this week, then advance to Chapter 2's observe-before-explain without pretending the feed will never tempt you again. Fluency is maintenance, not arrival.

You understand this. You know what to ask yourself tomorrow on the commute. You know what a caption offered as the study sounds like when it is not yet an evidence look. You know what a finished-looking chatbot précis without your attempt looks like. You can do this on the go and in a deeper sit. Start with one card tomorrow.

Chapter 2



Figure 4: Stranger-readable lab notebook with honesty card and observe/record stickies — still-life, no people, no logos, no readable UI.

Observe and record

You glance at a desk plant and already have a story: it needs water; it is fine; it is dying. The story arrived before the look. Or you open a weather app and read the narrative first — “unsettled afternoon” — before the numbers. Or you fill a neat table from memory because empty cells feel embarrassing.

This chapter slows that reflex. Observe before you explain. Leave a light record a stranger could almost read. Name what the object *is* on an honesty card so invent-data and generated “results” cannot sneak in wearing a lab coat.

Chapter 1 gave you the identity lock: evidence before feed. Chapter 2 gives you eyes and a notebook. Without a record, a look evaporates into opinion. Without honesty about what the object is, a screenshot of a feed can pretend to be your measurement, and a generated gel can pretend to be a result.

Sister pointers, then teach: *Science for Emerging Adults* owns parent kitchen checks and stranger notebooks for ages 15–18 — one pointer, then this adult self-study desk. *History Refresh for Adults* owns primary-before-feed — one pointer, then observe the object before the slogan. Stay on science looking. Then notice carefully.

You are still the learner on the go. Commute, desk, and weekend sits are context bands — not ages. The teacher beside you is this page: specific about time, materials, safety, and the move.

Why this matters

Here is the move in one breath: notice what you did not expect before you explain; measure lightly with units; label honesty; leave a note a stranger could almost read; write “not measured” instead of inventing. That sequence is observe and record for adults on the go.

Looking is not the same as narrating. Narrating is fast, social, and often useful at work. Looking is slower. It asks what you notice that you did not expect — before you explain. That pause is the move.

Adults need the pause because ordinary life rewards explanation. A manager wants a takeaway. A feed wants a caption. A chatbot will happily complete a blank data

table. Your Science Refresh Hour needs the opposite muscle: notice, measure lightly with units, label honesty, leave a note tomorrow-you could parse.

America's Lab Report warned that isolated labs are weak when they are not integrated with ideas and talk — and that fake constructed data is not a lab experience.³⁷ This chapter takes that warning into adult self-study. Inventing neat rows to look scientific trains the wrong habit. Filing an AI-generated figure as what you saw trains a worse one. A kitchen dissolve picture can teach observation if you label it a picture, not Chemistry credit.

What unlocks next: systems looking needs observations that are not already slogans. Fair tests need measurements with definitions you can detect. Denser claims need evidence from *this* table or labelled set. Feed hygiene from Chapters 1 and 3 needs a mind that still knows the difference between a number you measured and a number a caption asserted. Observe and record are how evidence gets onto the page when the feed is not the source.

You do not need a laboratory to practice. You need one object in view, a timer, a notebook, and the courage to leave a blank when you did not measure. RAMP waits in the wings for heat and chemicals; this week's tries stay ordinary and food-safe when kitchen appears at all. Public paths, grocery aisles, weather pages, and pharmacy labels (units only) are enough surfaces for a full week.

Money and work motivate: a label rate, a workplace measurement claim, a park you already walk. They do not replace eyes on the object. Noticing is not a sting. You are not auditing a colleague's notebook. You are rebuilding your own.

Rust is ordinary here too. Many adults last kept a lab notebook in eleventh grade. That gap is not a character flaw. It is unused practice. Place by whether you can observe before explain today — not by diploma year.

This is useful practice for adult looking, not a promise that every adult will see the same plant, weather page, or park path the same way.

³⁷National Research Council, *America's Lab Report* (2006, Singer, Hilton, Schweingruber, eds.): isolated labs weak; integrate look-try-talk; fake constructed data $\neq$ lab experience. Access date for URLs in these notes: 22 September 2026.

Understand it yourself

Everyday picture. A window plant or desk plant sits in ordinary light. For ninety seconds you write only what you can see: leaf color patches, soil moisture look, stem lean, light direction, any motion. No “photosynthesis.” No “it needs water” yet. Then one question the observation raises. That is observe before explain in a shape you can finish on lunch.

The same everyday picture works on a commute window: condensation, sky color bands, rain streaks, temperature feel on glass — still notice before weather-story words. Or on a grocery produce item: skin texture, layers, seeds, smell — structure before anatomy lecture.

Precise picture. Observe means noticing with a temporary ban on causal story. Record means leaving a stranger-readable trace: what you did, with what, when, what you measured, in what units, and what you did not measure. An honesty card names the object type: my measurement; labelled secondary; screenshot; AI-generated; kitchen picture; unknown. An operational definition says how you will *recognize* a fuzzy word in this try (“wilting” means leaves droop past horizontal when viewed from the side — your rule for today). A photo of setup is data context, not a staged proof after inventing results.

Light measure with units is enough: three temperatures of tap water soft; five transect stops; dose frequency on a label as designed rate (units arithmetic only — no medical advice). Table before graph: raw rows first, chart only if it helps. Uncertainty crumb: measure thrice; write “about X” with a simple spread; refuse fake precision that performs science without measuring.

Why “stranger-readable”? Because future-you is a stranger to today’s haste. If tomorrow you cannot recover what you did, the look did not become evidence. America’s Lab Report’s warning about constructed data is the twin: a pretty table with invented cells is not a record of a look. It is theater.

Prior ideas arrive early. School-age research traditions on theory–evidence coordination and prior conceptions are labelled continuity here — not an invented adult RCT.³⁸ Adults still meld theory and evidence when looking is rushed. The repair is procedural: timer, line on the page, explanations below the line.

³⁸Continuity from school-age research on prior ideas and theory–evidence coordination (*Taking Science to School* / Driver traditions labelled school-age origin): adults still meld theory and evidence when looking is rushed — labelled continuity, not an invented adult RCT.

Wrong answers you should be able to hear — including from yourself:

1. **“I’ll explain first — I already know what this is.”** Jumping to the name (“photosynthesis,” “front,” “dehydration”) before noticing trains caption-brain on a plant. Cover the story. Notice first. The explanation may still be useful — after the look.
2. **“Empty cells look unprofessional — I’ll fill what I meant to measure.”** Invent-data. Pretty graphs that hide missingness train fraud habits even when stakes are kitchen-table low. Write “not measured.” Say aloud why. AS-04 rehearses this on purpose.
3. **“This AI gel / sky photo / micrograph is close enough to file as what I saw.”** Generated figure as result. Label fiction. Refuse. A model summary is also not your observation log. AS-05 belongs in your refusal toolkit even if you run it lightly this week.
4. **“A screenshot of the weather story is my observation.”** The app story is a later telling. Numbers and icons can be observed; the narrative compression is secondary. Cover the story first (OBS-03). Honesty-label the screenshot.
5. **“My notes are too messy — better to rewrite from memory into something neat.”** Neat rewrite that invents missing numbers fails the stranger notebook test. Repair the header: units, time, setup. Do not invent. Messy with units beats neat fiction.

Five-minute warm-up before the try. Sit with one object in view (plant, window weather, cup of water). Set a ninety-second timer. Write only Observe bullets — color, motion, moisture, light, sound if any. No explanations. When the timer ends, draw a line. Below the line, write one question. Cover the question. Read the bullets. Ask: did any explanation sneak above the line? Cross it down below. That correction is the warm-up win. If nothing unexpected appeared, write that too: “Nothing unexpected yet — light was even; leaves still.” Unexpected includes the absence of change.

How to try it this week

Session shape follows the Science Refresh Hour. This week’s move: observe before explain; light record with honesty.

Warm-up retrieval (2–3 min). Say: “Notice before name. Record so a stranger could almost recover what I did. Label what the object is.” Retrieve one wrong twin (explain-too-fast; invent neat rows; screenshot-as-measurement; generated-as-result).

Short model — an evidence look (3–5 min). Run a silent OBS-01 on a plant or window for ninety seconds while you watch your own hand. Then show yourself a repaired header line: date, object, units, “not measured: ____.” Speak the locked opening once.

Your attempt (8–12 min). Parallel object. Observe-only notes. One question. Honesty tick (AS-01). No chatbot completion of the table. If you feel shame about mess, leave the mess and add units only.

One good self-question. Locked opening below. Wait before lookup.

Mixed practice / exit sticky (2–3 min). “Today I noticed ____ before I explained. Honesty: ____.” Tomorrow: one Observe line on the commute window or desk corner.

Context bands

Commute / lunch (~15–20). OBS-01 micro on a window plant photo you already have, or OBS-03 weather page without the story first. REC-04 one setup photo with caption if you measured anything. AS-04 invent-data refusal rehearsal on a hypothetical empty cell. Keep materials pocket-sized: notebook page, phone timer, one object.

Desk or shift (~25–40). OBS-04 pharmacy label rate soft (units only; privacy; no dosing changes). REC-01 stranger notebook test on yesterday’s notes. REC-02 table before graph if you have two or more measurements. REC-05 operational definition sticky for one fuzzy word. REC-03 uncertainty crumb on tap-water temperature soft — no boiling required. Strip any workplace identifiers if notes leave the building.

Weekend deep sit (~45–75). OBS-05 park transect soft (five stops) or OUT-01 return. Table of five rows + one pattern claim with a limit. Optional OUT-03 grocery aisle structure–function if park weather fails. SAFE-03 if any kitchen picture appears — label not-for-credit. Stop while a look remains; do not turn five stops into a scavenger mania.

First try-its this week

OBS-01 Notice Before Name (window / desk plant). Time 15–20 min. Materials: one living or physical object in view; timer; notebook. Safety: ordinary. The move: 90 seconds Observe-only notes (color, motion, moisture, light). No explanations yet. Then one question the observation raises. Exit: three Observe bullets + one question. Refuse jumping to “photosynthesis” before noticing. Band: micro or desk.

OBS-03 Weather Page Without the Story First. Time 15–20 min. Materials: local weather chart or radar screenshot; honesty label. Safety: none. The move: cover the headline story; read numbers/icons only; write three observations; uncover story; note compression/hype. Exit: observation list vs story mismatch sticky. Refuse filing the app story as your observation. Band: micro.

REC-01 Stranger Notebook Test. Time 15–25 min. Materials: yesterday’s notes or today’s try table. Safety: as per paired try. The move: hand notes to a future-you test — could a stranger recover what you did, with what, when, and what you measured? Add missing units, time, and setup line. Exit: one repaired header line. Refuse pretty rewrite that invents missing numbers. Band: desk; micro repair flash.

AS-04 Invent-Data Refusal Rehearsal. Time 10–15 min. Materials: tempting empty cell in a table (hypothetical). Safety: none. The move: practice writing “not measured” instead of inventing; say aloud why invent-data trains fraud. Exit: “Not measured” line. Refuse filling for pretty graphs. Band: micro.

How to fade help. Day one: timer and “no explain” sticky on the page. Day two: timer only. Day three: notice the urge to explain without the sticky; wait; then notice. When a tool offers to “clean up” your table, keep raw rows; ask only for format tips after you have real numbers. Fade means fewer scaffolds, not faster rescue from a model.

When to stop looking up. After three Observe bullets and one question, stop. Do not open a textbook chapter on plants to rescue the question. The question is a seed for later chapters, not a homework hunt tonight. Looking up becomes rescue when it finishes the noticing for you.

Talk-to-yourself box

Opening (locked): “What do I notice that I didn’t expect — before I explain?”

Follow-ups (pick three to five; wait alone after each):

1. Meaning: “What counts as *notice* here — what would be explaining in disguise?”
2. Representation: “What does my table or sketch show before any caption I might add?”
3. Alternative: “What else could these observations fit besides my first story?”
4. Check: “What did I not measure — and did I write ‘not measured’?”
5. Revise: “I may change my mind — where could someone go wrong by explaining too fast?”

How to wait alone. After you ask, cover any feed, key, or chatbot. A slow three — pause-before-lookup.³⁹ Look at the object or the blank row, not at the phone. If silence is hard, describe one color out loud, then wait again. Rescue at one second is still rescue.

Stuck silence usually means: the question was vague; wait-time was zero; you are searching for the “right science word”; you think messy notes mean you failed; the phone is replacing looking; you think a finished chatbot CER without a trial is already a check; anxiety wants a finished chart — shrink to one Observe line; cover the story; return tomorrow. Your job is to pose, wait, and point back at the object.

Exact wording: “Before I look up the answer or trust the feed: what do I notice, what claim can I defend from *this* evidence, and what limit must I name?”

Practice that actually builds it

What “light measure” means (and does not)

Light measure is not a survey of every instrument in a catalog. It is enough quantity to keep you honest: a unit on the page, a repeated trial, a range instead of fake precision. Three tap-water readings beat one theatrical number written to three

³⁹Rowe (1972) wait-time; Stahl (1994) think-time — classroom origin. Adult translation: pause-before-lookup / pause-before-feed. Not an adult science RCT.

decimals. Five transect stops with the same question beat twenty photos with no table.

Light measure is also not medical advice, not dosing, not a workplace audit, and not Chemistry credit. When OBS-04 touches a pharmacy label, you practice rate literacy — what the label states per day, what “per day” assumes — and you write a question for a clinician if needed. You do not change medication from this Hour. When SAFE-03 touches a kitchen dissolve, you practice observation and honesty labels. You do not claim transcript Chemistry.

Honesty layers you will keep using

Keep the honesty card vocabulary stable across weeks so it becomes automatic:

- **My measurement** — you looked or measured; units and setup exist.
- **Labelled secondary** — package info, agency page, abstract, reputable explainer — named.
- **Feed screenshot** — later telling; observe numbers/icons if present; do not file as your measurement of the world.
- **AI-generated** — fiction unless proven otherwise; never a result.
- **Kitchen picture** — food-safe look; not-for-credit.
- **Unknown** — you cannot yet say what it is; that tick is progress.

AS-01 is the drill. AS-05 (generated figure / gel refusal) can sit beside it on a desk day: create or find one labelled fiction image of a “result,” list tells, write a refusal rule for Keeping track. You do not need fancy forensics. You need the sentence: “Generated is not what I saw.”

Stranger notebook — worked repair pattern

Composite illustration: a page says “plants taller after sun.” No date. No which plants. No unit (cm? inches?). No how measured. Repair header: “22 Sep 2026 evening; two basil pots on south sill; height from soil to tip, cm, ruler; not measured: soil moisture, leaf count.” The claim may still be soft. The record is now almost recoverable. That is REC-01’s win.

If you cannot repair without inventing, write “cannot recover — restart OBS tomorrow” instead of fabricating. Restarting is integrity. Fabricating is theater.

Representation before tidy slogan

A table is a representation. A sketch of leaf positions is a representation. A caption without rows is a slogan wearing boots. Practice looking at your own representation before you tidy it into a sentence. Ask: which cell supports that sentence? If no cell does, the sentence is explaining ahead of the record — move it below the line or delete it.

This habit will feed Chapter 7's denser CER: evidence must be locatable in *this* table or labelled set. You are planting that habit now without writing full CER four-boxes as this week's identity.

Banks for this chapter: OBS-01...05, REC-01...05, AS crumbs, TALK support. Menus, not calendars.⁴⁰ Aim for at least one on-the-go and one deeper sit. You do not owe all ten OBS/REC items this week. You owe honest attempts on objects you can see, with honesty labels that tell the truth about what they are.

Practice that builds observation is not a keyword list. Weather pages, grocery aisles, and park paths *are* the practice — they do not replace the looking structure. The structure is: notice → light measure with units → honesty → stranger-readable trace → one question or soft claim with a limit.

On the go — commute / lunch

OBS-01 as named above. If no plant, use a physical object on the seat tray: bottle condensation, light on metal, ice melting in a cup — still Observe before name. Write three bullets. Draw the line. One question below.

OBS-03 as named above. Honesty-label the screenshot: labelled secondary / app story covered first. Compare your three number/icon observations to the story's verbs. Note one compression ("unsettled" for a 20% rain icon, or similar mismatch in your own words).

REC-04 Photo of Setup Is Data Context. Time 10–15 min. Materials: phone camera; today's setup. Safety: no photos of other people's private meds or documents without consent. The move: one photo of setup with a scale or reference; caption date, what changed, what stayed same. Exit: caption completeness check. Refuse staging a "perfect" setup photo after inventing results. Band: micro.

⁴⁰Named try-its OBS-01...05, REC-01...05, AS-01/AS-04/AS-05, SAFE-01/SAFE-03, OUT-01/OUT-03, TALK-01/TALK-02 from memo 02 teaching banks — menus, not calendars.

AS-01 Honesty Card Drill. Time 10–15 min. Tick: my data / labelled secondary / feed screenshot / generated / kitchen picture / unknown. Exit: card completed. Band: micro. Skip no card because it felt “obvious.”

AS-04 as named above. Keep a hypothetical empty cell on a sticky in your bag. When the urge to invent appears on a real table, rehearse the line.

Desk or shift

OBS-04 Pharmacy Label Rate Soft (no medical advice). Time 15–20 min. Materials: one OTC label you already use or a public Drug Facts example; not a diagnosis session. Safety: do not change medication based on this Hour; no dosing experiments. The move: observe dose/frequency/warnings as *data about a designed rate*; ask what “per day” assumes; write one literacy question for a pharmacist or clinician — do not self-prescribe. Exit: “Label states ; ***I must not conclude*** without ____.” Refuse medical advice Hour; comparing illicit substances. Band: desk (privacy). Units arithmetic only — mg per tablet, tablets per day as label literacy, not a care plan.

OBS-02 Grocery Structure–Function Soft. Time 15–25 min (aisle or home with produce). Materials: two produce items; phone note. Safety: food hygiene; no tasting experiments that waste or risk allergens beyond your norms. The move: observe structure (skin, layers, seeds); hypothesize function; check one labelled secondary *after* your hypothesis. Exit: “Structure X may relate to function Y because ____ (limit: ____).” Refuse survey dump of plant anatomy. Band: micro (store) or desk. OUT-03 is the same move in-store when you want a deeper grocery sit.

REC-02 Table Before Graph. Time 15–25 min. Materials: two or more measurements; paper or sheet. The move: build a labelled table (variable names, units, trials) *before* any chart; only then graph if useful. Exit: table with units; optional graph linked to table rows. Refuse chart without a table; chart that hides raw values. Band: desk.

REC-03 Uncertainty Crumb. Time 10–20 min. Materials: one repeated measurement (length, time, temperature of tap water soft). Safety: no boiling required; avoid burn risk. The move: measure thrice; record range or simple spread; write “about $X \pm \dots$ ” in ordinary words. Exit: one uncertainty sentence. Refuse fake precision to look scientific. Band: micro or desk. Tap water soft means ordinary faucet temperature sense with a kitchen thermometer if you own one — not

a chemistry titration.

REC-05 Operational Definition Sticky. Time 15–20 min. Materials: one fuzzy word from your question (e.g. “wilting,” “faster,” “cloudy”). The move: define how you will *recognize* it in this try (observable criteria); refuse dictionary-only definitions. Exit: operational definition in one sentence. Refuse explaining without a detection rule. Band: desk. Example shape: “For this try, ‘cloudy’ means I cannot read 12-point text through the glass at arm’s length.” Your rule; your try; named limit.

REC-01 as named above. Do this on any note you already wrote this week. The repair *is* the practice.

Weekend deep sit

OBS-05 Park Transect Soft (five stops). Time 25–40 min. Materials: notebook; five stops along a short path; optional phone photos. Safety: public path; weather-appropriate; no trespass; tick/allergy awareness. The move: same question at five stops (e.g. moisture, plant height soft, human impact signs); record without explaining mid-walk; explain after. Exit: table of five rows + one pattern claim with limit. Refuse scavenger mania with no return to one claim. Band: desk-length walk or weekend OUT.

OUT-01 Park Transect Return (when you can extend). Same path another day; compare tables; name what changed in conditions; keep the honesty card. Limit: two walks do not settle a season.

SAFE-03 Kitchen Picture Honesty Card (if kitchen appears). Time 10–20 min. Materials: one food-safe change-of-matter idea (dissolve, melt ice); honesty card. Safety: food-safe only; no oven-cleaner “chemistry”; no drain-cleaner mixes. The move: label the try **picture / not Chemistry credit**; run soft OBS/REC if desired; write what a supervised secondary lab path would add. Exit: “This is a picture because ____.” Refuse filing kitchen fizz as transcript Chemistry. Band: desk. If you cannot minimize a hazard, stop; SAFE-01 RAMP belongs before any heat or chemical in later weeks.

Incorrect example to diagnose (composite illustration, not a reported case). A notebook page shows a perfect rising graph of “plant growth” for seven days. No units. No setup line. No “not measured” days. A generated photo of a lush plant is taped beside it, labelled “Day 7 result.” Diagnose: invent-data; missing stranger

header; generated figure as result; explain-before-observe likely. Repair: AS-04 “not measured” on empty days; REC-01 repair header; label the photo fiction if generated; start OBS-01 over on a real plant for ninety seconds. Keep the repaired page in your looking file — the repair is the learning product.

Talk-to-yourself box (practice depth)

Reuse the locked opening. When stuck, point at the object: “Say three things you see that are not a theory.” If the phone replaces looking, put it face down after the setup photo. TALK-01 (one question, three seconds) and TALK-02 (cover the feed/model) support wait practice alone — use them as micros when OBS feels rushed.

Blocked for a new move: do not jump to fair-test redesign as this week’s identity; notice and record first. Mixed: when the object is a feed screenshot, honesty-label it secondary and observe the numbers — do not pretend you measured the sky. When Chapter 3’s feed cards appear, bring honesty cards with you.

Safety spine crumbs (ordinary this week)

Most OBS/REC tries this week need no RAMP card. Ordinary judgment is enough: public paths, food hygiene, privacy on labels, no trespass. When anything with heat or chemicals appears, stop and run SAFE-01 before step one: Recognize hazards; Assess who could be harmed; Minimize; Prepare for spill/burn/glass; decide go / redesign / lab-path neighbor / skip. SAFE-02’s stop sentence still applies: you stop if you cannot name a hazard, cannot minimize, or feel pressured to mix unknowns. Adult present to self means you are allowed to stop — including when a video makes unauthorized mixing look casual.

Kitchen $\neq$ Chemistry credit remains a house rule for the whole book. Say it once when SAFE-03 appears, then teach the observation.

Connecting back to Chapter 1 without diluting Chapter 2

If a feed card interrupts your Observe session, do not abandon the plant for outrage scroll. Either: (a) finish ninety seconds on the object, or (b) switch deliberately to a Chapter 1 FEED-01 micro with honesty label — then return to Observe another day. Mixing without naming the move creates muddle. Naming the move creates a strip.

Chapter 3 will thicken feed vs study. Bring your honesty card vocabulary with you. A press release is labelled secondary. A model précis is generated secondary until checked. Your transect table is my measurement — if you measured.

Materials kit (cheap, pocketable)

Optional kit for the week: small notebook, pen, phone timer, kitchen thermometer if you own one, ruler, honesty card template (six ticks), “not measured” sticky, goggles only if a later risky try needs them. You do not need a lab bench. You need the kit you will actually carry.

A week that is honest (labelled hypothetical composite)

Illustration only — not a reported diary. Monday: OBS-01 micro on desk plant. Tuesday: OBS-03 weather without story. Wednesday desk: REC-03 uncertainty on tap water soft + REC-01 repair. Thursday: AS-04 invent-data refusal. Friday: OBS-04 label rate soft (units only) or OBS-02 produce at home. Saturday: OBS-05 five stops if weather allows. Sunday: rest or one honesty tick — not a dump. If life deletes days, keep one Observe line. Named sessions beat guilt calendars.⁴¹

Exit sticky bank (pick one)

- “Noticed ___ before explaining ___.”
- “Honesty: my data / secondary / screenshot / generated / kitchen / unknown.”
- “Header repaired: units / time / setup.”
- “Not measured: ___.”
- “Tomorrow: 90 seconds Observe-only.”

If it isn’t clicking

Diagnostic 1 — Explaining too fast. Your pen writes causes before colors. The urge feels like fluency. It is prior story arriving early. Next move: ninety-second timer; “no because” rule above the line; one Observe line only if needed. Cover any story (weather headline, textbook name, chatbot précis). Return to OBS-01

⁴¹Dunlosky et al. (2013): practice testing and distributed practice as high-utility technique ratings — not an adult-science programme evaluation. Dose honesty: named sessions are the ambition.

tomorrow. If you catch a “because” mid-bullet, draw an arrow down below the line and keep going.

Diagnostic 2 — Shame about messy notes. You rewrite from memory into something that looks published. Shame says mess equals incompetence. Next move: REC-01 stranger test on the messy page; repair header only; celebrate “not measured.” Messy with units beats neat fiction. If a colleague might see the page, remember: stranger-readable is clarity, not performance calligraphy.

Diagnostic 3 — Phone replaces looking. You photograph, then scroll, then accept an app story as observation. The phone is a tool and a trap. Next move: cover the story (OBS-03); write three numbers/icons; only then uncover. Or OBS-01 with phone in another room for ninety seconds. Setup photos (REC-04) happen *after* the Observe bullets, with a caption — not instead of looking.

When to slow down: if every note is already a theory; if empty cells make you invent; if kitchen pictures want credit language. When to go ahead: if you can produce three Observe bullets, an honesty tick, and a header with units. When to ask a human: librarian for weather archive pages; colleague for a second stranger-notebook read (strip private data); community class or maker space if measurement tools are the real goal; pharmacist or clinician for label questions that go beyond units literacy — this Hour does not answer those.

Correct yourself kindly and clearly: “That sentence was explaining — move it below the line.” Then keep the attempt. “Never correct yourself” is not this book’s rule. Crushing the attempt is also not the rule.

If none of the three fit, check wait-time. If you asked the locked opening and peeked at one second, cover again. Wait a slow three. Write one unexpected detail only.

Tools, including AI

Keep this short. The full AI allows and bans live once in *The Science Refresh Hour*. Point back there. Do not turn this chapter into an AI policy paper.

After you try: blank table templates with *your* key for isomorphic practice; revision questions on *your* note; a labelled SCRIPT of observe prompts for you to say; diagnose work you already did (crop to paper — no face); labelled schematic put away before the try — never filed as “what I saw.”

You may not: invent an observation or data table; generate a gel, micrograph, sky photo, spectrum, or “lab photo” and treat it as a result; paste a blank packet for the tool to complete; put a model summary on the table as the observation; ask the tool to write your lab note as the only draft; use TTS or narrated rewrite to replace looking at today’s object.

OpenStax Biology or agency weather pages may explore *after* your Observe bullets — they check or deepen; they replace thinking when they finish the ninety seconds for you. Photo-to-key stays banned during the attempt. Workplace proprietary data stays out of public consumer tools.

One pointer only: the Bastani high-school math field result stays **OFF-AGE for science literacy** — not an adult science RCT; the lunch-break rule still holds: the model may prepare the next page; it may not do your looking when looking is the lesson.⁴²

What “done enough” looks like

Before you move on, check:

- ☐ I observed before I explained on at least one real object (OBS-01 or OBS-03 or OBS-05).
- ☐ I left a table or note with units a stranger could almost parse (REC-01 / REC-02).
- ☐ I filled an honesty card (AS-01) and practiced “not measured” (AS-04) instead of invent-data.
- ☐ Pharmacy or label work, if any, stayed on units — no medical advice.
- ☐ Kitchen picture, if any, is labelled not-for-credit (SAFE-03).
- ☐ My talk-to-yourself opening matched this chapter’s locked stem.
- ☐ I did not file a generated figure as a result or a model summary as an observation.
- ☐ I can point to one unexpected detail I would have missed if I explained first.

When “done enough” feels blurry, sort with one question: did I look at a real object and leave a recoverable trace, or did I only tidy words? Tidying words can wait. Looking cannot be retroactively invented without becoming invent-data.

⁴²Bastani et al. (2025), *PNAS*: Turkish HS math; GPT Base about +48% practice / –17% unaided; GPT Tutor about +127% / $\approx$ control. **OFF-AGE for science literacy**. Not an adult science RCT. Keep Math Refresher’s adjacent label separate from this book’s OFF-AGE tag.

If you already notice before naming with ease, deepen uncertainty crumbs and operational definitions this week, then advance toward Chapter 3's feed hygiene without dropping honesty cards. Fluency is maintenance. A generated figure will still tempt later — keep the refusal sentence warm.

If park access is hard, grocery produce (OBS-02) and weather pages (OBS-03) carry the same observe-before-explain muscle indoors. Place by life demand. The strip does not require a nature documentary weekend.

Placement by skill, not diploma year. A certificate is not the work. Revisit next Tuesday: one Observe line before any story. Space across days. Mix last week's plant with this week's weather page. Keeping track will hold dated notes, stranger-readable tables, and invent-data / generated-result refusal logs later — not a teen transcript and never Science Refresh for Adults I.

You understand this. You know what to ask yourself tomorrow on the commute. You know what explaining-before-noticing sounds like when it is your own voice. You know what invent-data and generated gels look like when they pretend to be records. You can observe before you explain — on the go and in a deeper sit. Leave the blank when you did not measure. That blank is part of the science. Start with ninety seconds tomorrow. Cover the story. Write what you see. Wait before you explain.

Chapter 3

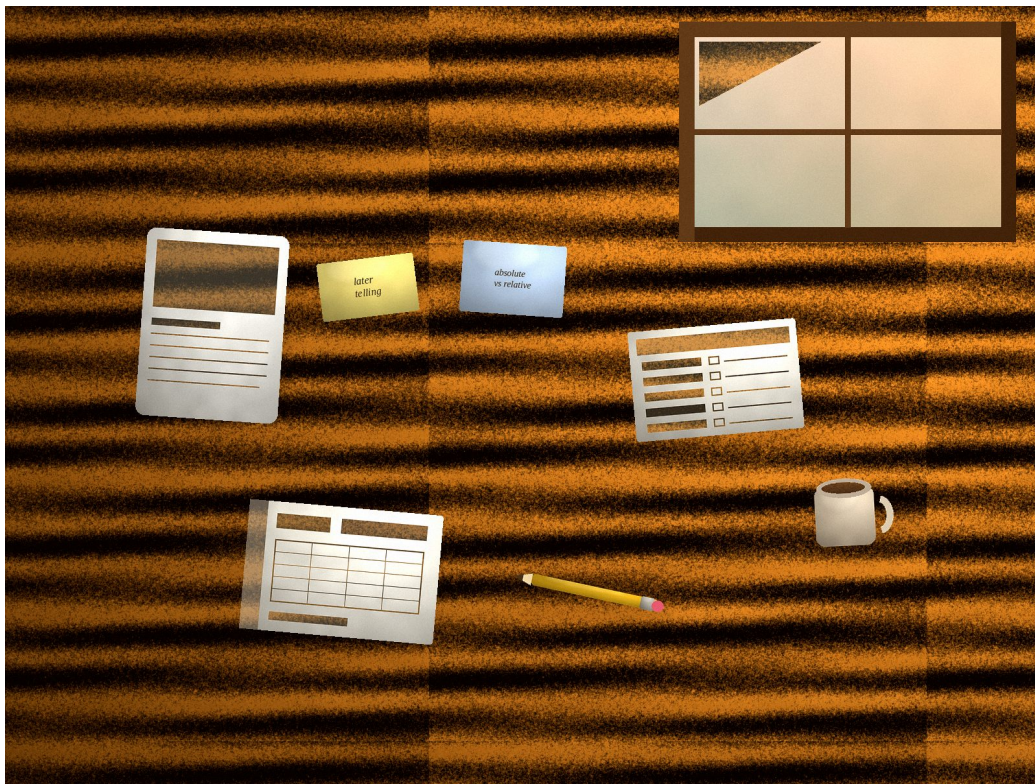


Figure 5: Described feed card, later-telling sticky, absolute-vs-relative sticky, honesty card — still-life, no product chrome, no logos.

Feed vs study

A card lands. It says a study shows something sharp. Your thumb moves. Or a podcast host says “researchers found,” and the voice is warm enough that the claim feels checked. Or a workplace slide ends with “the research is clear,” and the room treats the caption as peer review. Or a chatbot hands you a neat précis of “the findings,” and the neatness feels like having read.

This chapter is where the focus lock lives in full: evidence before feed summary. The card is a later telling until you name design, absolute risk, sample, and a limit — or mark them unknown — and until you know what you would open next. Chapter 1 named the identity. Chapter 2 gave you eyes and a notebook. Chapter 3 is hygiene for the archive you actually use: the phone, the meeting, the earbud, the model window.

You are the learner. Ordinary stakes only. No live controversy as curriculum. Noticing is not a sting.

Why this matters

Here is the move in one breath: treat the card as a later telling until design, absolute risk, sample, and a named limit are on the table — or marked unknown — and name what you would open next before you share, decide, or file the claim as settled.

Money, workplaces, and health leaflets motivate because they arrive as cards. They do not replace opening the evidence. A pharmacy leaflet risk line is a looking surface for absolute versus relative — not a dosing session. A workplace study claim is a looking surface for citation and sample — not a gotcha campaign. Climate, energy, and vaccine *topics* may appear only as who-measured / what-compared / what-the-abstract-can-settle surfaces — never as a culture-war pamphlet curriculum. Live controversy stays off this table.

Without feed hygiene, every other looking move collapses into slogan. You can observe a plant carefully and still treat a viral graph as settled science. You can draw a systems boundary and still confuse a press release with an abstract. You can write a denser claim and still paste a model summary as if it were your trial.

Adults’ default archive is a phone. That is not a moral failure. It is the environment.

Science literacy for adults on the go means portable hygiene moves: caption is not the study; workplace “study says” needs a methods crumb; press release verbs differ from abstract verbs; influencer graphs need axes and baselines; podcast hosts are storytelling secondaries; model summaries are not observations; outrage clicks can wait sixty seconds; absolute risk needs a denominator; sample is not population; design comes before slogan.

Gigerenzer’s risk-communication work is the named neighbour for absolute versus relative: a relative “100% increase” can be an absolute change of **1→2 per 7000** (an increase of **1 in 7000**); a “25% risk reduction” in mammography contexts is often misread as 25 or 250 fewer per 1000 when the absolute figure people need is closer to **~1 in 1000**.⁴³ NIH-style public communication checklists neighbour the same habit: association versus causation; absolute and relative together; study type; limitations, sample, conflicts.⁴⁴ Ioannidis (2005) is an essay/model neighbour about how positive predictive value depends on power, bias, and pre-study odds — not a census that “most papers are false” in every field.⁴⁵

Why struggle? Because the feed will still be there tomorrow. Hygiene is maintenance. Parallel to *History Refresh for Adults*’ primary-before-feed architecture — one pointer — then stay on science evidence; stay on science evidence rather than swapping history sourcing into CER.⁴⁶

This is useful hygiene for adult looking, not a promise that every adult will read every paper behind every card, and not a promise that every study replicates the same way in every home.

⁴³Gigerenzer (2009) WHO editorial and related PSPI pathway: relative “100%” pill scare as absolute **1→2 per 7000** (increase **1 in 7000**); mammography “25% risk reduction” often misread vs **~1 in 1000**. Access date for URLs in these notes: 22 September 2026. Useful communication research — not a promise every adult will misread the same way, and not medical advice.

⁴⁴NIH public-communication checklist neighbour: association vs causation; absolute and relative risk; study type; limitations, sample, conflicts — teach as hygiene prompts, not as a statute.

⁴⁵Ioannidis (2005), *PLoS Medicine*: PPV depends on power, bias, pre-study odds, multiple teams — essay/model neighbour, not a proven “most papers false” census for every field.

⁴⁶Parallel to *History Refresh for Adults* primary-before-feed / Civic Online Reasoning lateral grain — one pointer, then science evidence. Stay on science evidence rather than swapping history sourcing into CER.

Understand it yourself

Everyday picture. A headline shouts that a risk “doubles.” Your stomach drops. Everyday hygiene asks: Double what? Out of how many? If the card will not say, you mark absolute unknown and you do not share yet. The feeling can stay. The claim does not get to graduate.

Precise picture. A feed card, press release, host claim, influencer graph, workplace “study says,” or model précis is a *later telling*. The study — when it exists — lives in methods, results, limitations, sample, and design. Your job is to ask: Is this card the study or a later telling — and what would I open next? Next might be the abstract, a methods crumb, a reputable explainer, a second source (lateral hop), or an honest “cannot evaluate yet.”

Absolute risk puts a denominator under a change. Relative risk alone can sound huge when the baseline is tiny — Gigerenzer’s **1→2 per 7000** picture. Sample versus population asks who was measured and who you are tempted to speak for. Design before slogan asks whether comparison structure can support the verb on the card (“proves,” “suggests,” “associated with”).

Wrong answers you should be able to hear — including from yourself:

1. **“The feed card is the study — it says study shows.”** Caption-as-study. Cover the caption. Ask what you would need. Open next or mark unknown.
2. **“The model précis means I’ve read the findings.”** Model summary $\neq$ observation or study. Label generated secondary. Check two concrete claims against an abstract or your table. Never file as notebook.
3. **“Relative is enough — doubling is doubling.”** Without absolute and baseline, you cannot tell whether doubling is **1→2 per 7000** or something larger. Ask for both.
4. **“This sample speaks for everyone.”** Sample $\neq$ population. Write who was measured, N if findable, time window, generalization limit (AS-03).
5. **“I don’t have time for papers, so the host will do.”** Time is real. Hygiene scales: one methods crumb or one absolute sticky still beats zero. FEED-07’s wait is shorter than a rabbit hole.

Precise picture, continued: “What would I open next?” is load-bearing because hygiene without a next action becomes mood. Next action can be tiny. Opening

the first screen of an abstract counts. Writing “paywalled — library Saturday” counts. Writing “no DOI on card — cannot open” counts. Sharing before any of those fails the hour.

Representation before tidy slogan applies to feeds too. Look at the graph’s axes before the caption’s moral. Look at the abstract’s limitation sentence before the host’s certainty. Chapter 2’s observe-before-explain is the same muscle on a plant; Chapter 3 applies it to mediated claims.

Five-minute warm-up before the try. Pick one science card. Cover caption sixty seconds. Write: study or later telling? What would I open next? Absolute (out of how many)? Sample? Limit or unknown? Uncover. Notice what is still missing. Stop. Warm-up ends while a look remains.

How to try it this week

Session shape follows the Science Refresh Hour. This week’s move: feed vs study; evidence hygiene; lateral look soft.

Warm-up retrieval (2–3 min). Say: “The card is a later telling until design, absolute risk, sample, and a limit are named — or marked unknown. What would I open next?” Retrieve one wrong twin (caption-as-study; model-as-read; relative-without-absolute).

Short model — an evidence look (3–5 min). FEED-01 on one public card: cover, need-list, open abstract crumb or mark unknown. Speak exit: “Caption claims ; *evidence I checked .*”

Your attempt (8–12 min). Parallel card or workplace claim (identifiers stripped). Same exit line. No sharing before the hop. No pasting into a chatbot as first move.

One good self-question. Locked opening below. Wait.

Mixed practice / exit sticky (2–3 min). “Card = later telling / study surface / unknown. Next open: . *Absolute:* / unknown.”

Context bands

Commute / lunch (~15–20). FEED-01 caption $\neq$ study; FEED-07 click restraint; FEED-04 influencer graph honesty if a graph appears; AS-01 honesty tick;

absolute-risk sticky (Double what? Out of how many?).

Desk or shift (~25–40). FEED-02 workplace study-says soft; FEED-03 press release vs abstract; FEED-05 podcast host vs named source; FEED-06 model summary refuse; AS-02 methods crumb; AS-03 sample/population; WRITE-04 caption rewrite crumb optional.

Weekend deep sit (~45–75). Two-source feed vs study: card + abstract (or press + abstract). CER-02 crumb: evidence must be locatable. Optional OUT-04 library/database abstract hunt. Claim with limit: what the card cannot settle. Stop while a look remains.

First try-its this week

FEED-01 Caption Is Not the Study. Time 15–20 min. Materials: one science newsfeed/social card; screenshot OK if honesty-labelled. Safety: none. The move: cover caption; note image/graph; uncover; ask what evidence would be needed; open abstract/methods crumb or reputable explainer before sharing. Exit: “Caption claims ; *evidence I checked* .” Refuse sharing before the hop; refuse virality as corroboration. Band: micro.

FEED-02 Workplace “Study Says” Soft. Time 15–25 min. Materials: one Slack/email/meeting claim that cites a study. Safety: workplace confidentiality — strip identifiers; no undercover audit; no proprietary data in public tools. The move: ask for citation or abstract; note sample/population if findable; write limit; do not weaponize against coworkers. Exit: citation status + one limit. Refuse gotcha culture as science literacy. Band: desk/shift.

FEED-03 Press Release vs Abstract. Time 20–35 min. Materials: one press release + paper abstract on same claim (when findable). Safety: none. The move: compare verbs (proves / suggests / associated); list compressions. Exit: verb-shift sticky. Refuse treating press release as peer review. Band: desk.

AS-02 Methods Crumb or Mark Unknown. Time 15–20 min. Materials: science claim. Safety: none. The move: write methods crumb or explicitly “unknown — cannot evaluate yet.” Exit: crumb or unknown. Refuse assuming rigor from confident tone. Band: micro or desk.

How to fade help. Day one: four blanks template (design / absolute / sample / limit). Day two: blanks from memory after cover. Day three: name “what I’d

open next” without template. When a tool offers a study summary, put it away until your blanks exist.

When to stop looking up. After one abstract crumb, one absolute sticky, or one honest unknown — stop. Do not collect five outrage cards for “balance.” One parallel try is enough for a micro.

Talk-to-yourself box

Opening (locked): “Is this card the study or a later telling — and what would I open next?”

Follow-ups (pick three to five; wait alone after each):

1. Meaning: “What do I mean by *later telling* — who is telling after the measurement?”
2. Representation: “What does the table, figure, or abstract show before the caption?”
3. Alternative: “What else could explain these numbers if the host is polished?”
4. Check: “What can this evidence not tell me — absolute? sample? causal verb?”
5. Revise: “I may change my mind — where could someone go wrong treating this card as settled?”

How to wait alone. Cover the feed or model. Slow three — pause-before-feed.⁴⁷ Look at blanks or clock. Rescue at one second is still rescue. Forty minutes of scrolling is not a micro-session.

Stuck silence usually means: vague question; wait-time zero; outrage eating working memory; host polish substituting for methods; “no time for papers” as total shutdown; chatbot précis feeling like a check — shorten; one methods crumb; one absolute sticky; return tomorrow.

Exact wording: “Before I look up the answer or trust the feed: what do I notice, what claim can I defend from *this* evidence, and what limit must I name?”

⁴⁷Rowe (1972); Stahl (1994) — classroom origin → pause-before-feed. Not an adult science RCT.

Practice that actually builds it

Design before slogan — soft checklist

When a card uses strong verbs, run a soft design checklist in ordinary nouns:

- What was compared to what?
- What stayed the same (if anything)?
- Was this observational association language or experimental comparison language?
- Is causation being sold with association verbs?
- What limit does the abstract already admit?

You do not need to run a clinical trial to ask these. You need to refuse slogan verbs when design crumbs are missing. AS-02's "unknown — cannot evaluate yet" is a legitimate exit when the card will not say.

Press release vs abstract — what to listen for

Press offices compress. That is their job. Your job is to hear the compression. Common shifts (composite patterns, not a census): "proves" where the paper says "suggests"; "breakthrough" where the abstract says "preliminary"; missing absolute risks; missing N; missing conflicts. FEED-03 trains the ear. When only the press release is open, label it secondary and write what you still need. Do not pretend the press release is peer review.

Influencer graphs — ordinary tells

You do not need a statistics degree for FEED-04. Ask: Are axes labelled? Is zero on the axis truncated without note? Is the sample named? Is the time window named? Is the chart aesthetic doing the persuading? Write "not trustworthy enough to cite yet" when tells pile up. Aesthetic is not truth.

Podcast and model — two secondaries that feel like primaries

Hosts summarize for ear comfort. Models summarize for fluency. Both can be useful after you try — both fail when filed as the study or as your observation. FEED-05 and FEED-06 are parallel drills: label secondary; check two concrete

claims; exit with match/mismatch/cannot check. If a model invents a citation, that mismatch is the finding about the tool.

Workplace soft — integrity without theater

FEED-02 fails if it becomes a gotcha sport. Success looks like: citation requested; identifiers stripped in your personal notes; one limit written for *your* file; no public dunking on a colleague. Science literacy is not a weapon. If proprietary data is the only “evidence,” do not paste it into a public consumer tool — open your employer’s acceptable-use page instead.

Sister boundaries (one pointer each, then teach)

History Refresh for Adults: primary-before-feed and lateral reading grain — then science evidence hygiene. *Math Refresher for Adults*: absolute-versus-relative numeracy neighbour and rust-is-ordinary — then science; Bastani stays OFF-AGE here. *Critical Thinking for Adults*: myside and testimony — then science looking. *Science for Emerging Adults*: parent Hour systems and denser CER — then this adult phone-archive hygiene. Do not reprint those banks.

Materials for a feed week

Phone with one saved card, notebook, honesty card, timer, optional library login for abstracts, earbuds for FEED-05, chatbot window only after blanks for FEED-06 compare. You do not need media-studies software. You need the hold.

FEED-01...07 and AS are this chapter’s primary banks. WRITE-04 and CER-02 appear as crumbs. Menus, not calendars.⁴⁸ Aim for at least one on-the-go and one desk or weekend sit. Weather pages and grocery aisles remain good surfaces; this week the structure is feed vs study.

Absolute vs relative — sticky you will reuse

Time: 5–15 minutes. Materials: one relative headline (“doubles,” “100%,” “cuts by half,” “25% reduction”). Safety: none; no medical decisions from the sticky. The move: write “Double what? Out of how many?” If numbers exist, translate

⁴⁸Named try-its FEED-01...07, AS-01...03, WRITE-04, CER-02, OUT-04 from memo 02 banks — menus, not calendars.

to absolute change with denominator. If not, mark unknown. Keep Gigerenzer's picture available as a named example: relative 100% can be absolute **1→2 per 7000**; mammography "25%" talk is often misread versus **~1 in 1000**.⁴⁹ Exit: sticky with relative + absolute (or unknown). Refuse relative alone as settled. Band: micro.

This sticky is not a diagnosis of any condition. It is risk-numerate looking.

On the go — commute / lunch

FEED-01 as named above. Real card from your feed. Honesty-label screenshot.

FEED-07 Click Restraint Before Science Outrage. Time 10–15 min. Materials: timer; one tempting science-outrage card. Safety: emotional self-care — skip if flooded. The move: wait 60–90 s; write what you already know and what you need; then click only if still worth it. Exit: "I waited; I needed ____." Refuse instant outrage reward as the session. Band: micro.

FEED-04 Influencer Graph Honesty. Time 15–25 min. Materials: one viral science graph; axes visible or noted missing. Safety: none. The move: Axes? Baseline? Sample? Truncation? Write what would make the graph trustworthy enough to cite. Exit: trust checklist in ordinary words. Refuse aesthetic = true. Band: micro or desk.

AS-01 Honesty Card Drill. Tick object type before you argue. Feed screenshot ≠ my measurement.

Absolute-risk sticky as above.

Desk or shift

FEED-02 as named above. Ordinary stakes. Strip identifiers. No gotcha.

FEED-03 as named above. When press and abstract both findable, verb-shift sticky is gold. When abstract is paywalled, write "abstract not open — cannot evaluate verb yet" and stop or use library access another day (OUT-04).

⁴⁹Gigerenzer (2009) WHO editorial and related PSPI pathway: relative "100%" pill scare as absolute **1→2 per 7000** (increase **1 in 7000**); mammography "25% risk reduction" often misread vs **~1 in 1000**. Access date for URLs in these notes: 22 September 2026. Useful communication research — not a promise every adult will misread the same way, and not medical advice.

FEED-05 Podcast Host Claim vs Named Source. Time 20–30 min. Materials: 2–3 min podcast science claim; named paper/scientist/agency page. Safety: none. The move: label host summary as storytelling secondary; check one concrete claim. Exit: “Host said ; **source shows** / cannot check ____.” Refuse filing podcast as the study. Band: desk (earbuds OK on micro if source pre-saved).

FEED-06 Model Summary Is Not Observation. Time 15–20 min. Materials: chatbot “summary of the findings”; one real abstract or your table. Safety: none. The move: label model output generated secondary; check two concrete claims; never file as observation log. Exit: model vs source mismatch or confirm sticky. Refuse generated-primary; model-as-notebook. Band: micro or desk. Walters’ work on fabricated citations is a labelled neighbour for why fluent cites still need opening — about 55% fabricated citations in one GPT-3.5 condition and about 18% in a GPT-4 condition in that study’s settings.⁵⁰ Open the source.

AS-03 Sample Size / Population Soft. Time 15–25 min. Materials: abstract or graph. Safety: none. The move: who/what measured? N? Time window? Generalization limit. Exit: population limit sticky. Refuse one anecdote as population. Band: desk.

WRITE-04 Caption Rewrite (crumb). Time 15–20 min. Materials: one science feed caption. The move: rewrite the caption to match what the abstract can actually support, including a limit clause. Exit: rewritten caption + “cannot settle ____.” Refuse punchier-than-true.

CER-02 Evidence Must Be Locatable (crumb). Time 15–20 min. Materials: one claim you almost believed. The move: point to the cell, sentence, or figure panel that would support it — or write “not locatable.” Exit: locatable pointer or refusal. Band: desk.

Weekend deep sit

Two-source feed vs study. Time 45–75 min. Materials: one viral or workplace claim + abstract or agency page; optional press release as third surface. Safety: strip workplace identifiers; skip if emotionally flooded. The move: four blanks on the card; open abstract; list compressions; absolute sticky; sample limit; write what the card can settle and cannot. Exit: two- or three-column sticky (card / press

⁵⁰Walters (fabricated citations study): about 55% GPT-3.5 / 18% GPT-4 fabricated citations in that paper’s conditions — open the source. Labelled neighbour for FEED-06.

/ abstract). Optional OUT-04 abstract hunt via library tools. Optional denser claim crumb with named limit — not a full Chapter 7 essay.

Incorrect example to diagnose (composite illustration). A chat window produces: “According to Smith et al. (2024) in *Nature*, the intervention cut risk by 40%.” No link. No N. No absolute baseline. No methods. The adult files the chat text as “what the study found.” Diagnose: model summary as study; possible fluent citation without opening; relative without absolute; sample unknown. Repair: FEED-06 label generated secondary; open or fail to find Smith et al.; AS-02 unknown if unfindable; absolute sticky; refusal to share until openable. If the citation is fabricated, that is data about the tool — not about nature.

Talk-to-yourself box (practice depth)

Reuse the locked opening. When stuck: “Point at what you’d open next — abstract, methods, second source, or unknown.” If outrage rises, FEED-07 first. If host polish hypnotizes, FEED-05 check one concrete claim only.

Blocked for a new move: do not jump to AP-style multi-paragraph science op-eds as weekly identity. Mixed: when you have your own table from Chapter 2, you may claim from *that* table with a limit — still refuse to let a feed caption overwrite your units.

What “open next” can mean (menu)

- Abstract or methods crumb of the named paper
- Agency explainer that cites the paper
- Press release *beside* abstract (not instead)
- Second independent write-up (lateral hop once — not an open-web agent as only partner during the attempt)
- Your own labelled table if the claim is about something you measured
- “Unknown — cannot evaluate yet” as a valid next

Lateral hop once means you leave the card to check a source — parallel to Civic Online Reasoning grain in spirit, pointed once from History Refresh — then return. It does not mean handing the attempt to an unsupervised browsing agent.

Worked absolute sticky (composite illustration)

Composite card text (illustration, not a reported headline): “New habit doubles chance of outcome X.” Your sticky:

- Relative claim: doubles (100% increase).
- Baseline on card: not stated.
- Absolute: unknown — need rate before and after, out of how many.
- Gigerenzer neighbour in mind: doubling can be 1→2 per 7000.
- Next open: abstract methods / results for base rates; else mark cannot evaluate magnitude.
- Limit: this card cannot settle personal risk without denominators.
- Action: do not share yet.

That sticky is a complete micro-session product. You did not need to become an epidemiologist. You needed the denominator habit.

Worked FEED-03 verb-shift (composite illustration)

Press line: “Study proves treatment prevents Y.” Abstract line: “In a sample of N adults over T weeks, treatment was associated with lower Y (limitations: ...).” Your sticky: proves → associated; prevent causal verb unsupported yet; N and T now visible; limitations named. Exit: “Press overclaims relative to abstract.” That is literacy — not cynicism for its own sake.

Worked FEED-06 mismatch (composite illustration)

Model: “Lee et al. 2023 (Science) found a 50% improvement in Z.” You open search: no such Lee et al. in Science 2023 on Z — or a real Lee paper with different numbers. Sticky: generated secondary; citation unchecked/failed; do not file as study. Walters neighbour reminds fluent citations can be fabricated in known rates in that research’s conditions — open anyway.⁵¹

Connecting observe and feed without muddle

If you are mid OBS-01 on a plant and a feed card pings, name the switch: either finish Observe, or deliberately start FEED-01. Do not blend “I noticed the

⁵¹Walters (fabricated citations study): about 55% GPT-3.5 / 18% GPT-4 fabricated citations in that paper’s conditions — open the source. Labelled neighbour for FEED-06.

plant needs water because a study says.” That sentence mixes explanation-before-observe with caption-as-study. Separate moves. Separate stickies.

Chapter 2’s honesty card travels with you: feed screenshot tick before FEED-01; generated tick before FEED-06; labelled secondary for press and podcast.

Safety and ordinary stakes (feed week)

No RAMP needed for most FEED tries. Emotional self-care counts: skip FEED-07 if flooded; choose ordinary-stakes cards; avoid graphic harm as practice objects. Workplace confidentiality is a safety rule. No undercover audits. No culture-war stinging as a try-it. Public abstracts, labelled composites, pamphlets you already hold — those are the objects.

Dose honesty again

There is no adult RCT that proves five FEED micros mint lifelong hygiene. Named sessions are the ambition. Three to five micros plus one desk sit in a week, with an occasional deep two-source look, is honest. A massed Sunday of forty outrage cards is not hygiene; it is exposure without looking.

A week that is honest (labelled hypothetical composite)

Illustration only. Monday: FEED-01. Tuesday: FEED-07 + absolute sticky. Wednesday desk: FEED-02 or FEED-03. Thursday: FEED-06 model refuse. Friday: FEED-04 or FEED-05. Saturday deep: two-source card vs abstract. Sunday: rest or AS-01. If days delete, keep one caption cover. Named sessions beat guilt calendars.⁵²

Exit sticky bank

- “Later telling / study surface / unknown — next open ____.”
- “Absolute: ____ per ____ (or unknown).”
- “Sample/N: ____ — generalizes to ____ / not to ____.”
- “Verb on card: ; **verb abstract can support:** .”
- “I waited ____ s before clicking.”

⁵²Dunlosky et al. (2013): high-utility technique ratings for practice testing and spacing — not an adult-science programme evaluation. Dose honesty: named sessions are the ambition.

How to try — fading and parallel tries in more detail

After FEED-01 succeeds once, parallel try means a second card the same day or next, same four blanks, less template. After FEED-02, parallel means a different meeting claim with identifiers stripped — not a deeper fight with the same slide. After FEED-06, parallel means a fresh model précis on a different claim, still checked against an openable source.

Fade help from tools the same way: first, tool gives you questions to ask yourself after your blanks; later, you ask the questions without the SCRIPT; never let the tool fill blanks first. Struggle before rescue applies to feeds too. Grabbing the summary at one second is still rescue.

When to stop looking up includes stopping lateral hops. One hop that opens an abstract is hygiene. Ten hops that never return to a sticky is doomscrolling in a lab coat. Return. Write the exit line. Close the session while a look remains — the same stop rule as the weekend deep sit in the Hour.

If it isn't clicking — wait-time and anxiety extras

If diagnostic 1–3 do not fit, check wait-time and anxiety. Wait-time zero: you asked the locked opening and peeked. Cover again. Anxiety: absolute sticky with a calm card, not an outrage card; shorten to ten minutes; early win; no countdown timer as default rigor. This remains session design, not a therapy manual.

Correct a wrong reason: “I treated relative as enough — adding denominator blank now.” Keep the attempt. Silence in the face of your own wrong reason is not kindness.

Tools — what “after you try” looks like on a feed day

Concrete allow after try: “Explain this abstract’s limitation sentence to me in ordinary words; title is ____; my goal is absolute vs relative.” Concrete ban: “Summarize whether this Twitter card is true” as the first move with no blanks of yours. The difference is who does the looking.

Open *your* AUP at work when workplace examples appear. Do not invent a federal workplace statute. Hermes 4 is not Hermes Agent; Grok the model is not Grok Bot; agents are not chatbots — names stay separate as in the Hour box.

If it isn't clicking

Diagnostic 1 — Outrage scroll. You collect cards instead of opening one. Outrage feels like engagement. It is working-memory theft. Next move: FEED-07 once; one methods crumb or absolute sticky; stop. Put the phone face down after the sticky. Return tomorrow with one card only.

Diagnostic 2 — Trust in host polish. A warm voice or neat design substitutes for methods. Next move: FEED-05 or FEED-04 — check one concrete claim or one axis/baseline. Label storytelling secondary. Polish is not peer review.

Diagnostic 3 — “I don’t have time for papers.” Time scarcity is real. Total shut-down is optional. Next move: scale hygiene down — AS-02 “unknown — cannot evaluate yet” is a complete micro win; absolute sticky is five minutes; FEED-01 cover without a full paper still trains the hold. Library abstract hunt (OUT-04) waits for a weekend when credit or curiosity makes time.

When to slow down: if you cannot say whether the card is study or later telling; if relative headlines keep graduating without denominators. When to go ahead: if you can name what you’d open next and either open it or mark unknown without sharing first. When to ask a human: librarian for access; colleague for workplace citation path without gotcha; tutor if dense methods prose is the skill bottleneck.

Correct yourself without crushing the attempt: “That was caption-as-study — cover and retry.” Then parallel card.

If none fit, check whether a chatbot précis finished the attempt. Cover the model. Wait. Fill blanks unaided. Then compare.

Tools, including AI

Short. Point back to the Science Refresh Hour AI box. This chapter will not become an AI policy paper — even though feed/model hygiene is the topic.

After you try: explain an abstract *to yourself* from a named title and today’s micro-goal; revision questions after *your* four blanks or rewritten caption; labelled SCRIPT of hygiene questions; absolute vs relative prompting *after* you drafted a risk claim from named numbers; isomorphic practice with answers on *your* page.

You may not: treat a model summary as the study or observation; invent research

or trust fluent citations without opening; paste employer sheets into public tools; ask for write-my-science-op-ed as the only draft; park an unsupervised chatbot or agent as the only partner during the attempt; print a literacy certificate from a chat window; detector-grade your own unaided write-up (Liang’s average false positive rate about **61.22%** on human TOEFL essays in that study — do not treat a detector as a verdict on yourself).⁵³

Lateral hop once with *you* holding the browser is allowed after the try starts with blanks. An open-web agent finishing the attempt is not.

Bastani remains **OFF-AGE for science literacy**: unguarded practice help can look better and leave unaided performance worse in that high-school math field experiment — not an adult science RCT; lunch-break rule still holds.⁵⁴

What “done enough” looks like

- ☐ I treated at least one card as a later telling and named what I would open next.
- ☐ I opened a methods/abstract crumb or marked unknown before believing a caption.
- ☐ I named absolute risk or marked unknown (denominator habit).
- ☐ I named sample/population limit or marked unknown on one claim.
- ☐ I refused model summary as observation/study on at least one try (FEED-06).
- ☐ Workplace tries, if any, stripped identifiers and avoided gotcha culture.
- ☐ My talk-to-yourself opening matched this chapter’s locked stem.
- ☐ I can say what the card cannot settle in one plain sentence.

Support-for-science is still not this hour’s success metric. You can hold a card carefully without turning the hour into a campaign. NAS Conclusion 11 warned that raising literacy is not the foremost path to raising general support for science; specific issues interact with worldviews.⁵⁵ Teach looking because feeds and decisions

⁵³Liang et al.: about **61.22%** average false positive rate on human TOEFL essays in that study — do not detector-grade yourself.

⁵⁴Bastani et al. (2025), *PNAS*: Turkish HS math grades 9–11; GPT Base about **+48%** practice / **–17%** unaided; GPT Tutor about **+127%** / $\approx$ control. **OFF-AGE for science literacy**. Not an adult science RCT. Keep Math Refresher’s adjacent label separate from this book’s OFF-AGE tag.

⁵⁵National Academies, *Science Literacy* (2016), Conclusion 11 — raising literacy is not the foremost means to raise general support for science.

require it.

PIAAC and NSF/Pew numbers from Chapter 1 stay adjacent context — literacy mean 258, numeracy 249, process items about 60/50/66 — not a pretest you retake here to graduate. Teen NAEP Science and PISA stay refused as adult measures. The card in your hand is the measure of today's practice.

Placement by skill, not diploma year. Revisit next Tuesday: cover one caption; ask what you'd open next. Space across days. Keeping track will store feed honesty cards and refusal logs later. Never Science Refresh for Adults I. Optional CLEP stays a later bridge when credit is the goal — not this chapter's finish line.

If you already open abstracts before captions, deepen absolute denominators and sample limits this week, then carry FEED hygiene into systems and fair-test chapters without dropping the hold. Fluency is maintenance. A polished host will still tempt.

If access to papers is hard, agency explainers and library help are neighbours — still open *something* beyond the card, or mark unknown without pretending. Unknown is integrity. Caption-as-study is not.

When blurry, sort: did I open evidence or only rearrange the caption? Rearranging captions is WRITE-04 practice only after a hop. Opening evidence is the hour.

You understand this. You know what to ask yourself tomorrow on the commute. You know what a caption offered as the study sounds like when it is not yet an evidence look. You know what a finished-looking chatbot précis without your attempt looks like. You can treat the card as a later telling until evidence is named — on the go and in a deeper sit. Open one next thing tomorrow, or mark unknown with a clear head. Evidence before feed summary is a habit you practice, not a mood you wait for.

Chapter 4

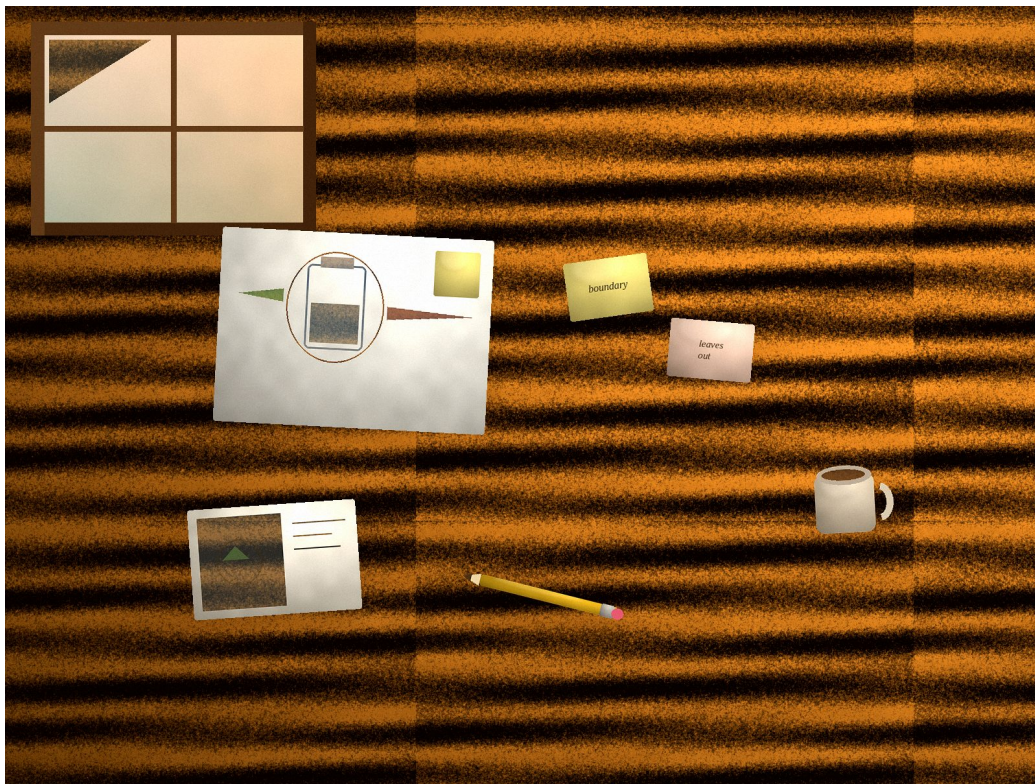


Figure 6: Systems-boundary sketch with boundary sticky and museum postcard — still-life, no people, no logos.

Systems looking (boundary, I/O, what model leaves out)

Why this matters

“Everything is a system” sounds wise until you try to draw it. Without a boundary, the word *system* is a slogan. It can mean the jar on the sill, the room, the city, the planet, and your feelings about the planet — all at once. That mash feels deep. It is not looking.

Systems looking, in the sense this book teaches, is smaller and kinder to the page. You draw a boundary on purpose. You name a few parts inside. You name what you left outside on purpose. You sketch one or two inputs and outputs — matter, energy, or information soft — with ordinary arrows. You ask what your model leaves out that would matter for a decision. Sometimes you notice a loop where an output later changes an input. Sometimes you shift scale and watch what appears or disappears. You refuse to treat the sketch as the world.

Chapter 2 taught you to notice before you explain. Chapter 3 taught you to treat a feed card as a later telling until design, sample, absolute risk, and a named limit are on the table. This chapter assumes you can look and that you can hold a later telling. It adds a drawable boundary. The habit is light and repeatable: same object, a closed line, insides named, outsides honest, one omission sticky. It is not a weekly encyclopedia of ecosystems. It is not every crosscutting concept from a standards map dumped onto Tuesday. It is not a museum scavenger hunt with no systems claim.

The NRC *Framework for K–12 Science Education* names systems and system models among the crosscutting concepts that help people organize looking across domains.⁵⁶ That Framework is a **map**, not an adult mandate and not a worksheet calendar. *Science for Emerging Adults* ages systems looking for a parent Science Hour with teens. One pointer, then teach adult self-study here — no parent talk-boxes, no transcript edge. You are the learner. A plant pot, an HVAC loop, a

⁵⁶National Research Council, *A Framework for K–12 Science Education* (2012): practices and crosscutting concepts, including systems and system models. Use as a **map** for adult systems looking — not an NGSS performance-expectation dump as weekly spine, not an adult programme evaluation, and not a mandate for working adults. Access date for URLs in these notes: 22 September 2026.

watershed map soft, a body-at-rest soft sketch (not a diagnosis), and a museum exhibit treated as someone's model are enough surfaces for the week.

What this unlocks is usable thickness on ordinary adult objects. A workplace "study says" slide about a process often hides a system claim without a boundary. A weather page compresses atmosphere into icons; the icons are a model with omissions. A grocery produce look becomes structure–function when you ask what stays inside the fruit's skin and what crosses it. A museum caption can be an exhibit-as-model critique: what did the designers leave out for visitors? Money, workplaces, and civic life motivate. They do not replace the looking structure. No live controversy as the object. Low-heat systems first so the move can land.

If you leave a boundary sketch, one input, one output, and one honest omission, the week did its job. Place by that strip, not by diploma year and not by how many system words you can recite.

You already practice a cousin of this move when you decide what a workplace slide is *about*. A process flowchart with no edge is a systems claim without a boundary. A wellness poster that says "your body is a system" without naming a subsystem is a slogan. A park map that never shows what lies outside the park still chose a boundary — the printed edge is someone's deliberate cut. Naming that cut is literacy, not fussiness.

Systems looking also protects you from a quiet overreach common in adult feeds: treating a local observation as a whole-Earth conclusion, or treating a whole-Earth slogan as if it measured your windowsill. Scale honesty is part of the boundary. When you say "for this look, the system is the pot and the near air," you are not denying the watershed. You are choosing a question-sized frame so your eyes and your notebook can finish the hour.

Rust shows up here as two twin habits. Adults who love big pictures may expand the boundary until nothing is outside. Adults who feel rusty may refuse to draw at all, waiting for a "real" diagram from a textbook. Both patterns are ordinary. Neither is a character flaw. Both have a repair: shrink to a jar, a room, a pot, or a single exhibit; name one outside on purpose; put the pen down before the feed or the chatbot finishes the sketch for you.

Understand it yourself

Everyday picture. A plant in a pot on a windowsill — composite illustration, ordinary home. Inside the boundary you choose: soil, roots, stem, leaves, pot, a patch of air near the leaves. Outside on purpose: the whole neighborhood, yesterday’s weather front, the factory that made the pot, your mood about gardening. Light and water cross the line as inputs soft. Water vapor and heat soft leave as outputs you can name without a lab. The everyday trap is to say “it’s all connected” and stop. The everyday repair is the closed line and the outside list.

Scale that picture to adult life: the HVAC loop in a small office (boundary = the suite, not the city grid); a fridge as a cold box with door openings as disruptions; a bus route as people and time moving through stops; a body-at-rest soft sketch after stairs (boundary = a subsystem you name, not a clinic chart — and not medical advice).

Precise picture. A system model is a deliberately incomplete picture of interacting parts. Usefulness depends on the question you wrote down. Boundary first answers: *what am I treating as inside for this look?* Inputs and outputs answer: *what crosses the line?* Feedback soft answers: *does an output later change an input — and does that tend to stabilize or amplify, in ordinary words?* What the model leaves out answers: *what omission would change a decision?* Scale shift answers: *what appears or disappears when I zoom?* The Framework’s systems language helps you organize; it does not finish the sketch for you.⁵⁷

You do not need lab jargon on the train. You need a closed line, a few named insides, two outsides left out on purpose, and a ban on “everything” as the only answer. Model worship is the twin error on the other side: treating your sketch as if it were the world. A model that cannot name an omission is not humility. It is theater.

Wrong answers you should be able to hear (including from yourself).

1. **No boundary — “it’s all interconnected.”**

Looking dissolves into atmosphere.

⁵⁷National Research Council, *A Framework for K–12 Science Education* (2012): practices and crosscutting concepts, including systems and system models. Use as a **map** for adult systems looking — not an NGSS performance-expectation dump as weekly spine, not an adult programme evaluation, and not a mandate for working adults. Access date for URLs in these notes: 22 September 2026.

What it usually means: slogan without a diagram.

What you try: Draw a closed line around one jar, pot, room, or exhibit. Name three insides. Name two outsides left out on purpose.

2. **Everything inside.**

The outside list is empty because the boundary grew until it swallowed the neighborhood.

What it usually means: fear of leaving something out; coverage itch.

What you try: Shrink. One room. One pot. Write “outside on purpose: ____ and ____.”

3. **The model is the world.**

Sketch treated as complete reality.

What it usually means: model worship; omission honesty skipped.

What you try: Finish with “My model leaves out , **so I cannot conclude .**”

4. **Scale mash.**

Cell, organism, and city heat soft appear in one claim without notice.

What it usually means: zoom forgotten; sentence sounds deep and settles nothing.

What you try: One scale-shift sticky: “At leaf scale I see ; **at plant scale that becomes / disappears as .**”

5. **Decorative arrows.**

Inputs and outputs drawn to look scientific, with no meaning you can say aloud.

What it usually means: representation without a looking move.

What you try: Label each arrow with matter, energy, or information soft — or erase it. Mark one arrow you are unsure about.

Five-minute warm-up. Pick one everyday system in view (jar plant, fridge door, HVAC vent, desk lamp and cord soft). Draw a boundary. Name three parts inside. Name two things outside left out on purpose. Cover any textbook diagram or feed. Say the locked opening aloud. Warm-up done.

Locked opening: **“Where is the boundary of this system — and what did I leave outside on purpose?”**

Notice how the everyday and precise pictures share one demand: the question comes before the biggest possible map. If your question is “why is this leaf drooping after three dry days,” the pot-and-near-air frame is usually enough to start.

If your question is “where does stormwater go after it leaves my block,” the pot frame is the wrong tool — and saying so is systems literacy, not failure. Adults sometimes feel foolish choosing a small frame. That feeling is coverage anxiety wearing a lab coat. Small frames finish. Giant frames stall.

A second precision point: inputs and outputs are claims about crossing, not decorations. “Energy in” written beside a wavy arrow does no work unless you can say, in a sentence, what you mean — light on leaves, heat through a wall, a notification that changes what you do next. Information soft counts when the system you chose includes a control loop (thermostat, app alert). It does not count when you add “information” to sound technical. Erase any arrow you cannot defend in ordinary words. Uncertainty is allowed; theater is not.

When prior ideas show up — for example, treating a body sketch as a clinic verdict, or treating a weather icon set as a complete atmosphere — label the prior idea and return to the boundary. Continuity from school-age misconception research notes that people often meld theory and evidence; adults are not magically cured of that meld.⁵⁸ Your repair is the same soft move as elsewhere in this book: another look, a named limit, no scolding.

If the only “system” you have is a feed card that says “the climate system” or “the immune system” with no diagram, honesty-label the card as a later telling, then draw *your* small boundary on an object you can see or a labelled public exhibit photo. Chapter 3 still applies: the caption is not the model. Your sketch is the attempt.

How to try it this week

Science Refresh Hour shapes apply. This week’s strip: boundary first; one input and one output; one omission; refuse slogan-without-diagram.

Session shape. Retrieve an observe-before-explain sticky or a feed-vs-study sticky from earlier chapters. Put one ordinary system on the table. Draw before you explain. Locked opening. Boundary + outsides. Optional I/O arrows. Exit sticky that names one omission. Mixed later: slip a feed “system” slogan into a clean boundary sit and ask whether the card drew a line.

⁵⁸Continuity: *Science for Emerging Adults* owns ages 15–18 parent Science Hour systems grain. This chapter ages the move into adult self-study — no parent/student split. One pointer, then teach.

Commute / lunch (15–20 min). Materials: phone note or pocket card; one system in view (bus route soft, coffee cup as temporary heat system soft, office suite HVAC feel). Safety: private; no public quiz that humiliates a colleague; no undercover workplace audit. Move: boundary + two outsides only — no essay. Optional: one input named in ordinary words.

Desk or shift (25–40 min). Materials: paper; plant pot, fridge, or room HVAC as object; pen. Safety: ordinary; no unauthorized opening of equipment panels; no medical diagnosis framing if you sketch body soft. Move: SYS-01 boundary first; then SYS-02 inputs and outputs; optional SYS-04 omission sticky the same sit or next day.

Weekend deep sit (45–75 min when needed). Materials: museum exhibit (or labelled exhibit photo); notebook; optional park watershed soft or street-to-city heat soft for scale. Move: SYS-06 exhibit-as-model critique; or SYS-03 feedback soft on thermostat / thirst / notification loop (non-pathologizing); or SYS-05 scale shift on leaf ↔ plant or street ↔ city soft. Dual-perspective sentence when prompted: “Another fair boundary for a different question would be ____.” Claim crumb with a named limit. Stop while a look remains.

Exact wording you can say to yourself.

- “Where is the boundary of this system — and what did I leave outside on purpose?”
- “What do you mean, *outside on purpose* — which two things did I refuse to swallow?”
- “What does the diagram show before the caption — insides, arrows, omissions?”
- “What else could explain this flow if my boundary is wrong?”
- “What can this model not tell me?”
- “You may change the boundary when the question changes.”

Fade help. Boundary template with labelled insides → blank closed line → I/O arrows you label → omission sentence without a worked example beside you. Stop looking up when you can point to a boundary and one omission without a model sketch filed as “what I saw.” If you peek, peek at your own prior sticky — not at a generated schematic you will pretend was observation.

Fade ladder, said once. Day one: closed line with labelled insides from a tiny template. Day two: blank closed line on a new object. Day three: arrows you must label or erase. Day four: omission sentence without a worked example in

view. Day five optional: scale-shift sticky or exhibit critique. If you skip a day, do not “make up” five diagrams on Sunday. One honest boundary beats a massed stack.⁵⁹

Stop looking up when... you can point at a boundary and an outside list without a downloaded schematic under your sticky notes. Peeking at your own earlier sticky is fine. Peeking at a generated “perfect HVAC diagram” you will later call observation is not.

Mixed practice seed. After two clean boundaries, drop one feed card that uses the word *system* without a diagram into the sit. Ask: did this card draw a line? What would I need to open next (Chapter 3) before I trust any causal promise? Keep the interloper short so systems looking stays the spine.

Composite desk, longer. You sketch the office suite: boundary = walls of the suite; insides = people soft as occupants not as diagnoses, vents, door; outsides on purpose = the building’s full mechanical room detail, the utility’s generation mix, the city’s heat island. Inputs: conditioned air, outdoor air on door openings, heat from equipment soft. Outputs: warmer return air soft, noise soft. Omission: “Leaves out measured CO₂ and actual filter status, so I cannot conclude air quality from the sketch.” That sentence is the win — not a freelance facilities report, and not a culture-war pamphlet about offices.

First try-it (on the go). Boundary + two outsides on something in your bag line of sight. Deeper sit: SYS-01 on a plant pot or HVAC suite, then SYS-02 arrows. If the deeper sit stalls, fall back to a jar-sized boundary and one outside — still a win.

Sample micro (composite). Lunch bench. Coffee cup with lid. Boundary: cup + lid + liquid. Insides: liquid, vapor soft above the surface, cup wall. Outside on purpose: the café’s whole HVAC, the bean farm, yesterday’s weather. Input soft: heat from the liquid; output soft: heat to your hands and air. Exit sticky: *Outside on purpose — farm and café system.* Close. No feed about “complex systems.”

Sample desk (composite). Plant pot at the window. Boundary around pot–plant–near air. Inputs: light, water you add. Outputs: vapor soft, heat soft. Omission sticky: “Leaves out soil chemistry detail and pest arrivals, so I cannot conclude why yellowing started from this sketch alone.” Next sit optional: one Observe line

⁵⁹Dunlosky and colleagues (2013): distributed practice rated high utility as a learning-technique rating — not an adult science-refresher programme evaluation. Space systems sits across days.

across two days (Chapter 2 grain) without turning the week into a Bio survey.

Worked pictures you can steal (then put away)

Picture 1 — Boundary first (SYS-01). Everyday system. Closed line. Three insides. Two outsides on purpose. Exit: diagram + outside list. Wrong twin: “everything is interconnected” with no diagram.

Picture 2 — Inputs and outputs (SYS-02). Add arrows. Matter / energy / information soft labels. One uncertainty sticky on an arrow you cannot defend. Refuse decorative arrows.

Picture 3 — Feedback soft (SYS-03). Thermostat, thirst, or notification loop — pick one non-pathologizing. Output affects later input. Stabilize or amplify guess in ordinary words, with a limit. Refuse therapy-as-Hour.

Picture 4 — What the model leaves out (SYS-04). Two omissions that would matter for a decision (cost, time, species, scale). Exit: “My model leaves out , *so I cannot conclude .*”

Picture 5 — Scale shift (SYS-05). Same phenomenon at two scales. What appears or disappears? Exit: one scale-shift sentence. Refuse mixing scales in one claim without notice.

Picture 6 — Museum exhibit systems (SYS-06 / OUT-02). Treat the exhibit as a system model someone built. Boundary? Flows? What is left out for visitors? Exit: exhibit-as-model critique sticky. Refuse whole-museum bingo.

Picture 7 — Wrong twin (labelled). Sticky: “The body is a system, so my fatigue proves the workplace is toxic.” Repair: “I sketched breath-after-stairs soft as a subsystem look — not a diagnosis and not a workplace audit; my model leaves out sleep, workload, and clinical causes, so I cannot conclude ____ from this sketch.”

Close the pictures. Build one boundary on a parallel object yourself. Keep the first pass under twenty minutes even at a desk. Speed here is a feature: you are drawing, not writing the seminar paper. If the parallel object is thinner than the worked pictures, good — thin objects are what life sends. One line. Two outsides. One omission. That is a complete sit.

When you finish, ask once: did I leave something outside on purpose? If the outside list is empty, shrink the boundary until two outsides appear. Empty outsides usually mean the line grew, not that the universe fit in your jar.

Practice that actually builds it

Blocked early: several boundary micros on jar / pot / room; then I/O arrows; then one omission sticky. Mixed later: feed “system” slogan interloper; scale shift after a clean boundary; museum exhibit after a home pot.

Named try-its below carry **time, materials, safety, the move**. Banks are menus, not calendars. You do not owe SYS-01 through SYS-06 in order this week.

Talk-to-yourself box

Exact opening question:

“Where is the boundary of this system — and what did I leave outside on purpose?”

Follow-ups (3–5):

1. Meaning: “What do you mean, *outside on purpose* — which two things did I refuse to swallow into the diagram?”
2. Representation: “What does the sketch show before any caption — closed line, insides, arrows, omissions?”
3. Alternative: “If I moved the boundary outward by one room / one block, what new parts appear — and what claim breaks?”
4. Check: “What can this model not tell me — cost, species, time scale, another input?”
5. Revise / diagnose: “Where could someone go wrong — slogan without diagram, everything inside, model-as-world, scale mash, decorative arrows? You may change your mind.”

How to wait alone. After you ask, cover the feed, the textbook schematic, and the chatbot. A slow three is a convention, not a sacrament — classroom think-time research is school-origin; your adult translation is pause-before-lookup / pause-before-feed.⁶⁰ Look at the blank page or the pot, not at a finished-looking model diagram. After you stop mid-reason, wait again before peeking. Rescue at one second is still rescue. Abandonment for forty minutes of scrolling is not a micro-session.

Stuck silence usually means: the question was vague (“think about systems”);

⁶⁰Rowe (1972) wait-time in science class; Stahl (1994) think-time conventions — classroom origin. Adult translation in this book: pause-before-lookup / pause-before-feed. Not an adult science RCT.

wait-time 1 was zero; you are searching for a vocabulary word instead of a line; you are guessing what a textbook diagram would want; the question was a recitation item in disguise (“define system”); you offered a keyword where a looking move was asked for; you think “science person” means leave; you think a finished-looking chatbot schematic is already a check; anxiety is eating working memory — shorten the session, do one jar boundary, come back tomorrow. Your job is to pose, wait, and point back at the object — not to fill the silence with a model summary filed as observation.

On the go — SYS-01 flash (15–20 min)

- **Time:** 15–20 min
- **Materials:** phone note or index card; one everyday system in view
- **Safety:** private; ordinary
- **The move:** Closed line. Three insides. Two outsides on purpose. Locked opening once.
- **Exit:** Outside list of two
- **Refuse:** Interconnected slogan with no diagram

Desk — SYS-01 then SYS-02 (25–40 min)

- **Time:** 25–40 min
- **Materials:** paper; plant pot, fridge, or HVAC-in-room soft
- **Safety:** do not open sealed equipment; no forced HVAC panel work
- **The move:** Boundary first. Then input/output arrows with one uncertainty sticky.
- **Exit:** Arrow set + “I am unsure about ____ because ____.”

- **Refuse:** Decorative arrows; filing a downloaded schematic as your observation

Desk — SYS-04 omission honesty (15–25 min)

- **Time:** 15–25 min
- **Materials:** yesterday's diagram
- **Safety:** none
- **The move:** List two omissions that would matter for a decision. Finish the limit sentence.
- **Exit:** “My model leaves out , *so I cannot conclude* .”
- **Refuse:** Model worship

Desk — SYS-03 feedback soft (20–30 min)

- **Time:** 20–30 min
- **Materials:** thermostat, thirst cue, or notification loop (non-pathologizing)
- **Safety:** avoid medical self-diagnosis framing
- **The move:** Loop sketch. Stabilize or amplify guess with limit.
- **Exit:** Loop + limit
- **Refuse:** Therapy session as science Hour

Desk or micro — SYS-05 scale shift (15–25 min)

- **Time:** 15–25 min
- **Materials:** leaf and whole plant; or street and city-heat soft public figure

- **Safety:** none
- **The move:** What appears or disappears when scale changes?
- **Exit:** One scale-shift sentence
- **Refuse:** Mixing scales in one claim without notice

Weekend — SYS-06 / OUT-02 museum exhibit systems (30–60 min)

- **Time:** 30–60 min on site (or 20 min from a labelled exhibit photo if travel is hard)
- **Materials:** one exhibit; notebook
- **Safety:** venue rules; no touching restricted objects
- **The move:** Exhibit as someone's system model. Boundary, flows, visitor omissions.
- **Exit:** Exhibit-as-model critique sticky
- **Refuse:** Whole-museum bingo; worksheet scavenger with no systems claim

Weekend optional — park or yard systems soft (OUT support)

- **Time:** 30–50 min
- **Materials:** notebook; short path you may legally walk
- **Safety:** public access; weather sense; tick awareness; no trespass
- **The move:** Choose one question (shade and understory soft; litter as an input soft). Draw a boundary for the patch you will treat as the system. Five Observe stops if useful (Chapter 2 grain). One systems claim with a limit — not an ecosystem vocabulary tour.

- **Exit:** Boundary + one claim + limit
- **Refuse:** Scavenger mania; treating the walk as a survey year of ecology

Transfer sentence bank (write one)

- “On the pot I left ____ outside; on the HVAC suite I left ____ outside — same move, new surface.”
- “The exhibit left out ____ for visitors; my home sketch left out ____ for time — both are models.”
- “At leaf scale I noticed ; *at plant scale that became / disappeared as .*”

What a stuck silence usually means (systems week)

The question was “think about systems” with no object; wait-before-lookup was zero; you hunted a vocabulary word instead of a line; you guessed what a textbook diagram wanted; you offered “interconnected” where a boundary was asked for; you thought “not a science person” means leave; you treated a finished-looking chatbot schematic as already a check; anxiety ate working memory — shorten, do one jar win, return tomorrow. Pose, wait, point at the object. Do not fill the silence with a model summary filed as seeing.

Incorrect example to diagnose (labelled composite)

A feed card reads: “New study: office air is a complex system — open windows to fix focus.” An adult draws nothing, nods at *complex system*, opens a window, and files the caption as settled.

Diagnosis: slogan-without-diagram; feed-first; model-as-advice without a boundary or omission.

Repair: Draw the suite boundary. Name inputs (outdoor air, HVAC supply soft) and outputs. List omissions (noise, security policy, pollen, measured CO₂ if any). Ask what the card’s “study” designed and sampled — Chapter 3. Insufficient evidence is allowed. Opening a window may be comfort; it is not automatic proof from a caption.

Mixed practice when the move is stable

Pair a clean pot boundary with a FEED interloper that uses the word *system* without a line. Pair scale shift with an omission sticky. Pair museum critique with a home HVAC sketch and ask which omissions each model chose. Keep CER crumbs light — Chapter 7 owns denser claim. Here, a one-line claim with a limit is enough: “With this boundary, light is an input I can name; I cannot conclude nutrient deficiency from the sketch alone.”

Life surfaces that carry the move (they do not replace it)

Weather pages, grocery aisles, museum captions, and workplace process slides *are* the practice when you bring a boundary. They do not replace the looking structure. A weather app icon set is a model with omissions. Produce skin is a boundary soft. A process slide that never draws the process is a slogan waiting for SYS-01.

If it isn’t clicking

Diagnostic 1 — Slogan without diagram.

You can say “system” and “interconnected” and still have a blank page.

Next move: Shrink to a jar, a pot, or a single room. Three insides. Two outsides. No vocabulary hunt. If anxiety spikes, one five-minute warm-up counts; come back tomorrow.

Diagnostic 2 — Too big a system.

Your boundary includes the neighborhood, the supply chain, and the biosphere, so nothing is outside and every arrow is decorative.

Next move: Force two outsides by shrinking. Write the question this look is for. A different question may earn a bigger boundary later — after this small one exists.

Diagnostic 3 — Model equals world / schematic as observation.

You downloaded a beautiful diagram and filed it as what you saw; or you refuse to omit anything.

Next move: Put the schematic away before the try (tools rules below). Draw from the object. End with an omission sentence. If you need a human, ask a colleague to point at your page and ask “what’s outside?” — not to finish the sketch for you. A librarian can help you find a museum day; a tutor or class can help if credit Chemistry or a survey restart is the real goal (Chapter 6 and Resources).

Correct a wrong reason without crushing the attempt — applied to yourself. “Never correct yourself” is not this book’s rule. Hearing “I put everything inside because I was afraid to leave science out” and answering “fear of omission is ordinary; two outsides are the honesty move” is the work. Short sessions. Early wins. No countdown timer as default rigor. This is not a therapy manual.

When to go ahead: you can point to a boundary and one omission on two different objects without a feed finishing the line. When to slow down: every sit still ends in interconnected slogans. When to ask a human: you need a supervised lab path, a credit decision, or access help — not because you failed a systems gene test.

Tools, including AI

Point back to the Science Refresh Hour AI box. Keep this short.

You may, after you have tried: explain a systems idea to yourself from a named OpenStax / Khan / museum / agency page (title, page, today’s micro-goal); ask for isomorphic practice with answers on **your** page — blank boundary cards + your key; request a labelled SCRIPT for questions to ask yourself; ask revision questions after *your* diagram or omission sentence; take a hint after an attempt; diagnose a sketch you already made (crop to page — no face). A labelled schematic may help you check *after* the try — put it away before the next unaided attempt. Never file a model schematic as “what I saw.”

You may not during the attempt: paste a blank systems worksheet and ask a tool to complete it; ask “draw my systems diagram” or “write my lab note” as the only draft; invent flows you did not look for; generate an exhibit photo or gel and treat it as observation; park an unsupervised chatbot as your only partner while you should be drawing; treat a model summary as the observation or as the study; paste proprietary employer process data into a public consumer tool; print a certificate of systems literacy.

One mechanism sentence, labelled **OFF-AGE for science literacy**: in a high-school math field experiment, an unguarded chatbot made practice look better and left students worse when the window closed.⁶¹ Not an adult science RCT. Not this

⁶¹Bastani and colleagues (2025), *PNAS*: Turkish high-school math field experiment (grades 9–11). Unguarded chatbot practice gains with unaided losses; tutor-guarded pattern different. Cited here as **OFF-AGE for science literacy** — not an adult science RCT; do not mash *Math Refresher for Adults* ON-AGE-adjacent label onto this book. Mechanism for the Hour: model may prepare;

science band. The lunch-break rule it supports: the model may prepare you and the next looking page; it may not do your boundary, arrows, omission honesty, or claim when those are the lesson.

Museum pages, park maps, and weather agency figures are neighbors for check and explore after the try. They replace thinking when they finish the attempt. Photo-to-key stays banned. Invent-data stays banned.

What “done enough” looks like

Before you move on:

- You drew a **boundary** on a real object and named **two things left outside on purpose**.
- You named **one input** and **one output** in ordinary words (or marked an arrow as uncertain).
- You wrote **one omission honesty** sentence: what the model leaves out and what you therefore cannot conclude.
- You refused at least one wrong twin: slogan-without-diagram, everything-inside, model-as-world, scale mash, decorative arrows, schematic-as-observation.
- Optional weekend: one museum exhibit-as-model critique or one scale-shift sticky.
- You can ask the locked opening and wait.

A science literacy certificate is a receipt someone else might invent — not the work, and not this book’s product. Optional CLEP edges remain optional. Place next work by whether “system” still arrives as a slogan with a blank page.

Life of the habit (seed). Revisit next Tuesday with a new object. Space systems sits — do not mass five diagrams on Sunday and call it done.⁶² Mix a FEED

model may not do the looking move when that move is the lesson.

⁶²Dunlosky and colleagues (2013): distributed practice rated high utility as a learning-technique rating — not an adult science-refresher programme evaluation. Space systems sits across days.

“complex system” interloper into a clean boundary week so the slogan stays honest. File the sketches in Keeping track. Chapter 5 will ask whether you can tell for sure — you will need the habit of naming what you held outside and what you changed, because a fair look is a kind of boundary on variables. Chapter 6 will put Bio / Chem / Phys / Earth soft on wheels as vehicles; the vehicle is not a survey year, and the boundary move still rides along.

If this week only yields one jar boundary and two outsides, that is a real win. Systems looking returns one closed line at a time.

If a coworker asks why you are doodling a box around a coffee cup, you can say you are practicing naming what you are leaving out. That sentence is adult science literacy in plain clothes — and it is enough reason to keep the pen moving this week.

Looking-move strip check (systems). Evidence before feed still holds when a card says “system.” Observe before explain still holds when you itch to label photosynthesis before noticing droop. Systems looking adds: boundary, I/O soft, omission honesty, scale notice. Fair tests (Chapter 5) will ask whether you can tell for sure — you will need to know what you held constant and what you left outside the comparison. Vehicles (Chapter 6) will carry these moves on Bio / Chem / Phys / Earth soft without turning the week into three transcript years.

If a colleague asks what you are doing with a doodle of a pot, you can say: “I am practicing naming what I am leaving out.” That sentence is adult science literacy in plain clothes.

Self-placement probe (skill strip, not diploma year): Can you draw a boundary before you explain? Can you name two outsides on purpose? Can you point to one input and one output? Can you say what the model leaves out? If yes, you are ready to mix systems looking into fair-test weeks and vehicle weeks. If no, stay here with thinner objects and shorter sits. A graduate can still slogan. A GED-holder can run a clean pot diagram. No diploma required to open this chapter; no diploma finishes it either.

Chapter 5



Figure 7: Confound-vs-fair side-by-side cards with can-we-tell sticky and notebook — still-life, no people, no logos.

Fair tests / can we tell for sure?

Why this matters

Adults meet comparisons that look like proof. A workplace slide shows Team A's numbers after a new tool and Team B's numbers without it — different teams, different weeks, different managers, and a caption that says the tool “worked.” A kitchen dissolve “proves” cold water is slower while the spoon, the stir count, and the grain size also changed. A feed “experiment” changes three things, measures one mood score, and calls the result science. Without a fair-look chapter, those surfaces get swallowed by Bio / Chem / Phys / Earth soft later — and the question *can we tell for sure?* never gets a home.

Fair looking, in this book, is a small adult habit. You name what you change. You name what you measure. You name what you try to keep alike. You ask whether anything else differed that could explain the result. You redesign when the first design cannot tell. You allow **insufficient evidence** as a valid exit — not as failure of nerve. You transfer the same rule to a new surface: grocery to weather, phone brightness to commute time, pharmacy-rate soft (units only) to a plant watering claim.

Chapter 2 taught you to notice before you explain. Chapter 3 taught you to treat a feed as a later telling until design and limit are named. Chapter 4 taught you to draw a boundary and say what you left outside. This chapter assumes those moves. It asks a sharper question about comparison: when two outcomes differ, can you tell it was the variable you named?

Classroom research on controlling variables supplies a useful *shape* — change one thing, measure one thing, hold others as steady as you can, ask whether the comparison is fair — often studied with elementary and middle-school samples.⁶³ That shape is portable to adult self-study when you keep adult nouns and adult surfaces. Child percentages are not your score. They are not an adult-home effect size. America's Lab Report continuity still holds: a try without an idea and a talk is weak theater; a try that invents numbers to look “significant” is not a lab experience.⁶⁴

⁶³Chen and Klahr (1999) and related control-of-variables (CVS) classroom research — **shape only**. Origin samples often elementary/middle school. Do not print child percents as adult scores. Not an adult-home effect size. Access date for URLs in these notes: 22 September 2026.

⁶⁴National Research Council, *America's Lab Report* (2006): isolated labs often weak for content

What this unlocks is usable caution on ordinary stakes. You can redesign a confounded kitchen picture without claiming Chemistry credit. You can press a workplace A/B claim without a public sting. You can end a sit with “insufficient evidence” and mean it as science, not as giving up. Money, workplaces, and health leaflets motivate. They do not replace the fair-look structure. No live controversy as the object. No medical advice. No dosing experiments.

If you leave a confound-versus-fair pair, a three-line card naming what you change / measure / hold, and one redesign sentence — or a clean insufficient exit — the week did its job. Place by that strip, not by diploma year.

Fair looking also protects you from a quieter adult habit: treating a single vivid before/after as a controlled comparison. A photo of a cleaner counter after a new spray proves little if light, time of day, and elbow grease also changed. A step-count spike after a new watch proves little if vacation week also arrived. The literacy move is not cynicism. It is asking what else differed before the causal verb hardens. That question travels from kitchen to commute to workplace slide without needing a lab coat.

Rust shows up here as twin speeds. Adults who argue for a living may be fast at confident causal talk and slow at naming what else differed. Adults who fear being “wrong” may treat “insufficient” as personal failure. Both patterns are ordinary. Neither is a character flaw. Both have a repair: slow the first pass; name the extras that changed; practice insufficient as a scientific outcome; redesign one step without inventing data to rescue the claim.

Understand it yourself

Everyday picture. Two plant pots. You water one daily and the other every three days — and you also move one to a brighter window, change the soil in one, and start them on different days. At week’s end one looks greener. You feel sure it was the watering. A stranger looking at your notes cannot tell. Too many things differed. The everyday repair is smaller: change watering only, or light only; measure one thing you defined; keep pot size, soil batch, start day, and window as alike as you can; then ask *can we tell for sure?*

Scale that picture to adult life: two dissolve cups where temperature and stir rate

mastery; integrate investigation with ideas and discussion; constructed fake data $\neq$ lab experience. Continuity for adults: redesign and talk beat kitchen theater-as-credit.

both change; a phone-brightness “focus” claim where sleep and caffeine also moved; a commute-time comparison across two weeks that also changed route and weather; a workplace tool roll-out compared across teams that never matched on workload.

Precise picture. A fair comparison tries to isolate one difference so an outcome difference can be attributed with honesty. In adult nouns: **what you change** (sometimes called the independent variable in school language — you may use ordinary words); **what you measure** (dependent); **what you keep alike** (constants). The locked question is not “did I prove it forever.” It is “can I tell for sure it was the variable I named — what else differed?” Redesign is the repair when the answer is no. Insufficient evidence is valid when methods are thin, samples are unmatched, or the feed card never says what was held constant. Inventing a tidy table to make the claim look fair is not redesign — it is invent-data, banned in the Science Refresh Hour.

You do not need a lab coat on the train. You need two sketches — confounded beside fair — and permission to stop at redesign when time is short. Running the confounded design and claiming proof is the wrong twin. Discovery-only mythology (“just explore and you’ll know”) is the other twin when it skips the fair-look question entirely.

A third precision point: “constants” are aims, not magic. Holding stir count alike is an aim you can check. Holding “everything about the workplace” alike is a fantasy. Fair looking names the few that matter for *this* question and admits the rest as limits. That honesty is closer to usable science than a confident caption over unmatched teams.

Wrong answers you should be able to hear (including from yourself).

1. Confounded but confident.

Many things changed; the caption still says “proves.”

What it usually means: causal hunger outran design.

What you try: List everything that differed. Circle which design allows a telling comparison. Redesign or say insufficient.

2. “We’ll control everything” without naming.

Vague control talk with no list.

What it usually means: fair-test vocabulary without a card.

What you try: Three lines — change / measure / keep alike — in ordinary words.

3. **Discovery-only mythology.**

Exploring is treated as automatic fair knowledge.

What it usually means: America's Lab Report warning ignored — try without idea and talk.

What you try: Keep curiosity; add one fair-look question before the claim.

4. **Inventing a “significant” result.**

Numbers appear that were never measured; or a generated gel/table is filed as the run.

What it usually means: invent-data / gel-as-result reflex.

What you try: Honesty card. Stop. Redesign on paper counts. Do not invent.

5. **Fear of “insufficient.”**

Thin methods get filled with vibes or tribal certainty.

What it usually means: insufficient treated as personal failure.

What you try: Write “insufficient evidence because ____” as the exit. List one next check. That is science.

Five-minute warm-up. On paper, sketch two designs for one ordinary question (sugar in water; plant light soft; phone brightness soft). Left: confounded (two things change). Right: fairer (one change). Ask the locked opening on each. Circle which can tell. Warm-up done.

Locked opening: **“Can I tell for sure it was the variable I named — what else differed?”**

If the only “experiment” you have is a feed card with no methods crumb, honesty-label it as later telling, list what you would need to tell for sure, and practice insufficient as the exit. Chapter 3 still applies. Fair looking does not require you to run a risky try at home.

How to try it this week

Science Refresh Hour shapes apply. This week's strip: confound versus fair; name change / measure / keep; redesign or insufficient; transfer to a new surface.

Session shape. Retrieve a systems-boundary sticky or a feed-honesty sticky. Put one ordinary comparison question on the table. Locked opening. Confound sketch beside fair sketch. Optional three-line IV/DV/constants card in adult nouns. Redesign sentence or insufficient list. Exit sticky. Mixed later: slip a workplace

“A/B worked” slide into a clean fair sit and ask what else differed.

Commute / lunch (15–20 min). Materials: phone note; one saved workplace or news “experiment” card with thin methods — ordinary stakes. Safety: private; no public quiz that humiliates a colleague; no undercover audit. Move: FAIR-04 soft — list what you would need to tell for sure; end with insufficient if needed. Optional: one “what else differed?” line only.

Desk or shift (25–40 min). Materials: paper; food-safe dissolve picture idea or two-cup sketch (no unauthorized chemicals); or plant-watering redesign on paper; RAMP if any heat. Safety: food-safe only for kitchen pictures; SAFE-03 honesty label — picture $\neq$ Chemistry credit; stop if you cannot minimize. Move: FAIR-01 confound vs fair; FAIR-02 name change / measure / keep; FAIR-03 redesign after confound caught (redesign alone counts).

Weekend deep sit (45–75 min when needed). Materials: two different everyday surfaces (grocery structure claim soft; weather “because of X” story; phone-sensor soft). Move: FAIR-05 transfer — same fair rule, new surface. Optional: run a food-safe picture once if RAMP is green, with REC table (Chapter 2 grain) and honesty label. Dual-perspective sentence: “Another fair reading of what else differed is ____.” Claim crumb with named limit. Stop while a look remains.

Exact wording you can say to yourself.

- “Can I tell for sure it was the variable I named — what else differed?”
- “What do you mean, *for sure* — which other differences could explain this?”
- “What does the table / sketch show was changed, measured, and held?”
- “What else could explain these numbers?”
- “What can this evidence not tell me?”
- “You may change the design — redesign counts even if you do not re-run today.”

Fade help. Side-by-side template with worked confound → blank side-by-side → three-line card from memory → redesign sentence without a sample beside you → insufficient exit on a thin feed card. Stop looking up when you can spot a confound and name a redesign without a model inventing a clean data table for you. If you peek, peek at your own prior sticky — not at invent-data.

First try-it (on the go). Insufficient list on one thin “study says” card. Deeper sit: FAIR-01 on dissolve or plant-watering sketch. If the deeper sit stalls, fall back to naming what else differed on a workplace slide — still a win.

Sample micro (composite). Train seat. Workplace slide: “After the new checklist, errors dropped.” Locked opening. What else differed — staffing, volume, week of month, who counted errors. Exit sticky: *Insufficient to tell — staffing also moved*. Close. No public confrontation.

Sample desk (composite). Question: does warmer water dissolve sugar faster — picture-level. Confounded sketch: warmer water *and* more stirring. Fair sketch: same sugar mass, same grain, same stir count, same cup, water temperature differs. RAMP if heat: recognize splash/burn; assess; minimize small volume; prepare spill plan; or skip heat and stay on paper redesign. Honesty: kitchen picture $\neq$ Chemistry credit. Exit: circle fair sketch; “Next time I hold stir constant.”

Worked pictures you can steal (then put away)

Picture 1 — Confound vs fair (FAIR-01). One ordinary question. Two sketches. Ask can we tell on each. Exit: circled fair design. Refuse running confound as proof.

Picture 2 — Name change / measure / keep (FAIR-02). Three-line card in adult nouns. Ordinary words OK. Exit: IV/DV/constants card. Refuse “control everything” without names.

Picture 3 — Redesign after confound (FAIR-03). Name the confound. Redesign. Run once only if RAMP green — or stop at redesign. Exit: “Next time I will hold ___ constant.” Refuse defending the confounded claim anyway.

Picture 4 — Insufficient is valid (FAIR-04). Thin methods card. List needs. Exit: insufficient + one next check. Refuse vibe-fill.

Picture 5 — Transfer (FAIR-05). New surface, same rule. Exit: one transfer sentence. Refuse “fair tests are only for lab coats.”

Picture 6 — Wrong twin (labelled). Claim: “I proved cold water is slower — I used a bigger spoon in the cold cup and stirred less, but I’m sure.” Repair: “Spoon size and stir differed too; I cannot tell for sure; next design holds spoon and stir; this kitchen picture is not Chemistry credit.”

Close the pictures. Build one confound-versus-fair pair on a parallel question yourself. Keep the first desk pass under twenty-five minutes. Mapping is the win. If life sends a thinner question than the pictures, good — thin questions are what Tuesday sends.

When you finish, ask once: did I invent numbers to make it look fair? If yes, strike them. Redesign on paper. Honesty card. Return.

Practice that actually builds it

Blocked early: several confound-versus-fair micros on paper; then three-line cards; then one insufficient exit on a feed. Mixed later: workplace A/B interloper; transfer to weather; kitchen picture with SAFE-03 after a clean paper redesign.

Talk-to-yourself box

Exact opening question:

“Can I tell for sure it was the variable I named — what else differed?”

Follow-ups (3–5):

1. Meaning: “What do you mean, *for sure* — am I asking for proof forever, or for a fairer comparison than the confound?”
2. Representation: “What does the sketch or table show was changed, measured, and held alike?”
3. Alternative: “What else could explain these numbers or this greener plant?”
4. Check: “What can this evidence not tell me — sample, time, unmeasured differences?”
5. Revise / diagnose: “Where could someone go wrong — confounded confidence, invent-data, fear of insufficient? You may redesign or exit insufficient.”

How to wait alone. After you ask, cover the feed, the answer key, and the chatbot. A slow three is a convention from classroom think-time research — school-origin; your adult translation is pause-before-lookup / pause-before-feed.⁶⁵ Look at the two sketches or the thin methods card, not at a generated “perfect” data table. After you stop mid-reason, wait again before peeking. Rescue at one second is still rescue. Abandonment for forty minutes of scrolling is not a micro-session.

Stuck silence usually means: the question was vague; wait-time 1 was zero; you are searching for the word *variable* instead of naming the difference; you are guess-

⁶⁵Rowe (1972) wait-time; Stahl (1994) think-time — classroom origin. Adult translation: pause-before-lookup / pause-before-feed. Not an adult science RCT.

ing what a lab report would want; the question was recitation in disguise; you offered a keyword where a fair look was asked; you think “science person” means leave; you think a finished-looking chatbot CER with no trial is already a check; anxiety is eating working memory — shorten the session, do one insufficient sticky or one confound circle, come back tomorrow. Your job is to pose, wait, and point back at the design — not to fill the silence with invent-data.

On the go — FAIR-04 insufficient flash (15–20 min)

- **Time:** 15–20 min
- **Materials:** one thin “study says” or A/B workplace card (ordinary stakes); phone note
- **Safety:** private; no sting; no proprietary paste into public tools
- **The move:** List what you would need to tell for sure. Practice ending with insufficient.
- **Exit:** Insufficient list + one next check
- **Refuse:** Filling gaps with vibes

Desk — FAIR-01 confound vs fair (20–35 min)

- **Time:** 20–35 min
- **Materials:** paper; one ordinary question (dissolve soft, plant soft, brightness soft)
- **Safety:** food-safe if kitchen; RAMP if heat; SAFE-03 honesty if picture
- **The move:** Sketch confound. Sketch fair. Ask locked opening on each. Circle which can tell.
- **Exit:** Circled fair design
- **Refuse:** Running confound as proof

Desk — FAIR-02 name the three (15–25 min)

- **Time:** 15–25 min
- **Materials:** planned try card or redesign from FAIR-01
- **Safety:** as planned
- **The move:** What I change / what I measure / what I keep alike — adult nouns.
- **Exit:** Three-line card
- **Refuse:** “Control everything” without naming

Desk or weekend — FAIR-03 redesign (20–40 min)

- **Time:** 20–40 min
- **Materials:** prior confound or labelled hypothetical confound log
- **Safety:** re-run only if RAMP still green; redesign-alone counts
- **The move:** Name confound. Redesign. Optional one run with REC.
- **Exit:** “Next time I will hold ____ constant.”
- **Refuse:** Defending confounded claim; invent-data to “fix” fairness

Desk — FAIR-05 transfer (20–30 min)

- **Time:** 20–30 min
- **Materials:** two surfaces (watering vs brightness; commute vs weather story)
- **Safety:** as relevant

- **The move:** Same fair rule, new surface, no new jargon required.
- **Exit:** One transfer sentence
- **Refuse:** Fair test as lab-coat-only

Incorrect example to diagnose (labelled composite)

A kitchen video says “Science proves ice water locks flavor.” The cook changes water temperature, soak time, and salt, films once, and captions “controlled experiment.” An adult files the caption as a study.

Diagnosis: confound + caption-as-study + kitchen-as-credit risk.

Repair: List what differed. Ask locked opening. Redesign on paper holding time and salt. Label picture not-for-credit. Open abstract or skip — Chapter 3. Insufficient until a fairer comparison exists. Do not invent a table of “trials.”

Mixed practice when the move is stable

Pair a clean FAIR-01 with a FEED interloper that says “experiment.” Pair insufficient exit with a workplace slide. Pair transfer after redesign. Keep CER crumbs light — Chapter 7 owns denser claim. Here, “I cannot tell for sure because ____ differed” is enough.

Life surfaces that carry the move (they do not replace it)

Workplace A/B slides, weather “because” stories, grocery “works better” claims, pharmacy label rates soft (units only — no medical advice), phone-sensor self-experiments, and kitchen pictures *are* the practice when you bring confound-versus-fair. They do not replace the looking structure. Noticing is not a sting. A fair try-it uses a public card, a labelled composite, a grocery aisle, or a pamphlet you already hold.

Weekend optional — OUT neighbor with fair press

- **Time:** 30–50 min

- **Materials:** park transect or grocery structure–function soft (OUT-01 / OUT-03); notebook
- **Safety:** public path; store etiquette; food hygiene
- **The move:** One comparison question. Ask what else differed across stops or items. One claim with limit — or insufficient.
- **Exit:** Fair-press sticky
- **Refuse:** Scavenger mania; nutrition-war sermon

If it isn't clicking

Diagnostic 1 — Overconfidence on confounds.

You still feel sure when two or three things changed.

Next move: Force a written list of differences before any causal verb. If the list has more than one change, you may not tell for sure. Circle redesign. Short session. Early win: one circled confound today.

Diagnostic 2 — Fear of saying insufficient.

Thin methods make you fill with certainty or abandon the hour in shame.

Next move: Teach yourself insufficient as a valid scientific exit. Write the sentence. List one next check. That is the product. Anxiety tip: no countdown timer as default rigor; one sticky; return tomorrow.

Diagnostic 3 — Invent-data or chatbot “fair table” as rescue.

A model offers a clean data table; you almost file it as your run.

Next move: Put the tool away during the attempt. Redesign on paper. Honesty card. Point back to the Science Refresh Hour AI box. Ban invent-data to “make it fair.”

Correct a wrong reason without crushing the attempt — applied to yourself. Hearing “I was sure because the caption said experiment” and answering “caption is not design; list what differed” is the work. When to go ahead: you can show confound versus fair and name a redesign or insufficient on two surfaces. When to slow down: every sit still ends in confident causal talk on confounds. When to ask a human: colleague, librarian, tutor, community lab — when you need a

supervised path or a second pair of eyes on a workplace claim you must handle carefully, not because you failed a fair-test gene.

Tools, including AI

Point back to the Science Refresh Hour AI box. Keep this short.

You may, after you have tried: explain fair-test ideas to yourself from a named OpenStax / Khan / museum page (title, page, today’s micro-goal); ask for isomorphic practice with answers on **your** page — blank confound/fair cards + your key; request a labelled SCRIPT for questions to ask yourself; ask revision questions after *your* redesign or insufficient sentence; take a hint after an attempt; diagnose a sketch you already made (crop to page — no face). Absolute-versus-relative prompting stays for risk claims after you drafted from named numbers (Chapter 3 grain) — not a substitute for fair design.

You may not during the attempt: invent a data table to make a confound look fair; ask a model to “write my lab report” or “fix to publishable” as the only draft; generate a gel / micrograph / “lab photo” as a result; paste a blank worksheet or employer sheet for completion (photo-to-key); park an unsupervised chatbot as your only partner during the attempt; treat a model summary as the observation or the study; paste proprietary employer A/B data into a public consumer tool; print a certificate of fair-test literacy.

One mechanism sentence, labelled **OFF-AGE for science literacy**: in a high-school math field experiment, an unguarded chatbot made practice look better and left students worse when the window closed.⁶⁶ Not an adult science RCT. Not this science band. The lunch-break rule it supports: the model may prepare you and the next looking page; it may not do your fair look, redesign, measuring, or claim when those are the lesson.

America’s Lab Report continuity: integrate look–try–talk; clear outcome; do not sell isolated kitchen theater as credit Chemistry.⁶⁷ Lab-path neighbor (SAFE-05)

⁶⁶Bastani and colleagues (2025), *PNAS*: Turkish high-school math field experiment. Unguarded chatbot practice gains with unaided losses. **OFF-AGE for science literacy** — not an adult science RCT; do not mash *Math Refresher for Adults* ON-AGE-adjacent label onto this book.

⁶⁷National Research Council, *America’s Lab Report* (2006): isolated labs often weak for content mastery; integrate investigation with ideas and discussion; constructed fake data $\neq$ lab experience. Continuity for adults: redesign and talk beat kitchen theater-as-credit.

when home cannot minimize — Chapter 6 and Resources.

What “done enough” looks like

Before you move on:

- You showed **confound versus fair** on one ordinary question.
- You named **what you change, what you measure, what you keep alike** — or you said **insufficient** with a reason.
- You wrote **one redesign step** (“Next time I hold ____”) or a valid insufficient exit with one next check.
- You refused at least one wrong twin: confounded confidence, invent-data, caption-as-experiment, fear-of-insufficient-as-shame, kitchen-fizz-as-credit.
- Optional weekend: transfer the fair rule to a new surface.
- You can ask the locked opening and wait.

A science literacy certificate is a receipt someone might invent — not the work. Optional CLEP edges remain optional. Place next work by whether “what else differed?” still arrives after every tidy causal caption. Place next work by whether “what else differed?” still arrives after tidy causal captions — not by whether you can recite the letters I, V, and D. Ordinary nouns that name the comparison are enough.

Life of the habit (seed). Revisit next week with a new comparison. Space fair sits — do not mass five redesigns on Sunday and call it done.⁶⁸ Mix an insufficient exit into a clean redesign week so the valid exit stays honest. File the cards in Keeping track. Chapter 6 will put Bio / Chem / Phys / Earth soft on wheels as vehicles; the fair-look question rides along. Chapter 7 will ask for a denser claim with a warrant and a named limit — you will need the habit of pointing at *this* design and saying what it cannot settle.

⁶⁸Dunlosky and colleagues (2013): distributed practice high-utility technique rating — not an adult science-refresher programme evaluation. Space fair sits across days.

If this week only yields one circled confound and one “I cannot tell for sure,” that is a real win. Fair looking returns one honest comparison at a time.

Self-placement probe (skill strip, not diploma year): Can you sketch confound beside fair before you explain? Can you name change / measure / keep in ordinary words? Can you exit insufficient without shame? Can you spot invent-data offered as a “fair” table? If yes, you are ready to mix fair looking into vehicle weeks and claim weeks. If no, stay here with thinner questions and shorter sits. A graduate can still crown a confound. A GED-holder can run a clean redesign card. No diploma required to open this chapter; no diploma finishes it either.

More desk pictures (steal, then close)

Picture A — Phone brightness soft (no medical claim). Question: does lower brightness change how long I stay on a reading app tonight? Confound: lower brightness *and* earlier bedtime *and* no caffeine. Fairer sketch: same bedtime window, same caffeine plan you already use, same article length soft, brightness differs; measure minutes on app with honesty that apps track imperfectly. Locked opening. What else could differ — fatigue, interest in the article. Exit: redesign holds bedtime; insufficient if you cannot hold interest. Refuse using this as sleep medicine advice.

Picture B — Commute time soft. Question: does leaving ten minutes earlier shorten door-to-desk time? Confound: earlier leave *and* different route *and* different weather. Fairer: same route, note weather as a kept-alike or as a recorded covariate you refuse to pretend vanished; leave time differs; measure minutes. Exit: “I cannot tell for sure across a storm day.” Transfer sentence: same rule as plant watering — name what else moved.

Picture C — Grocery “works better” soft. Two produce bags, different stores, different days, different handling — caption says one farm’s method wins. Locked opening. Insufficient until handling and day are addressed. Chapter 3: open a named page if you chase methods; do not invent a farm trial table.

Picture D — Pharmacy label rate soft (units only). A label states a rate. An adult “tests” skipping a dose “to see” — stop. Fair looking never authorizes dosing experiments. The literacy move is reading units and asking what a rate assumes — then taking clinical questions to a clinician. Exit: non-dosing sticky. This picture

exists to refuse a wrong try-it.

Composite week map (menu, not calendar)

- Mon micro: insufficient sticky on one thin feed “experiment.”
- Tue desk: FAIR-01 dissolve or plant sketch.
- Wed micro: three-line change / measure / keep on yesterday’s fair sketch.
- Thu desk: FAIR-03 redesign sentence; optional food-safe picture if RAMP green and SAFE-03 labelled.
- Fri micro: workplace A/B “what else differed?” private note.
- Sat optional: FAIR-05 transfer on weather or commute soft; or OUT grocery with fair press.

Skip days without “making up” five designs on Sunday. One honest redesign beats a massed stack.⁶⁹

Fade ladder for fair looking

Day one: circled confound on a worked pair. Day two: blank confound-versus-fair on a new question. Day three: three-line card without a sample card in view. Day four: insufficient exit on a thin methods card. Day five optional: transfer. Stop looking up when you can name what else differed without a model inventing a clean multi-trial table.

Stranger-readable fair card (REC support)

Header: date; question; what I change; what I measure (with unit); what I keep alike; what else might have differed; can I tell for sure? (yes with limit / no — insufficient / not yet — redesign). If a stranger cannot recover the comparison, repair the header before you dress the claim. Do not invent missing trials to look complete.

⁶⁹Dunlosky and colleagues (2013): distributed practice high-utility technique rating — not an adult science-refresher programme evaluation. Space fair sits across days.

Wrong twins gallery (diagnose, then repair)

1. **Caption says controlled; spoon sizes differ.** Repair: list differences; redesign.
2. **“I controlled everything” with empty list.** Repair: write three lines.
3. **Chatbot offers a data table for a try you did not run.** Repair: delete table; honesty card; paper redesign.
4. **Kitchen fizz filed as Chemistry credit.** Repair: SAFE-03; picture label; lab-path if credit is the real goal.
5. **Insufficient feels like failure, so you crown the confound.** Repair: write insufficient as the product; one next check.
6. **Public quiz of a coworker’s “bad experiment.”** Repair: private literacy; noticing $\neq$ sting.

Talk turns you can borrow (then put away)

Say aloud: “I changed . *I measured* . I tried to keep ____ alike.” Say: “What else differed was .” *Say: “I cannot tell for sure because .”* Say: “Next time I will hold .” *Say: “Insufficient evidence — next check is .”* These are not a single right recitation. They must point at your sketches or card. If a line cannot point, it is commentary seeking a home.

Worked incorrect CER crumb (keep light)

Claim: “Warmer water dissolves sugar faster.” Evidence: “My cold cup used a bigger spoon and fewer stirs.” Reasoning: “Because temperature matters.” Limit: missing.

Diagnosis: evidence describes a confound; reasoning floats; no limit.

Repair: “I cannot tell for sure from this try because spoon and stir differed; next design holds spoon and stir; kitchen picture $\neq$ Chemistry credit.” Chapter 7 will densify warrant; here the fair-look repair is enough.

Safety spine reminder for fair try-its

RAMP before heat or chemicals. Adult present to self. Food-safe kitchen pictures only. No drain-cleaner chemistry. No sealed-container fizz. No wall-outlet physics. Stop if you cannot minimize. Lab-path neighbor when home cannot meet the ceiling. Fair looking does not require a risky run — redesign on paper counts.

Why “insufficient” is not a personality test

Science literacy includes knowing when a comparison cannot settle a claim. Population process items about control groups are context for why adults need this language — not your pretest and not a proficiency ladder.⁷⁰ Ending with insufficient after a thin workplace slide is closer to usable literacy than crowning the slide to seem decisive. Place by the strip: can you say insufficient without shame and name one next check?

Bridge to vehicles and denser claim

Chapter 6 will ask which looking move rides on a Bio / Chem / Phys / Earth soft vehicle. Fair looking is one of those moves. A dissolve picture can carry FAIR; a plant pot can carry FAIR; a weather page can carry FAIR — without a survey year. Chapter 7 will ask for claim + evidence + warrant + named limit. You will need today’s habit: point at the design, say what else differed, and refuse invent-data as a fake warrant.

One more composite desk sit (steal, then close)

Thursday evening. Question on the card: does a thicker mulch layer slow how fast pot soil looks dry at the surface? Confounded first thought: thicker mulch *and* a bigger pot *and* a sunnier sill. Fairer sketch: same pot size, same starting wetness soft (defined as “surface dark/light by eye” — operational, imperfect), same sill, mulch thickness differs; measure days until surface looks dry by your rule; record weather as something that may still differ outdoors. Locked opening. Exit: redesign holds pot and sill; limit — eye judgment of “dry” is coarse; insufficient across a heat wave if you cannot hold outdoor heat alike. Honesty: gardening pic-

⁷⁰NSF Indicators / Pew ATP process items (control-group usefulness about 60% in reported contexts) — labelled adult process neighbour, not this chapter’s pretest and not a proficiency ladder.

ture, not a soils course credit. File the card. No inventing three neat trials you did not run.

Friday lunch. Thin workplace slide: “Pilot team using the new form made fewer mistakes.” What else differed — team seniority, ticket mix, who scored mistakes, week of quarter. Exit sticky: insufficient to tell; next check would need matched workload or a clearer scoring rule. Private note. Noticing is not a sting.

Saturday optional. Transfer: take the mulch fair rule to a “earlier leave shortens commute” soft question. Same bones — what you change, what you measure, what you keep alike, what else differed. One transfer sentence. Stop.

If anxiety spikes mid-list, shrink. One confound circled. One insufficient sentence. Come back tomorrow. Fair looking is a strip, not a personality renovation.

Carry one sticky into next week: a circled confound, a redesign sentence, or an insufficient exit with one next check. That is enough continuity for Chapter 6’s vehicles to ride on without turning Tuesday into a survey year.

Chapter 6



Figure 8: RAMP card, kitchen honesty label, Bio/Chem/Phys/Earth soft stickies — still-life, no people, no logos.

Life, matter, motion, Earth soft as Hour vehicles (not survey dump)

Why this matters

Adults refreshing science often ask the wrong first question: “Should I restart Biology?” Then Chemistry. Then Physics. Then a year of Earth. Coverage itch arrives dressed as responsibility. A survey map is useful. A survey year is not this book’s Tuesday.

This chapter answers with a different first question: **which looking move am I practicing, and which soft vehicle will carry it this week?** Life, matter, motion, and Earth soft are vehicles — not three transcript years stuffed into one chapter. A grocery structure–function look can carry Observe. A plant pot can carry Systems. A dissolve picture can carry Fair looking — labelled not-for-credit. A tossed key can carry change-of-motion noticing. A weather page can carry evidence-before-feed and weather-versus-climate soft honesty. Depth is the hour. Coverage is a map you may open in Resources when credit or a supervised path is the real goal.

Kitchen honesty and RAMP sit beside the vehicles. A food-safe fizz can motivate looking. It does not mint Chemistry credit. America’s Lab Report continuity still warns against isolated try-theater without idea and talk.⁷¹ When home cannot minimize hazards, a lab-path neighbor is the honest door — community college, supervised maker bench, museum lab day — not a certificate you invent after baking-soda night.

Prior ideas travel into adulthood. Taking Science to School notes that Newtonian construction is “by no means obvious even to high school or college students who have had courses in introductory mechanics.”⁷² Force-as-fuel / impetus neighbours — the sense that a push is still “in” a tossed object on the way up — show

⁷¹National Research Council, *America’s Lab Report* (2006): isolated labs often weak for content mastery; integrate look–try–talk; constructed fake data $\neq$ lab experience. Continuity for adults: kitchen picture $\neq$ Chemistry credit; embed light try in idea and talk. Access date for URLs in these notes: 22 September 2026.

⁷²National Research Council, *Taking Science to School* (2007): Newtonian construction “by no means obvious even to high school or college students who have had courses in introductory mechanics”; force/motion prior ideas; theory–evidence coordination improves into adulthood but adults still meld theory and evidence. Labelled continuity — not an invented adult RCT for this book.

up in adults too. Heat-versus-temperature soft confusion and cells-as-tiny-mouths continuity appear in school-age literatures (Driver and related lines); they are labelled continuity here, not an invented adult RCT.⁷³ Repair is another look plus a named idea — not shame for “not being a science person,” and not a scolding that you should have finished this at sixteen.

What this unlocks is usable sequencing without coverage panic. You pick one vehicle per week max. You name the looking move that rides on it. You refuse Bio-then-Chem-then-Phys as the weekly spine. You refuse AP free-response packets as Tuesday identity. You refuse kitchen theater as transcript Chemistry. You allow OpenStax or Khan *after* the try when you want a named section check. Money, workplaces, parks, museums, and kitchens motivate. They do not replace the looking move.

If you leave a named move-on-vehicle sticky, a kitchen honesty label when a picture was used, and a RAMP card when heat or chemicals appeared — or a lab-path plan when home cannot minimize — the week did its job. Place by that strip, not by diploma year and not by how many chapters of a survey text you guilt-skimmed.

Rust here often sounds like two voices: “I should cover everything” and “I’m not a science person so vehicles won’t help.” Both can be true as feelings and false as placement. Coverage itch is ordinary. So is anxiety. Neither finishes the hour. One move on one vehicle does.

Understand it yourself

Everyday picture. Produce aisle: two fruits. You notice skin, layers, seeds (Observe). You guess a function, then check one labelled secondary after (not before). Or: ice melting in a cup — you label the try a **picture / not Chemistry credit**, notice change, refuse to file fizz as a course. Or: a key tossed gently upward — you notice change of motion; you refuse the automatic story that an upward “force fuel” stays in the key for the whole climb. Or: weather app — you read numbers before the story; you refuse one cold week as climate disproof theater.

⁷³Driver, *Children’s Ideas in Science* (1985) and alternative-frameworks line: heat/temperature, force/motion, matter — **school-age documentation**. Adult book may name soft confusion; do not invent adult RCT effect sizes from school samples. Cells-as-tiny-mouths continuity labelled from biology misconception literatures / sister science memos — repair with systems looking, not shame.

Precise picture. A vehicle is a domain soft enough to carry OBS, SYS, FAIR, FEED, or a light CER crumb — without becoming an encyclopedia. Biology soft: interacting systems, plant pot, variation in a sample, one ecosystem question, inheritance claim soft as map not wet lab. Chemistry soft: properties before names, stoppable reaction picture, conservation weighing honesty, dilution sense without stock acids, amount meaning before mole drill. Physics soft: change-of-motion, energy transfer storyboard, wave pattern soft, household-safe circuit limits or skip, rate from a simple graph. Earth soft: sky pattern looking up, water in the system, weather-to-climate evidence, rock/land process soft, human-impact claim with named evidence. Framework disciplinary ideas are a **map**, not an adult mandate and not a PE dump as weekly spine.⁷⁴

Kitchen $\neq$ credit. RAMP: Recognize hazards, Assess who could be harmed, Minimize (lowest volume, substitute, PPE, clear space), Prepare (spill/burn/glass plan), then go / redesign / lab-path / skip. Stop if you cannot minimize. Force-as-fuel and heat-vs-temperature get soft repair: another look, ordinary words, labelled school-age research origin — no invented adult trial numbers.

Wrong answers you should be able to hear (including from yourself).

1. Bio-then-Chem-then-Phys dump as the week.

Coverage theater returns.

What it usually means: depth replaced by tour anxiety.

What you try: One vehicle. One looking move. Stop.

2. Kitchen = Chemistry credit.

Fizz filed as transcript.

What it usually means: path honesty fail.

What you try: SAFE-03 honesty card. Label picture. Point Resources / lab-path if credit is the goal.

3. Force-as-fuel shame.

You notice the tossed-key idea and scold yourself as stupid.

What it usually means: prior idea treated as character flaw.

What you try: Another look at change of motion; name the idea softly; TSTS continuity labelled — adults and college students struggle too.⁷⁵

⁷⁴National Research Council, *A Framework for K–12 Science Education* (2012): practices, crosscutting concepts, disciplinary core ideas as **map** — not NGSS PE dump as adult weekly spine, not adult programme evaluation.

⁷⁵National Research Council, *Taking Science to School* (2007): Newtonian construction “by

4. **AP unit / FRQ as Tuesday identity.**

Adult hour becomes exam packet.

What it usually means: edge mistaken for spine.

What you try: Vehicle + looking move. AP is not adult default (Resources).

5. **Gel / sky / micrograph generated as “what I saw.”**

Model image filed as result.

What it usually means: invent-result reflex.

What you try: Ban. Observe your object. Honesty card.

Five-minute warm-up. Write today’s looking move (Observe / Systems / Fair / Feed hygiene / light CER). Pick one vehicle word: Bio / Chem / Phys / Earth soft. Name one object you already meet (produce, pot, cup, key, weather page). Say the locked opening. Warm-up done.

Locked opening: **“What looking move am I practicing on this Bio / Chem / Phys / Earth soft vehicle — and what does a picture try not earn as credit?”**

If credit Chemistry is your real goal this month, honesty requires a different door: supervised lab path, community-college survey, optional CLEP when appropriate — not a denser kitchen. This chapter will not pretend otherwise.

How to try it this week

Science Refresh Hour shapes apply. This week’s strip: one vehicle max; name the looking move; kitchen honesty + RAMP when needed; refuse survey dump and refuse picture-as-credit.

Session shape. Retrieve last week’s fair sticky or systems sticky. Choose vehicle + move before you open a textbook. Locked opening. Short try on the vehicle. Honesty / RAMP as needed. Exit sticky naming move + vehicle + what the picture does not earn. Mixed later: feed card that treats kitchen fizz as “study” — apply Chapter 3 + honesty.

Commute / lunch (15–20 min). Materials: phone note; weather page, grocery list soft, or tossed-key thought look. Safety: ordinary; no public quiz; no sun-staring.

no means obvious even to high school or college students who have had courses in introductory mechanics”; force/motion prior ideas; theory–evidence coordination improves into adulthood but adults still meld theory and evidence. Labelled continuity — not an invented adult RCT for this book.

Move: EARTH weather numbers before story; or PHYS change-of-motion sticky from memory of a safe toss at home earlier; or BIO structure–function one produce item if you are in a store (OUT-03 soft).

Desk or shift (25–40 min). Materials: plant pot; food-safe dissolve or ice melt; paper; RAMP card if heat/chemicals; SAFE-03 honesty card. Safety: food-safe only; goggles if splash; no drain-cleaner chemistry; no sealed containers; stop if unsafe. Move: BIO-02 pot as system; or CHEM-01 properties before names; or CHEM-02 stoppable picture with honesty; or PHYS-01 change-of-motion with clear space; or PHYS-02 energy storyboard on a kettle you observe without unsafe heating experiments.

Weekend deep sit (45–75 min when needed). Materials: park path; museum exhibit; night sky soft; or lab-path planning card. Safety: public access; tick awareness; venue rules; night walk sense; neighbor rules override home habits. Move: OUT-01 park transect with one claim limit; OUT-02 museum systems; OUT-06 sky/weather station; SAFE-05 / OUT-05 lab-path neighbor planning if credit Chemistry is real. Dual-perspective: “Another fair vehicle for this move would be ____.” Stop while a look remains.

Exact wording you can say to yourself.

- “What looking move am I practicing on this Bio / Chem / Phys / Earth soft vehicle — and what does a picture try not earn as credit?”
- “What do you mean, *vehicle* — which domain soft is carrying today’s move?”
- “What does the object / table / page show before the caption or textbook section?”
- “What else could explain this change of motion / melt / weather snap?”
- “What can this picture not earn as credit — and what would a supervised path add?”
- “You may change vehicles next week; you may not invent a survey year tonight.”

Fade help. Move+vehicle template filled → you write both blanks → honesty card without a sample → RAMP without a worked chemical list beside you → unaided vehicle try with looking move named first. Stop looking up when you can name move and vehicle before opening OpenStax. If you peek, peek at your sticky — not at a generated lab photo.

First try-it (on the go). Weather numbers before story, or one produce structure–

function soft. Deeper sit: pot as system *or* dissolve picture with SAFE-03 *or* change-of-motion card. If credit anxiety floods, write a lab-path neighbor plan card instead of forcing unsafe home chemistry.

Sample micro (composite). Bus. Weather app. Cover the story. Three number/icon observations. Uncover story; note hype. Exit: *EARTH soft + Observe/Feed hygiene — app story is not my observation.* Close.

Sample desk (composite). Kitchen. Ice melt in open cup. SAFE-03: “This is a picture because it is food-safe change-of-matter looking, not supervised Chemistry credit.” OBS notes. Optional light CER with limit. RAMP: recognize wet floor/slip; assess; minimize small volume; prepare towel. Exit sticky: *Chem soft vehicle + Observe — picture ≠ credit.*

Sample desk (composite) — force soft. Clear floor. Soft toss of a soft object upward (not at people/glass). Notice slowing, stop, return. Write: change of motion happened; “force as fuel staying inside on the way up” is a common idea — TSTS notes Newtonian construction is hard even after intro courses.⁷⁶ Another look: what interactions with Earth/air might we name in ordinary words later? No formula dump. Exit: *Phys soft + Observe — prior idea noticed without shame.*

Worked vehicle pictures (steal, then put away)

BIO-01 Body as interacting systems soft. Boundary subsystem soft after stairs — no diagnosis. Inputs/outputs. Limit: not medical advice. Refuse WebMD spiral.

BIO-02 Plant pot as system. Boundary pot–plant–window. Optional multi-day OBS. One factor claim with limit. Refuse photosynthesis cookbook without record.

BIO-03 Variation in a sample soft. Leaves or beans. Variation sentence. Refuse one leaf as the species.

BIO-04 Ecosystem one question. Park/yard. One question. OBS table. Claim with limit. Refuse vocabulary tour.

⁷⁶National Research Council, *Taking Science to School* (2007): Newtonian construction “by no means obvious even to high school or college students who have had courses in introductory mechanics”; force/motion prior ideas; theory–evidence coordination improves into adulthood but adults still meld theory and evidence. Labelled continuity — not an invented adult RCT for this book.

BIO-05 Inheritance / variation claim soft. Map not wet lab. Privacy. “I can claim ____ from observation; I cannot claim ____ without ____.” Refuse kitchen DNA theater.

CHEM-01 Properties before names. Salt/sugar food-safe. Property table. Honesty card. Refuse mystery bottles.

CHEM-02 Reaction you can stop (picture). Acid+bicarb soft small volume OR skip. Predict; run; stop; REC; CER+limit; honesty. Refuse scaling up; refuse drain cleaners.

CHEM-03 Conservation weighing honesty soft. Scale if available. Gas escape vs “matter destroyed.” Measurement limit. Refuse magic-trick disappearance without honesty.

CHEM-04 Dilution sense. Food coloring / juice soft. Dilution sentence. Non-dosing sticky. Refuse stock acids at home.

CHEM-05 Amount meaning before drill. Dozen analogy / label amounts. “Amount means ____ here.” Refuse mole worksheet dump as vehicle.

PHYS-01 Change-of-motion card. Soft toss / roll. Ordinary pushes/pulls. Refuse force-as-fuel slogan without noticing change; refuse formula-first.

PHYS-02 Energy transfer storyboard. Kettle / brake / charge soft — observe existing devices safely. Invisible path question. Refuse “energy destroyed” without limit talk.

PHYS-03 Wave pattern look soft. Sink ripple / string. Pattern claim with limit. Refuse full wave unit dump.

PHYS-04 Circuit household-safe or skip. Battery kit only — no wall outlets. Or lamp diagram without opening. Refuse outlet hacking.

PHYS-05 Rate from a simple graph soft. Public graph. Slope as rate in ordinary words. Axes named. Correlation $\neq$ full mechanism.

EARTH-01 Sky pattern looking up. Observe first; app after. No sun staring. Limit sticky.

EARTH-02 Water in the system soft. Watershed / drain / rain. SYS boundary. No floodwater entry.

EARTH-03 Weather-to-climate evidence soft. Snap story + longer-record figure from reputable agency page. “This snap alone cannot show ; *longer record suggests* (limit ____).” Refuse one-storm proof theater.

EARTH-04 Rock / land as system soft. Local look or museum photo. Process-over-time question. No cliff edges; no illegal collecting.

EARTH-05 Human impact claim with named evidence. Litter transect / air page / advisory — one. CER+limit. Separate understanding from pamphlet sermon. Hygiene if handling trash.

Close the pictures. Pick **one** for this week. Name the looking move before you start. Menu $\neq$ calendar.

Practice that actually builds it

Blocked early: several move+vehicle naming micros; then one BIO or EARTH soft desk; then one CHEM picture with honesty *or* PHYS change-of-motion. Mixed later: feed “kitchen science proves” interloper; force-as-fuel soft repair after a clean toss; lab-path plan after a SAFE-03 honesty sit.

Talk-to-yourself box

Exact opening question:

“What looking move am I practicing on this Bio / Chem / Phys / Earth soft vehicle — and what does a picture try not earn as credit?”

Follow-ups (3–5):

1. Meaning: “What do you mean, *vehicle* — is this carrying Observe, Systems, Fair, Feed hygiene, or a light claim?”
2. Representation: “What does the object, table, or page show before the caption or the OpenStax section?”
3. Alternative: “What else could explain this melt, droop, toss path, or weather snap?”
4. Check: “What can this picture not earn as credit — and what would supervised work add?”
5. Revise / diagnose: “Where could I go wrong — survey dump, kitchen-as-credit, force-as-fuel shame, generated gel as result? You may change the

vehicle next week.”

How to wait alone. Cover the feed, the textbook answer, and the chatbot. Classroom think-time is school-origin; your translation is pause-before-lookup / pause-before-feed.⁷⁷ Look at the pot, the cup, the sky, or your sketch — not at a generated micrograph. After you stop mid-reason, wait again. Rescue at one second is still rescue. Forty minutes of scrolling is not a micro-session.

Stuck silence usually means: the question was “study Biology” with no move; wait-time 1 was zero; you hunted vocabulary instead of a looking move; you guessed what a survey text wanted; the question was recitation in disguise; you offered a keyword where a vehicle+move was asked; you think “not a science person” means leave; you think a finished chatbot lab report is already a check; anxiety — shorten, do one honesty card or one tossed-key sticky, return tomorrow. Pose, wait, point at the object.

On the go — vehicle flash (15–20 min)

- **Time:** 15–20 min
- **Materials:** weather page / one produce item / memory of safe toss
- **Safety:** ordinary; store etiquette; no sun stare
- **The move:** Name move + vehicle. One Observe or Feed-hygiene crumb.
- **Exit:** Move+vehicle sticky
- **Refuse:** Survey dump on the train

Desk — BIO-02 pot as system (20–40 min)

- **Time:** 20–40 min (+ optional micro check-ins later)
- **Materials:** plant; notebook

⁷⁷Rowe (1972) wait-time; Stahl (1994) think-time — classroom origin. Adult translation: pause-before-lookup / pause-before-feed. Not an adult science RCT.

- **Safety:** gloves if sap irritant
- **The move:** SYS boundary. Optional OBS across days. Claim+limit on one factor.
- **Exit:** Boundary + factor claim with limit
- **Refuse:** Photosynthesis cookbook without record

Desk — CHEM picture + SAFE-03 (20–35 min)

- **Time:** 20–35 min
- **Materials:** food-safe melt/dissolve/fizz soft; honesty card; RAMP if needed
- **Safety:** food-safe; eye protection if splash; small volumes; no sealed containers; stop if unsafe
- **The move:** Label not-for-credit. Predict; run or redesign; REC; limit. Write what a supervised lab would add.
- **Exit:** “This is a picture because ____.”
- **Refuse:** Filing as Chemistry credit; drain-cleaner chemistry

Desk — PHYS-01 change-of-motion (15–25 min)

- **Time:** 15–25 min
- **Materials:** soft object; clear space; notebook
- **Safety:** no throwing at people/glass
- **The move:** Observe change of motion. Soft note on force-as-fuel idea without shame.

- **Exit:** Change-of-motion sticky
- **Refuse:** Formula dump first; self-scolding as the product

Desk — EARTH-03 weather-to-climate soft (20–35 min)

- **Time:** 20–35 min
- **Materials:** one snap story + one longer-record figure from a reputable agency page
- **Safety:** none
- **The move:** Distinguish event from longer pattern evidence. FEED hygiene on the snap.
- **Exit:** Snap-alone cannot show ____ / longer record suggests ____ (limit)
- **Refuse:** One-storm proof/disproof theater; culture-war pamphlet as only object

Weekend — OUT park / museum / sky (30–60 min)

- **Time:** 30–60 min
- **Materials:** notebook; path or exhibit or sky
- **Safety:** public access; venue rules; night sense; ticks
- **The move:** One looking move on Bio or Earth soft vehicle. One claim with limit.
- **Exit:** CER crumb + limit or systems critique
- **Refuse:** Whole-museum bingo; photos-only tourism

Weekend planning — SAFE-05 lab-path neighbor (20–40 min plan)

- **Time:** 20–40 min planning (visit another day)
- **Materials:** local contacts; RAMP for neighbor try
- **Safety:** neighbor rules override home habits
- **The move:** Name what home cannot safely provide. Schedule. Do not pretend the visit finished Chemistry.
- **Exit:** Neighbor plan + credit claims you will *not* make
- **Refuse:** Museum demo day as lab course

Incorrect example to diagnose (labelled composite)

An adult films baking-soda vinegar in a bottle, generates a “gel photo” with a chatbot for the notebook aesthetic, captions “Chem Unit 3 complete,” and pastes a model lab report as the only draft.

Diagnosis: kitchen-as-credit; generated result; write-lab-report as only draft; survey-unit identity.

Repair: Delete gel. Honesty card: picture $\neq$ credit. Keep OBS/REC of the real cup if food-safe and RAMP-green. Ban model report as only draft. If credit is the goal, write SAFE-05 lab-path plan. Open OpenStax *after* your looking try if you want a named section — put it away before the next unaided attempt.

Heat vs temperature soft (labelled continuity)

A weather page says air temperature. A stove claim says “heat.” Adults sometimes meld them. Driver-line school-age work documents heat/temperature compounding; this book names the soft confusion when a stove, fever talk soft, or weather page appears — without inventing an adult RCT from school samples.⁷⁸ Repair:

⁷⁸Driver, *Children’s Ideas in Science* (1985) and alternative-frameworks line: heat/temperature, force/motion, matter — **school-age documentation**. Adult book may name soft confusion; do not invent adult RCT effect sizes from school samples. Cells-as-tiny-mouths continuity labelled

temperature as a reading on a scale soft; heat as energy transfer story soft on PHYS-02 — another look, ordinary words, no shame.

Cells-as-tiny-mouths soft (labelled continuity)

A grocery or textbook diagram tempts “cells eat like tiny mouths.” Continuity from biology misconception literatures and sister science memos — labelled. Repair with systems inputs/outputs (Chapter 4) on a plant or body soft sketch — not vocabulary punishment, not a wet-lab DNA night.

Mixed practice when stable

Pair CHEM honesty with a FEED “kitchen proves” card. Pair PHYS change-of-motion with force-as-fuel soft repair. Pair EARTH weather-to-climate with Chapter 3 lateral hop. Pair BIO pot systems with Chapter 4 omission sticky. Keep denser CER for Chapter 7 — here a light claim with limit is enough.

Life surfaces that carry vehicles (they do not replace looking)

Grocery aisles, plant pots, weather pages, park paths, museum exhibits, safe tosses, and food-safe kitchen pictures *are* the practice when a looking move rides them. They do not replace OBS/SYS/FAIR/FEED structure. Noticing is not a sting. Credit goals use lab-path and Resources — not denser fizz.

If it isn’t clicking

Diagnostic 1 — Coverage itch.

You keep opening survey tables of contents and feeling behind.

Next move: Write one move + one vehicle for today. Put the TOC away. Depth is the hour. Coverage is a map. If credit is the real goal, skip to Resources / lab-path planning — honesty, not more kitchen.

Diagnostic 2 — Credit anxiety / kitchen-as-credit drift.

You want Chemistry credit and keep escalating home mixes.

Next move: SAFE-03 honesty. Stop unauthorized mixing. SAFE-05 lab-path

from biology misconception literatures / sister science memos — repair with systems looking, not shame.

neighbor plan. RAMP or skip. America's Lab Report continuity: picture without idea+talk is weak even in schools; it is not a credit course.⁷⁹

Diagnostic 3 — Prior-idea shame or formula-first freeze.

Force-as-fuel or heat/temperature soft shows up and you either scold yourself or flee into equation packets.

Next move: Another look. Soft label: common idea; TSTS notes Newtonian construction is hard even after intro courses.⁸⁰ Driver-line school-age continuity for heat/temperature — not your personal failure and not an invented adult RCT.⁸¹ Short session: one toss sticky or one temperature-versus-heat sentence. No AP packet as identity.

Correct a wrong reason without crushing the attempt — applied to yourself. “Never correct yourself” is not this book's rule. Hearing “I filed fizz as Chem credit because I needed progress” and answering “progress is honesty + looking move; credit needs a supervised path” is the work. Anxiety: short sessions, early wins, no countdown timer as default rigor — not a therapy manual.

When to go ahead: you can name move+vehicle, label a picture not-for-credit, and run RAMP or skip. When to slow down: every sit becomes a survey guilt spiral or an unsafe mix. When to ask a human: librarian for OpenStax access; community-college advisor for lab-path; clinician for health questions (this book does not diagnose); tutor for supervised practice — not because you failed a science gene test.

⁷⁹National Research Council, *America's Lab Report* (2006): isolated labs often weak for content mastery; integrate look-try-talk; constructed fake data $\neq$ lab experience. Continuity for adults: kitchen picture $\neq$ Chemistry credit; embed light try in idea and talk. Access date for URLs in these notes: 22 September 2026.

⁸⁰National Research Council, *Taking Science to School* (2007): Newtonian construction “by no means obvious even to high school or college students who have had courses in introductory mechanics”; force/motion prior ideas; theory-evidence coordination improves into adulthood but adults still meld theory and evidence. Labelled continuity — not an invented adult RCT for this book.

⁸¹Driver, *Children's Ideas in Science* (1985) and alternative-frameworks line: heat/temperature, force/motion, matter — **school-age documentation**. Adult book may name soft confusion; do not invent adult RCT effect sizes from school samples. Cells-as-tiny-mouths continuity labelled from biology misconception literatures / sister science memos — repair with systems looking, not shame.

Tools, including AI

Point back to the Science Refresh Hour AI box. Keep this short.

You may, after you have tried: explain a named OpenStax / Khan / CK-12 / museum / agency section to yourself (title, page, today’s micro-goal); isomorphic practice with answers on **your** page; labelled SCRIPT for self-questions; revision questions after *your* note or light claim (not rewrite as only draft); hint after attempt; diagnose work you already did (crop to page — no face); absolute/relative prompting after you drafted from named numbers; labelled schematic put away before the next try — never filed as what you saw.

You may not during the attempt: generate gel / micrograph / sky / spectrum as result; invent observation or data table; ask “write my lab report / science op-ed” as only draft; photo-to-key blank worksheets or employer sheets; treat model summary as observation or study; unauthorized mixing instructions as a substitute for RAMP; paste proprietary workplace data into public tools; park unsupervised chatbot as only partner during the attempt; print a science literacy certificate; detector-grade your unaided write-up.

One mechanism sentence, labelled **OFF-AGE for science literacy**: in a high-school math field experiment, an unguarded chatbot made practice look better and left students worse when the window closed.⁸² Not an adult science RCT. Not this science band. Do not mash *Math Refresher for Adults* ON-AGE-adjacent label onto this book. Lunch-break rule: model may prepare you and the next looking page; model may not do looking, measuring, recording, claim, lab report, or science op-ed when those are the lesson.

OpenStax Biology / Chemistry / College Physics / Astronomy and Khan / CK-12 are free digital backbones *after* the try. Podcasts and documentaries are neighbors, not studies. Lab-path neighbors when home cannot minimize. CLEP Biology / Chemistry optional edges — sitting is not finishing this book.

What “done enough” looks like

Before you move on:

⁸²Bastani and colleagues (2025), *PNAS*: Turkish high-school math field experiment. Unguarded chatbot practice gains with unaided losses. **OFF-AGE for science literacy** — not an adult science RCT; do not mash Math Refresher ON-AGE-adjacent label.

- You named **which looking move** rode on **which Bio / Chem / Phys / Earth soft vehicle** this week.
- If you used a kitchen picture, you labelled it **not-for-credit** (SAFE-03) and wrote what a supervised path would add.
- You ran **RAMP** (or skipped / redesigned / lab-path) when heat or chemicals appeared.
- You refused at least one wrong twin: survey dump, kitchen-as-credit, force-as-fuel shame, AP-as-Tuesday, generated result as observation.
- Optional: one OUT park/museum/sky sit or one lab-path plan card.
- You can ask the locked opening and wait.

A science literacy certificate is not the work. Optional CLEP remains optional. Place next work by whether coverage itch still replaces move+vehicle naming — and by whether picture-as-credit still sneaks in.

Life of the habit (seed). Next Tuesday: new vehicle or same vehicle, new move. Space sits — do not mass Bio+Chem+Phys on Sunday.⁸³ Mix a kitchen-honesty refuse into a week that went well so honesty stays sharp. File move+vehicle stick-ies in Keeping track. Chapter 7 will densify claim with warrant and named limit — you will need a vehicle try that actually happened, not a survey guilt pile and not a chatbot CER with no trial.

If this week only yields one produce Observe and one honesty card, that is a real win. Vehicles return one looking ride at a time.

Self-placement probe (skill strip, not diploma year): Can you name move before domain FOMO? Can you label a picture not-for-credit without despair? Can you soft-repair force-as-fuel without shame? Can you point to lab-path when credit is real? If yes, you are ready to deepen claim craft in Chapter 7 and to keep vehicles as carriers. If no, stay here with thinner vehicles and shorter sits. A graduate can still survey-dump. A GED-holder can run a clean pot+Observe week. No diploma required to open this chapter; no diploma finishes it.

⁸³Dunlosky and colleagues (2013): distributed practice high-utility technique rating — not an adult science-refresher programme evaluation. Space vehicle sits across days.

Carry one sticky forward: move + vehicle + (if needed) picture $\neq$ credit. That is enough continuity for denser claim weeks without turning the book into three transcript years.

More desk composites (steal, then put away)

Composite A — Produce structure–function soft. Two produce items. Observe skins, layers, seeds. Hypothesize function. Check one labelled secondary *after*. Exit: structure X may relate to function Y because ____ (limit ____). Vehicle: Bio soft. Move: Observe. Refuse nutrition-war sermon. Refuse anatomy survey.

Composite B — Dilution sense soft. Juice or food coloring series. Two dilutions. Observe. Connect “more dilute” to everyday labels without dosing experiments. Exit: dilution sentence + non-dosing sticky. Vehicle: Chem soft. Move: Observe + light claim. Refuse stock acids.

Composite C — Energy storyboard soft. Phone charging or kettle already in use — observe, do not jury-rig. Storyboard transfers. Ask where energy goes that you cannot see easily (thermal soft). Exit: storyboard + invisible-path question. Vehicle: Phys soft. Move: Systems-ish energy story. Refuse “energy destroyed” without limit talk. Refuse opening sealed devices.

Composite D — Water path soft. Storm drain / rain observation / local watershed map soft. SYS boundary for water path. Human infrastructure as part of system. Exit: water-path sticky. Vehicle: Earth soft. Move: Systems. Refuse culture-war pamphlet as only object. Refuse entering floodwater.

Composite E — Lab-path honesty evening. You want CLEP Chemistry or a transcript Chem. Write SAFE-05: what home cannot minimize; neighbor to call; credit claims you will not make from kitchen night. Exit: plan card. Vehicle: Chem soft as *path planning*, not fizz. Refuse calling a museum demo a lab course.

Week menu (not a calendar)

- Mon: name move + vehicle sticky (micro).
- Tue: BIO pot or produce Observe.

- Wed: PHYS change-of-motion or EARTH weather numbers-before-story.
- Thu: CHEM picture with SAFE-03 *or* skip to OpenStax section *after* a properties table.
- Fri: FEED interloper that treats kitchen as “study.”
- Sat optional: OUT park / museum / sky; or SAFE-05 planning.

One vehicle day is enough. Two domains on one night is coverage itch wearing a lab coat.

Fade ladder for vehicles

Day one: you fill move+vehicle blanks on a template. Day two: blank sticky before any text opens. Day three: honesty card without a sample. Day four: RAMP without a worked list. Day five optional: OUT with one claim limit. Stop looking up when move+vehicle are named before OpenStax. Peek at your sticky, not at a generated gel.

Stranger-readable vehicle card

Date; looking move; vehicle (Bio/Chem/Phys/Earth soft); object; honesty label if picture; RAMP go/redesign/skip/lab-path; one observation or boundary crumb; one limit; what this does *not* earn. If a stranger cannot recover what you practiced, repair the header.

Wrong twins gallery

1. OpenStax TOC as the hour → close TOC; name move+vehicle.
2. Fizz filed as Chem I → SAFE-03; lab-path if credit is real.
3. Force-as-fuel self-scold → another toss look; TSTS continuity soft.
4. Generated sky as EARTH observation → delete; look up (safely).

5. AP FRQ packet as Tuesday identity → put away; vehicle try.
6. Heat/temperature meld on weather page → soft rename; no shame.
7. “I’m not a science person” as exit → shorten session; one Observe win.

Talk lines you can borrow

“Today’s move is ; **the vehicle is** soft.”

“This is a picture because .”

“**I recognize** hazard; I will minimize by ____ — or I skip / redesign / lab-path.”

“A common idea is force-as-fuel; what I notice about change of motion is .”

“**This snap alone cannot show** ; longer record suggests ____ (limit ____).”

“You may change vehicles next week.”

Bridge to denser claim

Chapter 7 will ask for claim + evidence from *this* table or labelled set + warrant + named limit. Vehicles this week should leave a scrap a stranger could point at — OBS bullets, SYS boundary, fair redesign, weather numbers list — not a survey guilt pile and not a chatbot CER with no trial. Keep the scrap.

Why one vehicle beats three survey nights

Framework disciplinary ideas span years in schools.⁸⁴ Adults on the go do not gain literacy by skimming three TOCs on Sunday. They gain it by riding one looking move on one ordinary object until the move is audible in the next grocery aisle and the next feed card. Parallel to *History Refresh for Adults* using civic texts as objects without a Constitution-chain dump: vehicles without survey dump.⁸⁵

⁸⁴National Research Council, *A Framework for K–12 Science Education* (2012): practices, crosscutting concepts, disciplinary core ideas as **map** — not NGSS PE dump as adult weekly spine, not adult programme evaluation.

⁸⁵Continuity: *Science for Emerging Adults* owns ages 15–18 parent Science Hour (systems + denser CER; kitchen honesty). This chapter ages vehicles into adult self-study — no parent/student split, no AP weekly spine. One pointer, then teach. Parallel: *History Refresh for Adults* civic-texts-as-objects without Constitution-chain dump — vehicles without survey dump.

CLEP / OpenStax edges (honest)

If your real goal is credit or a placement exam, say so on a sticky. OpenStax after the try can deepen a named section. CLEP Biology / Chemistry are optional edges with fees named in Resources — sitting is not finishing this book. Kitchen night still does not become Chemistry credit because an exam date exists. Lab-path neighbor when technique must be supervised.

Carry forward one sticky only: looking move + soft vehicle + (if needed) picture $\neq$ credit. That single line is enough continuity for denser claim weeks without converting Tuesday into three transcript years. If coverage anxiety returns, reopen the sticky before you reopen a table of contents.

A last ordinary picture: you stand in a grocery aisle, name Observe, pick Bio soft, notice skin and seeds on one fruit, write a limit, and leave without a nutrition sermon and without a survey table of contents. That is vehicle literacy. Kitchen fizz, if you ever use it, stays labelled not-for-credit. Force-as-fuel soft, when it appears on a tossed key, gets another look — not a character verdict. Lab-path stays available when credit Chemistry is the real aim. You can do this without becoming three transcript years in one busy week. Start small.

Chapter 7

Talking and writing a denser claim



Figure 9: A still-life: four-box sticky labelled Claim / Evidence / Reasoning / Limit beside a notebook table with units and a pencil. No people. No logos. No readable UI.

Why this matters

What do I claim — which evidence from *this* table or labelled set shows it — and what does this claim not settle?

That is not a timed Advanced Placement free-response question. It is not a request for a fluent paragraph a machine could have written. It is not a science op-ed costume. It is the question that turns looking into a sentence another person can check. A claim without locatable evidence is a vibe. Evidence named with no claim is a caption. Reasoning that only restates the numbers is a shrug in fancy clothes. A claim with no limit pretends one table finished the fight. Science talk for an adult on the go is a claim that points. Science writing for an adult on the go is the same claim, slower, on a dated page — with a warrant and a named limit.

You already practiced evidence before feed, observe and record, systems boundary, fair look, and soft Bio / Chem / Phys / Earth vehicles. This chapter is the exit strip for the teaching arc: denser claim craft after looking — not instead of looking. Writing here is claim + evidence from *this* table or labelled set + reasoning via a scientific idea (warrant) + named limit or alternative. It is not a weekly AP free-response dump. An occasional weekend multi-source stretch is enough when paragraphs are stable.⁸⁶

What this idea unlocks is usable science speech and a short written claim you could almost show a stranger. On the commute you can point to a row, name a limit, or wait before the feed. At a desk you can fill four boxes, underline the idea in the reasoning, and refuse a chatbot CER filed as your work. On a weekend you can weave two labelled evidence surfaces into a short argument with a warrant and a limit — still not a five-paragraph costume.

This is worth the struggle because ordinary adult life rewards fluency and punishes pointing. You can talk for five minutes about a workplace “study says” slide without touching a number. A feed card finishes the moral for you. A chatbot offers a tidy CER that feels like homework done. Looking without a short claim leaves you fluent at noticing and silent at deciding what the evidence can support. A short claim without looking is theater. The hour needs both — in that order.

⁸⁶AP Biology / Chemistry / Physics / Environmental Science free-response culture is a high-school exam object. This book refuses AP FRQ dump as the adult weekly spine. Occasional weekend multi-source stretch (WRITE-05) is enough when four-box paragraphs are stable. Access date for pathways named in these notes: **22 September 2026**.

You do not need to be a scientist. You do need to hear a claim with no locatable evidence, a chatbot CER with no trial, a warrant that only restates the numbers, a missing limit, and AP FRQ as weekly identity — and to ask the locked question, then wait. Hearing those wrong turns is the work. Filling the silence with a perfect paragraph from a window is not.

Picture two Fridays. In one, the notebook holds a tidy “study shows” paragraph that could have come from any feed card — or from a chat window. In the other, a short paragraph names *this* table or labelled set, points at a row or figure, warrants why that row counts through a scientific idea in ordinary words, and admits what the claim does not settle. Both pages look like writing. Only one is science looking on paper.

If you used *History Refresh for Adults*, you already practiced claim + named source + warrant + limit on civic pieces. One pointer — then science: evidence is a row, a figure, a labelled abstract crumb, or a stranger-readable table you filled — not a credit line alone.⁸⁷ McNeill and the IQWST CER frame aged for adult self-study are the spine here: claim, evidence, reasoning — and this book’s denser exit adds a named limit.⁸⁸ That is a useful study pattern, not a promise that every adult will see the same result. Then teach.

This week you can learn to fill four boxes from *this* evidence, speak a thirty-second CER+limit before you write, press yourself with “where in the data?”, and name what would make you change your mind. Today you can write: “I claim ____ because [row / figure / labelled set] shows . ***This counts because [scientific idea in ordinary words]. This claim does not settle .***” A path through ordinary adult science life includes this strip. A fake certificate does not. Optional CLEP sits later if credit is your goal. For now: one good question, one pointer into the table, one cautious sentence.

Looking without a claim evaporates by Tuesday. A claim without a limit overreaches by Wednesday. Denser CER is the adult exit product of this book’s teaching arc — usable on a workplace slide, a science-news card, a museum exhibit

⁸⁷*History Refresh for Adults* holds claim + named source + warrant + limit on civic pieces — one pointer, then science. Evidence here is a table row, figure, labelled abstract, or stranger-readable measurement — not a credit line alone. Do not mash Wineburg sourcing into CER.

⁸⁸McNeill / IQWST CER frame (claim, evidence, reasoning) — middle-school curricular origin labelled; adult denser exit adds a named limit / alternative. Fade scaffolds. Useful pattern; not a promise every adult will see the same result. Solo check: cover the evidence; does the claim still stand?

note, or a park transect table. Money, workplaces, and civic life motivate. They do not replace the looking move.

Denser claim craft is also how you protect a fair look and a systems boundary from evaporating into slogans. Chapter 5 asked whether you can tell for sure. Chapter 4 asked where the boundary is. This chapter asks you to *say* what those looks support — and to stop the sentence before it claims the world. The warrant is the bridge: without it, the numbers float; without the limit, the bridge pretends to reach farther than the evidence built.

Understand it yourself

Everyday picture. You have an opinion about a weather page, a grocery structure–function look, a workplace “study says” slide, or a park transect. You can talk for five minutes. Someone asks, “Which row shows that?” You gesture at the air. The adult repair is CER-02 and TALK-03: finger or cursor on the supporting row or figure; if you cannot point, demote the claim to a question. Then CER-01: four boxes — claim, evidence, reasoning, limit. Then CER-03: underline the scientific idea in the reasoning (not a restatement of the numbers). Then CER-05: what result would make you change your mind.

Another everyday picture: a chatbot offers a fluent CER paragraph about “what the data mean.” You paste it into your note as your draft. The refuse is WRITE-style honesty: strike any sentence that only rests on the model; replace from *your* table or labelled set; redline. The model may ask revision questions **after** your draft. It may not be the only draft.⁸⁹

Precise picture. A *claim* here is a sentence about the natural or measured world that could be wrong and that *this* evidence can speak to. *Evidence* is locatable — a row, a cell, a figure panel, a labelled secondary with an honesty card, a stranger-readable table you filled — something another person can find. A *warrant* (reasoning) ties the evidence to the claim through a scientific idea in ordinary words:

⁸⁹Science Refresh Hour AI box (adult-held): unaided first; revision questions after *your* draft; ban write-my-lab-report / write-my-science-op-ed as the only draft; refuse model summary as observation or study; ban invent-data / generated gel as result. COPPA under-13 does not apply — not a companion licence.

why *this* row counts for *that* claim (rate, conservation soft, system boundary, sample vs population, absolute vs relative) — not a restatement of the numbers. A *limit* is one honest sentence about what this evidence cannot tell or what the claim does not settle (sample, design, time, place, measurement uncertainty, alternative explanation). *Competing claims, same data* (CER-04): list supporting and opposing rows for each rival; prefer better fit + clearer limit — or “tie / insufficient.” *Falsify* (CER-05): write what would make you change your mind. *Oral then written* (CER-06): speak thirty to forty-five seconds, then write; notice what writing forced you to fix.

Talk is not a luxury before writing. Authentic questions — questions where you have not already scripted the answer for yourself — are the spirit of the talk box.⁹⁰ Materials do not discuss themselves. Wait before the feed when the urge to share or look up arrives (TALK-01, TALK-02). Stuck silence gets a diagnosis, not forty minutes of scrolling. Accountable to the evidence soft (TALK-03): where in the data? If nothing could change your mind, you are not doing denser CER this hour.

A composite illustration (labelled as such — not a reported workplace): you meet a slide that says “the study shows a large improvement.” Your first draft is a three-paragraph rant about management. The adult sequence is different. Open the named abstract or table the slide gestures at (if public). CER-01 four boxes. Star the row that actually speaks to the improvement. CER-02 pointer. CER-03 — why that row counts (absolute numbers, sample, design crumb). CER-05 / limit — what the claim does not settle (relative-only framing, confound, population not yours). Optional CER-04: a rival claim that still fits some rows. You still may disagree with the workplace use of the slide. You first refuse vibe citations and model summaries as the floor.

Without a teacher, adults skip warrants (“the numbers are obvious”) and skip limits (“everyone knows”). CER-03 and CER-05 exist because *you* are the easy audience to con.⁹¹ Practice alone is not enough if practice means rewriting the same summary. The solo check stays: cover the evidence; does the claim still stand with no pointer? If yes, the table was decoration.

⁹⁰Authentic / accountable talk neighbors (via sister streams): questions without a prescribed recitation answer; debt to the evidence (“where in the data?”). Borrow as craft. Do not treat professional-development checklists as an adult science RCT.

⁹¹McNeill / IQWST CER frame (claim, evidence, reasoning) — middle-school curricular origin labelled; adult denser exit adds a named limit / alternative. Fade scaffolds. Useful pattern; not a promise every adult will see the same result. Solo check: cover the evidence; does the claim still stand?

Wrong answers you should be able to hear

1. **Claim without locatable evidence.** “It works.” No row, no figure, no labelled set. Repair: “Which evidence from *this* table shows it?” Point or demote to a question (CER-02).
2. **Chatbot CER with no trial.** Fluent four-box voice; précis filed as your measurement or as the week’s evidence. Repair: close the window; write from the table in view; tool only after a real draft, as revision questions — not a rewrite (WRITE refuse; Hour AI box).⁹²
3. **Reasoning that only restates the numbers.** “This supports my claim because the number is higher.” Repair: “What scientific idea links *that* row to *that claim*?” Underline one honest idea or mark “idea not yet earned” (CER-03).
4. **No limit / overreach.** “This proves it for everyone.” Repair: name sample, design, time, place, or alternative — one sentence (CER-05; WRITE-03 limit-first).
5. **AP FRQ as weekly identity / write-my-lab-report as practice.** Timed free-response every Wednesday; multi-document costume; “write my lab report / science op-ed” as the only draft. Repair: WRITE-01 / WRITE-05 as occasional weekend stretch when four-box paragraphs are stable; refuse AP FRQ dump as spine; ban write-lab-report as only draft.⁹³

A sixth worth hearing: **evidence salad** — three copied numbers and no sentence of your own. Repair: WRITE-01; claim; pointer; warrant; limit — then one paragraph.

Five-minute warm-up

Before any try-it, cover this page and work aloud:

⁹²Science Refresh Hour AI box (adult-held): unaided first; revision questions after *your* draft; ban write-my-lab-report / write-my-science-op-ed as the only draft; refuse model summary as observation or study; ban invent-data / generated gel as result. COPPA under-13 does not apply — not a companion licence.

⁹³AP Biology / Chemistry / Physics / Environmental Science free-response culture is a high-school exam object. This book refuses AP FRQ dump as the adult weekly spine. Occasional weekend multi-source stretch (WRITE-05) is enough when four-box paragraphs are stable. Access date for pathways named in these notes: **22 September 2026**.

- Point to one row or figure in a table you already hold. Write one claim that would *weaken* if that row vanished (CER-02).
- Fill four boxes on a sticky: Claim / Evidence / Reasoning / Limit (CER-01).
- Underline the scientific idea in the reasoning — or mark “idea not yet earned” (CER-03).
- Add one falsify sentence: “I would change my mind if ____.” (CER-05).
- Write the locked opening on a sticky. Cut any second dataset you do not need for a first claim.

That is enough for tomorrow’s micro-session.

How to try it this week

Use the Science Refresh Hour shapes you already have. This week’s idea is talk and a denser claim after looking. Struggle before rescue. Try unaided first. Knowledge first: know claim / evidence / warrant / limit well enough to hear vibe citations, then practice on tables and labelled sets you already meet.

Session shape across bands

Commute / lunch (about 15–20 minutes).

Warm-up retrieval (3–4 minutes): one claim stem; one pointer stem; one limit stem.

Short model (3–4 minutes): a solved four-box crumb under a table you already hold — then cover it.

Your attempt (5–6 minutes): CER-02 point to the row, CER-05 falsify/limit, TALK-01 wait three seconds, or TALK-02 cover the feed before a CER crumb.

One good self-question: the locked opening; wait a slow three.

Mixed glance (2 minutes): cover the evidence — does the claim still stand? If yes, the table was decoration.⁹⁴

Exit sticky (1 minute): “I claim ____ because [pointer]; this does not settle ...”

⁹⁴McNeill / IQWST CER frame (claim, evidence, reasoning) — middle-school curricular origin labelled; adult denser exit adds a named limit / alternative. Fade scaffolds. Useful pattern; not a promise every adult will see the same result. Solo check: cover the evidence; does the claim still stand?

Desk or shift (about 25–40 minutes).

Warm-up mixed from last week’s vehicle or fair-look strip plus one pointer (5 minutes).

Look first (8–12 minutes): open the table or labelled set; short observe/record if needed; star one row.

Short model / fade a step (5 minutes): CER-01 four-box + CER-03 warrant; cover the model and try yourself.

Two tries plus one interloper (10–12 minutes): one clean CER+limit; one limit-first rewrite (WRITE-03); one interloper that is a chatbot CER tempting you to file it as yours (refuse).

Self-question + wait + written reason (3 minutes).

Exit ticket: claim / locatable evidence / reasoning idea / named limit.

Weekend deep sit (about 45–75 minutes when needed).

Diagnostic wrong twin (10 minutes): claim without locatable evidence — repair from a starred row.

Longer write (20–30 minutes): WRITE-01 one-paragraph CER+limit *or* WRITE-05 weekend mini-argument (two evidence surfaces, ~400–700 words soft) with warrant and limit.

Competing claims or falsify (10 minutes): CER-04 same data, two rivals *or* CER-05 falsify sentence.

Revision pass (10–15 minutes): oral then written (CER-06) on *your* draft — notice what writing forced you to fix; one counter soft if available.

Done-enough checklist.

Schedule three micro cues for next week (pointer; warrant idea; limit sentence).

Spacing across days beats a Sunday dump.⁹⁵

Exact wording you can say to yourself

When you are about to summarize, share, or “just write”:

“What do I claim — which evidence from *this* table or labelled set shows it — and what does this claim not settle?”

Then wait. Cover any answer you drafted. Look at the table, the figure, or the labelled abstract — not at the feed or a chatbot précis. After you answer, ask one follow-up from the box below. Then try a parallel item.

⁹⁵Dunlosky et al. (2013): practice testing / distributed practice high utility — technique ratings, not an adult-science RCT. Spacing across days is honest ambition.

Other lines that help:

- “Point to the row — or demote the claim to a question.”
- “This counts because [scientific idea in ordinary words].”
- “This claim does not settle ____.”
- “What result would make me change my mind?”
- “What else could explain these numbers?”
- “I do not need a pretty paragraph. I need a pointer, a warrant, and a limit.”

First try-it this week

On the go — evidence must be locatable (15 minutes).

Materials: blank note or sticky; one table, figure, or labelled public abstract crumb you already hold.

Safety: public or labelled composite; no proprietary employer paste into a public tool; no live controversy as the object; pharmacy labels stay units-only — no medical advice.

The move: Write a claim. Point to the row/figure/sentence. If you cannot point, demote to a question (CER-02). Exit: pointer or demotion. Refuse “studies show” without a study.

Deeper sit — four-box CER+limit (30–40 minutes).

Materials: one notebook table or labelled figure; four-box template; paper.

Safety: as per data source; RAMP if your own try involves heat or chemicals; kitchen pictures labelled not-for-credit.

The move: Fill Claim / Evidence / Reasoning / Limit (CER-01). Underline the scientific idea in reasoning (CER-03). Add falsify sentence (CER-05). Optional: speak thirty seconds first, then write (CER-06). Exit: complete four-box + one paragraph (WRITE-01). Refuse claim without locatable evidence; refuse skipping limit.

How to fade help

First week of this strip: full frame — “I claim ____ because [pointer] shows . ***This counts because*** . This claim does not settle ____.”

Next: blank sticky — you supply the four parts.

Then: one spoken CER+limit, plus a written limit, no frame.

Stop looking up when you can refuse a vibe citation and write a limit without peeking at this chapter’s wording. When a tool is tempting mid-attempt, point

back to the Science Refresh Hour: unaided first; revision questions after *your* draft only.

Talk-to-yourself box

Opening (locked):

“What do I claim — which evidence from *this* table or labelled set shows it — and what does this claim not settle?”

Follow-ups:

1. What do you mean by the claim — can you say it in one sentence? (*meaning*)
2. What does the table / figure / labelled set show before your interpretation? (*representation*)
3. What else could explain these numbers — is there a rival claim that still fits some rows? (*alternative*)
4. What can this evidence *not* tell me — what does the claim not settle? (*check*)
5. You may change your mind — where could someone go wrong with a decorative number or a chatbot CER? (*revise / diagnose*)

How to wait alone. After you ask yourself, cover the answer or the feed. A slow three is a convention, not a sacrament — classroom-origin think-time translated as pause-before-lookup / pause-before-feed.⁹⁶ Look at the table, the boundary sketch, or the clock — not at the phone’s answer key. Rescue at one second is still rescue. Abandonment for forty minutes of scrolling is not a micro-session.

Stuck silence usually means: the question was vague; wait-time 1 was zero; you are searching for a vocabulary word; you are guessing what a textbook would want; the question was a recitation item in disguise; you offered a keyword where a looking move was asked for; you think “science person” means leave; you think a finished-looking chatbot CER is already a check; anxiety is eating working memory — shorten the session, do one pointer sticky or one absolute-risk sticky from

⁹⁶Stahl (1994) think-time / Rowe (1972) wait-time — classroom-origin; adult translation = pause before lookup / before feed. A slow three is a convention, not a sacrament. Not an adult science RCT.

earlier chapters, come back tomorrow. Your job is to pose, wait, and point back at the evidence — not to fill the silence with the right answer from a model summary.

Practice that actually builds it

Named try-its below. Banks are menus, not calendars. Pick by life demand. At least one on the go and one deeper sit this week.

On the go

CER-02 — Evidence must be locatable (15–25 min).

Materials: a draft claim you wrote earlier.

Safety: none beyond privacy of your note.

The move: Point to row/figure/sentence. If you cannot point, demote claim to question. Exit: pointer or demotion. Refuse “studies show” without a study.

CER-05 — Falsify prompt (15–20 min).

Materials: your current claim.

The move: Write what result or evidence would make you change your mind. Exit: falsify sentence. Refuse unfalsifiable vibe claims as science.

TALK-01 — One question, three seconds (10–15 min).

Materials: timer; one notebook page.

The move: Pose the locked opening. Wait 3–5 s. Write. Wait again. Edit once. Exit: timed sticky. Refuse instant lookup.

TALK-02 — Cover the feed / model (10–15 min).

Materials: paper cover over phone browser.

The move: Physical cover. Attempt CER crumb. Then uncover to check. Exit: attempt vs checked difference. Refuse “just checking quickly” mid-wait.

WRITE-04 — Caption rewrite (15–25 min).

Materials: misleading science-feed caption + evidence surface.

The move: Rewrite caption to match evidence + limit. Exit: honest caption. Refuse snark-only rewrite without accuracy. Ordinary stakes only — no culture-war pamphlet as the object.

Desk or shift

CER-01 — Four-box build (20–35 min).

Materials: one notebook table or labelled figure; four-box template.

Safety: as per data source.

The move: Fill four boxes. Reasoning must name a scientific idea in ordinary words. Limit is mandatory. Exit: complete four-box. Refuse claim without locatable evidence; refuse skipping limit.

CER-03 — Reasoning names the idea (15–25 min).

Materials: CER-01 draft.

The move: Underline the scientific idea in reasoning (not a restatement of the numbers). If missing, add one honest idea or mark “idea not yet earned.” Exit: underlined idea or honesty mark. Refuse fancy words that do not link.

CER-06 — Oral then written (15–25 min).

Materials: voice memo optional; then paper.

Safety: privacy if recording.

The move: Speak a 30–45 s CER+limit. Then write. Notice what writing forced you to fix. Exit: written CER+limit. Refuse AP rubric theater as weekly spine.

WRITE-01 — One-paragraph CER+limit (20–35 min).

Materials: four-box notes → paragraph.

The move: Convert boxes to one paragraph. Limit must appear. Exit: paragraph. Refuse AP FRQ length as weekly requirement.

WRITE-02 — Methods crumb sentence (10–20 min).

Materials: your try.

The move: Write what you did in 2–4 sentences a stranger could repeat roughly. Exit: methods crumb. Refuse invisible procedure.

WRITE-03 — Limit-first draft (15–25 min).

Materials: overconfident draft claim.

The move: Rewrite starting with the limit, then the claim. Exit: limit-first paragraph. Refuse hype verbs (proves/destroys) without warrant.

TALK-03 — Evidence press on yourself (10–20 min).

Materials: draft claim.

The move: Ask “where in the data?” aloud. Answer only with a pointer. Exit: pointer or demotion. Refuse circular “because it’s true.”

Weekend deep sit

CER-04 — Competing claims, same data (25–40 min).

Materials: one dataset/table; two rival claims.

The move: For each claim, list supporting and opposing rows. Prefer the claim with better fit + clearer limit. Exit: winner with limit — or “tie / insufficient.” Refuse declaring a liar; refuse culture-war framing.

WRITE-05 — Weekend mini-argument (45–75 min).

Materials: two labelled sources or your try + one secondary.

Safety: as needed; RAMP if heat/chemicals; honesty card on kitchen pictures.

The move: Short argument: claim, both evidences, reasoning, limit, one counter soft. Exit: 1–2 page max or 400–700 words soft. Refuse multi-document timed AP as identity.

TALK-05 — Disagreement without contempt (20–30 min).

Materials: two science accounts that differ.

Safety: emotional — skip outrage bait; ordinary stakes.

The move: Steel-man each. Name evidence each uses. Refuse contempt language. Exit: difference map without villain sticky. Refuse culture-war seminar.

Talk-to-yourself box (practice reuse)

Reuse the locked opening and the five follow-ups from “How to try it this week.” Wait alone the same way. If stuck silence returns, shorten: one pointer, one limit, stop. Come back tomorrow.

One incorrect example to diagnose

Incorrect (composite illustration — not a reported workplace):

Claim: “The new schedule improves focus for everyone.”

Evidence: “Studies show it.”

Reasoning: “Because the slide said improvement.”

Limit: (blank)

What went wrong: no locatable evidence; reasoning restates the slogan; no limit; feed/slide treated as the study; ready to paste a chatbot polish as the draft.

Repair: Open the named table or abstract if public. Point to a row (CER-02). Fill four boxes (CER-01). Underline a scientific idea — e.g. sample size, absolute vs

relative change, or what else differed (CER-03; fair-look crumb from ch05). Name a limit: population, design, time window, or “insufficient for everyone” (CER-05). Wait before the feed (TALK-02). If a tool helps, only after *your* draft — revision questions, not rewrite.

Mixed practice when the move is warm

Mix last week’s honesty card or fair-look sticky with this week’s four-box. Insert one interloper: a fluent model CER with no table underneath. Your job is to refuse it as your work, then write from evidence you can point to. Weather pages, grocery aisles, and feeds *are* the practice surfaces — they do not replace the looking structure.

Fade the scaffold without dropping the limit

Week one of denser CER often needs the full frame on the sticky. That is not weakness. The frame is training wheels for a move that workplaces and feeds reward you for skipping. When the four parts appear without peeking, fade: blank sticky, then spoken CER+limit, then a one-paragraph WRITE-01 with the limit still mandatory. What you never fade is the pointer and the named limit. A fluent paragraph that cannot point and cannot name what it does not settle is still a vibe — prettier, not denser.

If you notice yourself writing “studies show” again, stop mid-sentence. Demote. Ask for the row. The adult repair is boring on purpose. Boring is how the looking move survives a busy Tuesday.

Transfer surfaces (ordinary stakes only)

Use the same four-box on surfaces you already meet — not on a culture-war pamphlet and not on a medical-advice spiral:

- A weather page figure: claim about a trend crumb; pointer to axes; warrant about rate or pattern; limit about one station or one week.
- A grocery structure–function look: claim about a part–job soft observation; pointer to what you saw; warrant in ordinary systems language; limit — not a nutrition sermon.

- A museum exhibit systems postcard: claim about boundary or flow; pointer to the labelled diagram; warrant; limit — exhibit is a model that leaves things out.
- A park transect table: claim about one factor you recorded; pointer to your rows; warrant; limit — sample of stops, not the whole park.
- A public absolute-vs-relative risk sticky from a named figure: claim restated in absolute terms; pointer to the numbers; warrant; limit — population and design not fully on the card.

Pharmacy labels stay units-only. No dosing. No live controversy as the Hour object. Composite workplace slides stay ordinary stakes and redacted — never a sting, never proprietary paste into a public tool.

If it isn't clicking

Three diagnostics. No shame. Correct a wrong reason without crushing the attempt — applied to yourself.

1. Blank page, then phone.

You open a blank CER frame and immediately open a chatbot or the feed.

Next move: Physical cover (TALK-02). Write anything honest for sixty seconds — even “I do not know which row matters.” Then CER-02: demote to a question if needed. One pointer sticky is a win. Shorten the session. Anxiety eats working memory; early wins beat a countdown timer as “rigor.”

2. “I already know” too fast.

You write a confident claim without opening the table.

Next move: Ask for the row. Cover the claim. Point. If you cannot point, demote. Run CER-05 falsify once. Speed without a pointer is performance, not denser CER.

3. Pretty paragraph, missing warrant or limit.

You can write fluently, but the reasoning only restates numbers, or the limit is missing.

Next move: WRITE-03 limit-first. CER-03 underline the idea — or mark “idea not yet earned.” Delete fancy words that do not link. Optional: speak thirty seconds (CER-06) before rewriting — speech often exposes the missing idea.

When to slow down. If every claim needs a second dataset and a weekend essay, you may be chasing AP theater. Stay with four boxes and one paragraph until the limit appears without peeking.

When to go ahead. If four-box + limit is stable on commute and desk objects, try WRITE-05 once on a weekend — still not weekly AP FRQ.

When to ask a human. A colleague who already writes clear science memos, a librarian who can help you find an abstract, a tutor, a community-college writing center, or a supervised lab-path neighbor when your claim needs a technique home cannot minimize. You still own the looking question. “Never correct yourself” is not this book’s rule — hearing your own wrong reason, then asking a better question, is the work.

Tools, including AI

Optional helpers. Keep this short. Point back to the Science Refresh Hour AI box.

After you have tried, you may:

- Explain this claim structure *to yourself* from a named lesson / OpenStax page / study abstract (title, page, today’s micro-goal).
- Extra isomorphic practice with answers on **your** page (blank CER frames + your key).
- A labelled SCRIPT after you tried — a short spoken or written sequence for *you* to say, not a paragraph to copy as your lab note or science op-ed.
- Revision questions after *your* claim / lab note / science op-ed / science memo draft — not a rewrite as the only draft.
- A hint after an attempt. A hint is not the finished paragraph.
- Diagnose work you already did (crop to paper or note — no face).
- Discussion questions you vet and ask yourself.
- Absolute vs relative risk prompting **after** you drafted a risk claim from named numbers.

You may not during the attempt:

- Paste tonight’s worksheet, blank data table, lab packet, or employer sheet and ask the tool to complete it (photo-to-key).

- Ask “write my lab report / science op-ed” or “fix to publishable” as the only draft.
- Invent an observation or a data table, or generate a gel / micrograph / sky photo / spectrum / “lab photo” and treat it as a result.
- Put a model summary on the table as the observation or as the study.
- Park an unsupervised chatbot as your only partner during the attempt.
- Detector-grade your own unaided write-up.⁹⁷
- Invent research or trust a fluent citation without opening the paper.⁹⁸
- Paste proprietary employer data into a public consumer tool.
- Print a certificate of science literacy from a chatbot.

One mechanism sentence, labelled OFF-AGE for science literacy: in a high-school math field experiment, an unguarded chatbot made practice look better and left students worse on the test with the window closed.⁹⁹ **Not an adult science RCT. Not this science band.** (*Math Refresher for Adults* may call Bastani ON-AGE-adjacent for math — do not mash.) The lunch-break rule it supports: the model may prepare you and the next looking page; it may not do your looking, measuring, recording, claim, lab report, or science op-ed when those are the lesson.

Labelled OpenStax / Khan / museum / agency pages check or explore **after** the try; they replace thinking when they finish the attempt. Photo-to-key stays banned during the attempt.

What “done enough” looks like

Short looking-move checklist before moving on. Placement by skill, not diploma year. A “science literacy certificate” is a receipt this book never invents — and is not the work.

⁹⁷Liang et al. (2023): detector false-positive rate **61.22%** on human TOEFL essays — adult-side caution against detector-grading your own unaided write-up.

⁹⁸Walters & Wilder (2023): GPT-3.5 ~**55%** / GPT-4 ~**18%** fabricated citations in tested conditions — open the source; do not trust fluent citation lists.

⁹⁹Bastani et al. (2025), *PNAS*: Turkish high-school math, grades 9–11, nearly 1,000 students; unguarded chatbot patterns reported in that study — **OFF-AGE for science literacy**; not an adult science RCT. Do not mash *Math Refresher for Adults* ON-AGE-adjacent label. Lunch-break rule only.

You are done enough with this chapter's strip when you can honestly check most of these:

- ☐ I wrote one claim that points to locatable evidence from *this* table or labelled set.
- ☐ I pointed to a row / figure / sentence the claim depends on — or I demoted the claim to a question.
- ☐ I wrote one reasoning sentence that names a scientific idea in ordinary words (not a restatement of the numbers).
- ☐ I named one limit or alternative (what this claim does not settle).
- ☐ I waited before the feed or chatbot at least once this week (TALK-01 / TALK-02).
- ☐ I refused a chatbot CER / model summary as my only draft at least once.
- ☐ Optional: I ran competing claims on the same data (CER-04) or a falsify sentence (CER-05).
- ☐ Optional weekend: one multi-source paragraph or mini-argument with warrant and limit — still not weekly AP FRQ.
- ☐ My exit sticky could almost be shown to a stranger: claim / pointer / warrant / limit.

Life-of-the-habit tip (not a ninth teaching chapter). Revisit next Tuesday: one CER-02 pointer on the commute; one four-box at the desk; space across days.¹⁰⁰ Mix last week's table. Stop while a look remains. Dated denser CER notes and looking-move strip logs live in Chapter 8. Optional CLEP or college survey edges are bridges when credit is the goal — not the spine, and never titled Science Refresh for Adults I.

Self-placement probe (skill strip, not diploma year): Can you cover a feed until design and absolute risk are named, open a table before a caption, hold a rival claim for one sentence when prompted, and name one limit? If yes on most days this week, the teaching arc's exit strip is moving. If not, stay with the missing move — often CER-02 or CER-05 — before you chase a longer essay.

Work product a stranger can read: a dated denser CER with locatable evidence, warrant, and named limit — inside a skill title a stranger understands (the looking move, the life object, or a college course name if credit is the path). Never Science Refresh for Adults I. Never a fake certificate of science literacy. Optional CLEP

¹⁰⁰Dunlosky et al. (2013): practice testing / distributed practice high utility — technique ratings, not an adult-science RCT. Spacing across days is honest ambition.

score report belongs only if you chose that bridge and an institution recognizes it — sitting the exam is not finishing this book.

You do not need to be a scientist. You do need to hear a claim with no pointer, a pointer with no claim, and a fluent CER offered as your work without a trial — and to ask the locked question before you nod. You can do this on the commute and in a deeper sit.

Chapter 8



Figure 10: Dated keeping-track file with looking-move strip, honesty card, denser CER — still-life, no diploma seal, no people, no logos.

Keeping track

There is often **no** “Science Refresh for Adults I” as a job title, and this book will not invent one. A stranger — a colleague who missed the huddle, a tutor you might hire later, a college registrar, a future you on a Tuesday — already reads work they understand: a dated looking note, a stranger-readable table with units, a feed honesty card, a denser CER with warrant and named limit, a skill named in ordinary language, an optional official score report. Never Science Refresh for Adults I. Never Emerging Adults Science I. Never Certified Science Literacy. Never a fake certificate printed by a chatbot.

This chapter is the file you keep. The teaching chapters rebuilt looking moves. Here you place yourself by the looking-move strip, log what you practiced, leave a dated line someone else could almost follow, and — only if credit is your goal — decide honestly whether an optional CLEP Biology or Chemistry sitting is worth the fee.

Life of the habit lives here as *trace*, not as a ninth teaching chapter. Same kind of object next Tuesday, new date. That is enough.

What you are making

You are making a **looking-move strip log**, a **self-placement habit**, and a **dated looking note** (often with a stranger-readable table, a feed honesty card, or a denser CER a stranger could follow). Stacked over months, that file is the honest adult work product. You are not making a national diploma. You are not making a teen Records transcript. You are not manufacturing Carnegie units unless you separately enroll in a credit course or sit CLEP for an institution that asked for it.

Titles a stranger already understands:

- the **looking move** (evidence before feed; observe→record; systems boundary; fair look; denser CER + limit; Bio/Chem/Phys/Earth soft as vehicles)
- the **life / work artifact** (weather check; grocery look; museum systems; workplace study-says honesty; park transect; pharmacy label rate soft — units only)

- a **college course name** only if the work maps (Biology survey; Chemistry survey; College Physics; Astronomy)
- an **optional CLEP** exam name if you sat it and the receiving college cares (CLEP Biology; CLEP Chemistry)

Combining objects is allowed. Combining does not mint a franken-title. An OpenStax chapter plus a museum exhibit note plus a dated denser CER is still the looking-move file — not “Online Science Refresh for Adults I.”

Here is an illustration, labelled as such, not a reported workplace or clinical case:

22 September 2026. Band: desk. Object: workplace “study says” slide (public abstract title stub; no proprietary paste). Move: evidence before feed, then denser CER. Honesty card: labelled secondary (press/slide), not my measurement. Claim: the absolute change in the abstract’s table is smaller than the slide’s relative framing suggested. Evidence: Table 2, rows for baseline and follow-up rates. Warrant: absolute vs relative risk framing. Limit: I have not opened the full methods; sample may not match our workplace population. Minutes: 30. Refusal log: did not file a chatbot CER as my draft. Next cue: same abstract — methods crumb Tuesday.

That is readable by a tutor who was not in the room. It is not a badge. Place by looking move, not by brand.

The looking-move strip log

Keep a simple table or checklist you will actually reopen. Columns that earn their keep:

	Band							
	(com-							
	mute /							
	desk /	Object						
	week-	(feed						
	end /	card /						
	mu-	table /						
	seum /	ex-	One		Honesty			
	library	hibit /	claim		/			
	/ na-	park /	Move	One	refusal		Next	
Date	ture)	weather.	+ typed	pointer	limit	note	Minutes	cue

Looking moves this book uses (match the teaching chapters):

1. Evidence before feed (design / absolute risk / sample / limit — or your own table — before the caption)
2. Observe and record (honesty card; stranger-readable note with units)
3. Feed vs study (caption is not the study; lateral look soft)
4. Systems looking (boundary; inputs/outputs; what the model leaves out)
5. Fair look (“can we tell for sure?”; confound vs fair; insufficient evidence valid)
6. Vehicles soft (Bio / Chem / Phys / Earth carry one looking move — not survey dump; kitchen $\neq$ Chemistry credit)
7. Denser CER + warrant + named limit
8. Survey / CLEP edges (optional — breadth only when credit is the plan)

You do not need every move every week. A shift worker may live in evidence-before-feed, one Observe line, and one claim sentence on commute cards. A returning student may add systems, fair look, and denser CER deep sits. Placement is by life demand and looking skill, not by a podcast that said “everyone needs AP

Bio.”

Rebuild week vs maintenance week. A rebuild week opens a rusty or new move: more honesty cards, a deeper sit, blocked practice before mixing. A maintenance week revisits last month’s feed check or park note with three micros and one mixed interloper — no heroics. Adults who only rebuild and never maintain will re-rust. Adults who only maintain and never rebuild will not open the next gate. Name both in the log (“R” or “M” in the Next cue column is enough).

Invent-data / generated-result refusal log. Keep a short column or sticky when you refuse: inventing a row; treating a generated gel / micrograph / sky photo as a result; filing a model summary as the observation or as the study; pasting proprietary employer data into a public tool. The refusal is part of the work. It is not a confession booth.

Life of the habit, here: the note is how the habit leaves a trace. “I used to treat the feed caption as the study; now I open the table first” is growth. It is not a diploma ceremony.

Self-placement by looking move (not diploma year)

A college graduate can still treat a headline as the study. A GED-holder can source an abstract cleanly. **No diploma was required** to open this book, and no diploma year places the next module.¹⁰¹

Use short probes — mirrors, not oracles. Vendor placement tools (OpenStax chapter previews, Khan practice, CLEP practice guides, community-college placement, commercial adaptive platforms) place by content familiarity; they are not IES-certified RCTs of this book.¹⁰² NSF Indicators / Pew process items and PIAAC literacy/numeracy describe a national stock. They are not your pretest, not your posttest, and not a grade for your notebook.¹⁰³ Teen NAEP Science and PISA are

¹⁰¹Place by looking-move strip, not “too old,” not diploma year. Continuity with adult-math / History Refresh grain. Access date for sources in these notes: **22 September 2026**.

¹⁰²Vendor placement tools (OpenStax previews, Khan practice, CLEP practice, commercial adaptive systems, college guided self-placement) are mirrors. IES / What Works Clearinghouse does not certify them as RCTs of this book.

¹⁰³PIAAC 2023 U.S. adults 16–65: literacy mean **258**; numeracy **249**; Level 1 or below literacy **28%** / numeracy **34%** — adjacent stock, **no science domain**. NSF Indicators / Pew ATP process

not the adult measure — refuse the mash.

Probe ladder (self-check, not a published test):

- **Evidence before feed.** Cover the caption. Name design / absolute risk / sample / limit — or open your own table — before you decide what the card settles. If you still leap to the tidy story, stay here.
- **Honesty card.** Label my measurement / labelled secondary / screenshot / AI-generated / kitchen picture. Fail shape: generated image or invent-data filed as result.
- **Axes / units / sample.** Read what was measured and on whom before the slogan.
- **Claim + warrant + limit.** One sentence claim; pointer into *this* evidence; why it counts; one limit.
- **Systems boundary soft.** Ordinary object: jar, plant pot, HVAC soft, watershed soft — boundary; what left outside on purpose.
- **Fair look soft.** “Can I tell for sure it was the variable I named — what else differed?” Insufficient evidence is valid.
- **Survey / CLEP edges (optional).** Biology or Chemistry breadth — only for adults whose study plan needs credit.

Rule of thumb: if you cannot yet hold a feed card until design and absolute risk are named, you are not ready to skip to a 115-question survey sit. If a kitchen fizz still arrives as Chemistry credit, you are still in honesty-card work, whatever your diploma. Patch the hole in front of the next real task. You may be clean on workplace absolute-risk stickies and rusty on systems boundaries in the same month — that is ordinary, not a character flaw.

items (control **60%**; hypothesis **50%**; updating **66%**) — labelled process neighbour, not a proficiency ladder and not this book’s pretest. Refuse teen NAEP Science / PISA as adult measures.

Dated looking note / stranger-readable table / feed honesty card / denser CER

Keep what a stranger can use.

A dated looking note. Object named. Band named. Move practiced. One claim with a pointer. One limit. The date is 22 September 2026 or whatever day it actually was — not “Q3 science.”

A stranger-readable table with units. Operational definition sticky when needed. Uncertainty crumb when honest. Photo of setup is data context, not a generated result.

A feed / press / study honesty card. Layer labelled; one claim checked; one mismatch or agreement noted. Caption is not the study until evidence is opened.

A denser CER with warrant and named limit. Four boxes or one paragraph. Locatable evidence. Reasoning names an idea. Limit mandatory.

Optional museum / library / nature visit notes. Observe or systems soft on an exhibit, shelf, or park transect; one claim; one limit; optionally pair later with a labelled secondary. A museum hour is not a transcript line and not a certificate.¹⁰⁴

Optional invent-data / generated-result refusal line. What you refused and why — one sentence.

What not to keep as the proof: a random PDF “certificate of completion” from a site that is not transcript credit; a LinkedIn badge; a chatbot fluency seal; a franken-title; a teen Biology/Chemistry/Physics Records export mashed onto adult self-study. Those can be receipts at best. They are not the work.¹⁰⁵ Prefer: dated keeping-track file; community-college noncredit completion letter on letterhead if you earned one; official CLEP Biology or Chemistry score report if you chose that path; actual college transcript.

How much is enough? A line after a micro-session is enough for a first week. A

¹⁰⁴Library / museum / nature visits motivate; pair with a dated keeping-track note. Admission and free days are local — do not invent a national free-museum guarantee. A museum hour is not a transcript line.

¹⁰⁵Fake certificates and invented course titles (Science Refresh for Adults I / Emerging Adults Science I / Adult Science I / Certified Science Literacy) are refused as stranger-readable proof. Prefer dated looking notes, letterhead completions, official score reports, transcripts. Never a teen Biology/Chemistry/Physics Records reprint from *Science for Emerging Adults*.

stack of dated looking strips over a quarter is enough for a later reader. A binder labelled “Science Refresh for Adults, Level 2” is not more lawful than the line. Frequency the evidence actually supports at the commute grain for *this* book is unknown — there is no adult commute science RCT at this exact grain. The honest pitch is often enough that the question is familiar, and dated enough that next Tuesday can find last Tuesday.

If writing itself is the wall, keep the parts anyway: move, object, claim, pointer, limit, date. A human colleague or librarian who already writes clear notes can help with the writing shape. You still own the looking question.

Workplace honesty: a completed OpenStax chapter is preparation, not a license. A Khan streak is practice, not a hiring credential. A podcast season is motivation, not evidence. Redact proprietary employer details before you share a work sample.¹⁰⁶ Kitchen pictures stay labelled not-for-credit — they are not Chemistry credit.

What employers and colleges already read

Short work samples beat badges. Prefer:

- a one-page before/after of a headline check (relative framing → absolute numbers + limit)
- a redacted slide verification (public abstract pointer; no proprietary paste)
- a museum or nature visit note with a claim and a limit
- a stranger-readable table from an ordinary try
- a denser CER paragraph a tutor could follow

Colleges read transcripts, placement results, and — when you chose the path — official CLEP score reports and receiving-college credit policy. They do not read “Certified Science Refresh for Adults.” Community-college noncredit letters are local objects — read the letterhead for what they actually certify.¹⁰⁷

¹⁰⁶Workplace proprietary-data caution: redact employer sheets and confidential details before sharing a sample or pasting into a public consumer tool. Pharmacy-label examples stay units-only — no medical advice.

¹⁰⁷Community-college noncredit “certificates of completion” are local objects — read the college page for what they actually certify.

Optional CLEP Biology / Chemistry

CLEP is optional credit-by-exam when a specific institution will grant credit. It is not required to refresh science literacy for work, health-news hygiene, or civic looking. **Sitting CLEP $\neq$ finishing this book.**¹⁰⁸

CLEP Biology (opened 22 September 2026). Approximately **115** questions / **90** minutes; some pretest items unscored. Roughly equal weight: molecular and cellular biology; organismal biology; population biology. ACE recommendation: credit-granting score **50 / 6** semester hours — **institutions set policy**. Exam fee **\$97** plus test-center or remote fee. Remote proctoring **\$30** on College Board Quick Reference Guide 2026–27.¹⁰⁹

CLEP Chemistry (opened 22 September 2026). Approximately **75** questions / **90** minutes. Structure/states of matter, reactions, stoichiometry, equilibrium, kinetics, thermodynamics, descriptive and experimental chemistry. Online TI-30XS MultiView scientific calculator and periodic table in software. Experimental chemistry about **9%** of content — exam questions *about* lab, **not** a lab year. ACE **50 / 6** hours — institution may differ. Fee **\$97** + fees; remote **\$30** on the same Quick Reference.¹¹⁰

Prep honesty. Confirm the receiving college's credit policy **before** paying. OpenStax + practice nearer the sitting can help; do not invent pass rates or "most colleges accept." Neither CLEP is a kitchen lab substitute. If credit needs a supervised lab, schedule a lab-path neighbor (community-college lab, supervised co-op, museum bench) — schedule honesty, not guilt.

AP science sittings are high-school objects with coordinator logistics — not the adult refresher default. One sentence if you already teach or sit near a school, then

¹⁰⁸Sitting CLEP $\neq$ finishing this book. No diploma required to open this book.

¹⁰⁹College Board CLEP Biology and CLEP Chemistry pages opened **22 September 2026**: fee **\$97.00** each plus administration or remote-proctor fee. Quick Reference Guide for Educators **2026–27**: remote proctoring **\$30**. ACE notes score **50 / 6** semester hours — institutions may differ. Biology **~115 Q / 90 min**; Chemistry **~75 Q / 90 min**; Chemistry **~9%** experimental-chemistry content *about* lab. Confirm live fees before you budget.

¹¹⁰College Board CLEP Biology and CLEP Chemistry pages opened **22 September 2026**: fee **\$97.00** each plus administration or remote-proctor fee. Quick Reference Guide for Educators **2026–27**: remote proctoring **\$30**. ACE notes score **50 / 6** semester hours — institutions may differ. Biology **~115 Q / 90 min**; Chemistry **~75 Q / 90 min**; Chemistry **~9%** experimental-chemistry content *about* lab. Confirm live fees before you budget.

stop.¹¹¹

DANTES funding for eligible service members — confirm locally. Omit unverified coach prices.

You try this week

Open one page in a notebook or a plain file. Date it. Name the looking move you touched. Name the object (feed card, table, exhibit, park, weather page). Write one claim with a pointer and one limit. Mark rebuild or maintenance. Note one refusal if you refused invent-data, a generated result, or a model summary as observation. Write one next cue for Tuesday.

If you have three dated lines already, glance back: which move reappears? That move may need a maintenance week. Which move never appears though life keeps asking for it — a caption you keep swallowing, a kitchen picture you keep filing as credit? That move may need a rebuild week.

If credit is on the table, open *one* receiving-college page and write the CLEP rule in your notebook before you spend \$97. If credit is not on the table, do not let an exam brochure become this week's homework.

That is the record. It is not a diploma. It is not a teen transcript. You can stop when the next life demand is met — and you can return without starting from shame.

¹¹¹AP Biology / Chemistry / Physics / Environmental Science are high-school exam objects — not the adult refresher default. Do not confuse AP score reports with CLEP.

Chapter 9



Figure 11: Three fit cards (\$0 library, OER, optional CLEP) with library card and museum systems postcard — still-life, no logos, no readable UI.

Resources

This chapter names programs a working adult actually meets. It does not rank them. It does not sell them. Fit is cost shape, time shape, social shape, credential shape, depth, safety/lab shape, and tone — whether a commute, a desk, a museum hour, a nature walk, or a weekend sit can use the object this week.

Library, museum, and nature **\$0** first. A free digital survey text before any paid platform. Confirm a current price on the live page before you buy; this chapter does not invent list prices it did not open. Access date for the opened pages named below: **22 September 2026**.

No opened WWC report crowns OpenStax, Khan, CK-12, or any podcast for adult self-study of science. School-labelled looking shapes transfer as *shape* — not as a kitchen franchise and not as a diploma.¹¹²

Never a fake certificate as the year's proof. Never Science Refresh for Adults I as a credit title. Combining two honest objects does not mint one. Podcasts and documentaries are **neighbors, not studies**. AP science is **not** the adult default.

How to read a program

Six questions do the work.

Cost shape. Nothing tonight? Free digital text? Local museum admission or free day? Local adult-ed fee? Tuition? Exam fee (\$97 plus proctor for CLEP)? Subscription?

Time shape. Asynchronous twenty minutes? Commute audio? Scheduled evening class? Gallery hour? Park transect? One sitting after weeks of prep?

Social shape. Alone? Librarian? Instructor? Cohort? Docent? Supervised lab?

Credential shape. None? Local completion letter? Transcript credit? Optional CLEP score report?

¹¹²Institute of Education Sciences / What Works Clearinghouse: no opened adult self-study crowning report for OpenStax Biology/Chemistry/College Physics/Astronomy, Khan Academy science, or CK-12 in this research stream. School-labelled looking research transfers as shape only. Access date for sources in these notes: **22 September 2026**.

Depth shape. Feed honesty card? Survey semester? Lab-path Chemistry? Astronomy soft?

Safety / lab shape. Observation-only? RAMP-needed try? Community lab when home cannot minimize?

Tone shape. Patient refresh? College-credit pace? Contest / AP grain?

What a stranger can still read is the looking move you practiced or the college course you actually took. Rank never. Fit first.

Library, museum, and nature — \$0 first

Library hour — fit card. Cost: **\$0**. Mode: science shelf, quiet desk, printing when needed, research help, sometimes databases or interlibrary loan when a survey title is not on the shelf. Credential: none. Strength: quiet, print, community, ILL. Weakness: variable hours; thin science selection in some branches. Evidence: no national “average library science minutes” study for working adults opened here — quality and hours still vary by branch.¹¹³

Museum / science-center / nature-center hour — fit card. Cost: local admission, free day, or parking (local objects — **omit** a national free-museum guarantee). Mode: outing. Credential: none. Strength: exhibits force systems looking and Observe before caption. Weakness: travel time; one-visit tourism without a dated strip teaches little. Pair with a keeping-track note.¹¹⁴

Nature / park / night-sky — fit card. Cost: **\$0** when public access. Mode: park transect, yard look, sky pattern after Observe (app only after notes). Credential: none. Strength: real objects; dated note possible. Weakness: weather; tick/traffic awareness; easy to walk without a claim sentence. Safety: public access; no cliff edges; no collecting where prohibited; no staring at the sun.

Kitchen \$0 without platforms. Pencil, paper, phone note, public-domain PDF, library print, food-safe picture labelled not-for-credit. Fit when bandwidth or login

¹¹³Public library as first **\$0** social fit — hours and holdings vary. Do not invent national adult science-minutes averages.

¹¹⁴Museum / science-center / nature-center admission and free days are local. Pair Observe or systems looking with a dated keeping-track note. A museum hour is not a transcript line.

fatigue is the blocker. Still skill-based. Title: the looking move or the life object — never Science Refresh for Adults I. Kitchen change pictures are **pictures**, not Chemistry credit.¹¹⁵

Adult use. Pick one short item. Run evidence before feed, Observe–record, systems soft, or denser CER. Write one claim and one limit. Come back Tuesday.

Not. A diploma. A certificate. A substitute for opening evidence.

OpenStax — free digital backbones

Fit card. Cost: **\$0** digital (low-cost print editions exist per OpenStax higher-ed pattern — confirm live print price if you need paper). Mode: college OER survey texts. Credential: none unless a college adopts a text inside a credit course. Strength: stable definitions and problem sets that stay put. Weakness: college density; requires self-pacing; not a looking franchise by itself.¹¹⁶

Opened / catalogue consensus titles for this book:

- **Biology 2e** — two-semester majors scope; evolutionary lens explicit in the preface; free web/PDF; CC BY-NC-SA.
- **Chemistry 2e** — two-semester general chemistry survey.
- **College Physics 2e** — algebra/trig-based (not calculus).
- **Astronomy 2e** — intro non-majors; one- or two-semester.

Adult use. Read a section. Work a few problems. Keep a dated error/looking log. Pair with one looking strip (feed honesty, table pointer, museum systems, park note). Do not treat chapter completion as a license.

¹¹⁵Kitchen pictures labelled not-for-credit; ACS RAMP when heat/chemicals; no unauthorized mixing; lab-path neighbor when home cannot minimize. America's Lab Report integrity neighbor: fake constructed data $\neq$ lab experience.

¹¹⁶OpenStax (Rice University nonprofit OER): Biology 2e preface opened 22 September 2026 (evolutionary lens; free web/PDF; CC BY-NC-SA); Chemistry 2e, College Physics 2e, Astronomy 2e — catalogue / preface consensus. Live details pages can be script-thin — open the chapter you need. Not a diploma by itself.

Not. A diploma by itself. “WWC-approved OpenStax kitchen self-study” — no such opened report. Never the credit title on a résumé.

Khan Academy and CK-12 — free practice neighbors

Khan Academy science — fit card. Cost: \$0. Mode: videos and problem sets across middle/high school science and cosmology/astronomy; AP-labelled tracks exist for exam prep — **AP track is not the adult default spine.** Credential: none. Strength: diagnose a gap, drill it, move on. Weakness: easy to streak without a looking note; not employer ID; not CLEP; not WWC-proven for adult self-study in any report opened this stream.¹¹⁷

CK-12 FlexBooks — fit card. Cost: \$0 digital. Mode: adaptable Biology / Chemistry / Physics / Earth texts. Credential: none. Strength: lighter digital prose when OpenStax feels heavy. Weakness: live feature cards thin on some accesses — open the chapter you need; still not a diploma.¹¹⁸

Pairing shape. Khan for interactive reps; OpenStax for longer problem sets and definitions that stay put; CK-12 when a lighter digital text fits. None replaces a community-college transcript if you need credit. None replaces the talk-to-yourself wait before the feed or the chatbot.

Adult use. Borrow the practice. Log the strip. Do not export “Khan” or “CK-12” as a credit line.

Not. A diploma. A weekly AP dump as the spine of this book. A replacement for eyes on evidence.

¹¹⁷Khan Academy science: free practice; AP-labelled tracks exist — **not adult default.** Not WWC-proven for adult self-study in reports opened this stream. Live hub can be script-heavy — catalogue/blog consensus.

¹¹⁸CK-12 FlexBooks: free digital adaptable science texts — catalogue consensus; live pages thin on some accesses. Not a diploma.

Podcasts and documentaries — neighbors, not studies

A host's tidy summary, a documentary voiceover, and a viral "science fact" card are **later tellings** until you open a named abstract, a labelled table, or your own stranger-readable note and check the claim.

Science podcast — fit card. Cost: often **\$0** with ads or library-app access (local; **omit** unverified subscription prices). Mode: commute audio. Credential: none. Strength: topic interest; repeated exposure to names and ideas. Weakness: host framing is a later telling; easy to confuse narrative polish with evidence; hard to pause for Observe or CER while driving (safety: audio-only awareness is not a looking session behind the wheel).¹¹⁹

Documentary — fit card. Cost: library, broadcast, or platform access (**omit** unverified prices). Mode: weekend sit. Credential: none. Strength: motivation; maps of questions to ask. Weakness: voiceover confidence is not methods.

Adult move after listening or watching (not while driving): write one claim the host or narrator made; open one short abstract, agency figure, or labelled secondary; mark match / tension / unknown. Do not list a podcast season as a peer-reviewed evidence log. Do not cite "as heard on [show]" as study evidence in a keeping-track file without the underlying source. This chapter does not crown any named show.

Not. A study. CLEP. An abstract substitute.

Community college — noncredit, credit, and lab-path

Noncredit / adult-ed science — fit card. Cost: local (omit invented national averages). Mode: evening or weekend class; sometimes a completion letter on letterhead. Credential: local letter — often not transferable credit. Strength: human instructor; cohort; structure. Weakness: schedules; fees vary. Read the college page for what the letter actually certifies.¹²⁰

¹¹⁹Podcasts and documentaries motivate and map questions; they are later tellings until evidence is opened. Not CLEP. Not studies. Omit unverified show or platform prices.

¹²⁰Community-college noncredit / adult-ed / library lecture fees are local — omit invented national averages. Noncredit completion letters are local objects; read the college page.

Credit survey course — fit card. Cost: tuition local (omit unverified). Mode: transcript course; lab section when required. Credential: transcript. Strength: real credit when you need it. Weakness: time load; placement rules. Not a kitchen certificate.

Lab-path neighbor — fit card. Cost: local. Mode: community-college lab, supervised co-op, museum bench. Credential: as the program states. Strength: regulated techniques home cannot minimize (RAMP ceiling). Weakness: scheduling. Use when credit Chemistry or a real technique is the goal — schedule honesty, not guilt.¹²¹

Adult use. Choose by credential need. Keep looking-move practice beside the course — a transcript without evidence-before-feed is still brittle on Tuesday's feed card.

Not. This book's required spine. A substitute for unaided looking.

Optional CLEP Biology / Chemistry

Fit card. Cost: \$97 per exam plus test-center or remote fee; remote proctoring \$30 on College Board Quick Reference Guide 2026–27 (opened 22 September 2026). Mode: credit-by-exam if the receiving institution accepts. Credential: official score report + institution policy. Strength: optional bridge when a college asked for it. Weakness: sitting $\neq$ finishing this book; neither exam is a kitchen lab year.¹²²

CLEP Biology: ~115 questions / 90 minutes; ACE note score 50 / 6 semester hours — institution may differ; molecular/cellular, organismal, population ~equal weight.

¹²¹Kitchen pictures labelled not-for-credit; ACS RAMP when heat/chemicals; no unauthorized mixing; lab-path neighbor when home cannot minimize. America's Lab Report integrity neighbor: fake constructed data $\neq$ lab experience.

¹²²College Board CLEP Biology and CLEP Chemistry pages opened 22 September 2026: \$97.00 each plus administration or remote-proctor fee. Quick Reference Guide for Educators 2026–27: remote proctoring \$30. Biology ~115 Q / 90 min; Chemistry ~75 Q / 90 min; ACE notes 50 / 6 hours — institutions may differ; Chemistry ~9% experimental content about lab. Confirm live fees before budgeting. Sitting $\neq$ finishing this book.

CLEP Chemistry: ~75 questions / **90** minutes; ACE **50** / **6** hours; online TI-30XS MultiView and periodic table in software; ~**9%** experimental-chemistry *content about* lab — not a lab year.

Adult use. Confirm receiving-college policy **before** paying. Prep with OpenStax + looking strips nearer the sitting. Do not invent pass rates or “most employers require CLEP Biology.”

Not. This book’s spine. Chemistry credit from a kitchen picture. AP by another name.

AP and dual enrollment — not adult default

AP Biology / Chemistry / Physics / Environmental Science — fit card. Mode: high-school exam objects with coordinator logistics. Credential: AP score report. Adult use: one sentence if you already teach or sit near a school — then stop. **Not** the adult refresher default.¹²³

Dual enrollment — fit card. Mode: local eligibility / transcript. Credential: college credit when the program says so. **Not** the adult default refresher spine. Fees and rules are local — omit unverified.

Homeschool vendor neighbours and labelled paid paths

Apologia / BJU / Abeka / Elemental / Berean — fit card. Christian HS packaging and conversational/textbook paths. **Not the adult default.** Fence full HS survey packaging, transcript titles, and NCAA stack to *Science for Homeschooling* / *Science for Emerging Adults*. An adult may still read a clear chapter; do not sell a boxed HS year as the Science Refresh path.¹²⁴

¹²³AP Biology / Chemistry / Physics / Environmental Science and dual enrollment are high-school / local objects — **not** the adult refresher default.

¹²⁴*Science for Homeschooling* and *Science for Emerging Adults* own HS survey packaging, transcript titles, and related machinery. One pointer if you also guide a teen — then return to adult

Home Science Tools — kits-for-use soft. Vendor path for supplies and kits usable with many curricula; some products are HS-edge grain. Adult fit: occasional supplies for a supervised try or museum kit analog — not a ranked curriculum. **Omit** unverified kit prices.¹²⁵

Great Courses / streaming survey lectures — labelled vendor. Motivation and map, then evidence on the table. **Omit** unverified 2026 list prices. Not a diploma. Not a looking substitute.

Safety pointers (neighbors)

ACS **RAMP** (Recognize, Assess, Minimize, Prepare) is the safety spine when a try leaves the page.¹²⁶ NSTA and Flinn are labelled safety neighbors for school/lab culture — borrow caution, not a curriculum dump. Default banks in this book stay observation-safe: reading, caption looking, outdoor observation, museum, library. Heat, chemicals, glass, electricity beyond battery toys, living specimens — stop if you cannot minimize. No unmarked bottles. No bleach mixes. No drain cleaners. No alcohol flames. No sealed pressurized fizz bottles. Curiosity is not a permit for unauthorized mixing.

Optional coverage maps (not ranks)

Choose by fit — each map still uses looking moves:

- **Feed-era science hygiene + light content:** honesty cards and abstract-before-headline alternating with short OpenStax/Khan sections.
- **Biology refresh:** OpenStax Biology 2e or Khan HS biology + looking strips; optional CLEP Biology.

self-placement. This book is not a reprint. Apologia / BJU / Abeka / Elemental / Berean = neighbours, not adult default.

¹²⁵Home Science Tools kits-for-use soft; omit unverified 2026 kit prices. Great Courses / streaming survey = labelled vendor neighbours — motivation then evidence.

¹²⁶ACS RAMP; NSTA / Flinn as labelled safety neighbors. Adult present to self; stop if unsafe.

- **Chemistry refresh:** OpenStax Chemistry 2e + looking strips; optional CLEP Chemistry; lab-path if credit needs lab.
- **Physics / Astronomy soft:** College Physics 2e or Astronomy 2e + museum/planetarium notes.
- **Local/public science:** museum, nature center, library databases — deep on place.

“Finish the textbook” is not done-enough. Done-enough is by move strip (Chapter 8).

Pairing shape

Library shelf for survey context; OpenStax / Khan / CK-12 for free digital practice; museum for systems and exhibit captions; nature for Observe and transect; quiet desk for the denser CER; podcast or doc for motivation *after* or *beside* — never instead of — evidence. Lab-path when home cannot meet the safety ceiling. CLEP only when an institution asked. None replaces the talk-to-yourself wait. None replaces unaided looking.

Fit table (seed, not a ranking)

Path	Cost shape (opened / local)	Credential	Adult note
Library / museum	\$0 / local	None	First social fit;
/ nature	admission		dated note
OpenStax	Free digital;	None by itself	Section +
Bio/Chem/Phys/Astr	low-cost print		problems + looking log

Path	Cost shape (opened / local)	Credential	Adult note
Khan / CK-12	\$0	None	Practice neighbor; AP track not default
Podcasts / docs	Often \$0 / local	None	Neighbors, not studies
CC noncredit / credit / lab	Local tuition	Letter or transcript	Lab-path when needed
CLEP Bio / Chem	\$97 + fees; remote \$30	Institution-set credit	Optional edge
AP / dual enrollment	HS / local	Score / transcript	Not adult default
Homeschool vendors	Paid (omit unverified)	None for adults	Neighbours only

You try this week

Name *why* tonight's object fits — usually library / museum / nature **\$0** or OpenStax / Khan / CK-12 before paid. Open one free section or one exhibit/park object that matches the looking move in front of you. Do one unaided parallel try on a feed card, table, weather page, or denser CER crumb. Date the line.

If credit is on the table, open *one* receiving-college page and write the CLEP or placement rule in your notebook before you spend \$97. If credit is not on the table, do not let an exam brochure become this week's homework.

If a podcast filled the commute, write the host's claim before you sleep — then open an abstract or labelled figure on the next desk sit. The file is still the proof. Fit first. Rank never. Start free when free will do. Return to the looking-move strip when a catalog page starts to feel like shopping.

A Note on Sources

Studies named in the chapters are listed in Notes, in one series at the back. That is where the full citations live, so the teaching pages can stay a teaching voice.

Some items the research behind this book did not open stay out of the teaching voice. I did not invent a coefficient, a statute, or a product feature to fill a hole. If a study is in the chapter, it is in the notes. If we could not open it, it is not used as a finding here.

Program hours, exam fees, and product descriptions in the resources chapter are taken from the companies' or agencies' own pages. Access date 22 September 2026.